# User Manual

# DA16200 DA16600 AT Command

## UM-WI-003


### Abstract

*This document describes the use of the AT commands to configure the DA16200 and DA16600.*

# Contents

**Abstract** ..... **1**

**Contents** ..... **2**

**Figures** ..... **4**

**Tables** ..... **4**

**1 Terms and Definitions** ..... **6**

**2 References** ..... **7**

**3 Introduction** ..... **8**

**4 AT Command Development Environment Configuration** ..... **8**

4.1 How to Connect the DA16200 (DA16600) Board ..... 8

4.2 Configure the Serial Port for UART ..... 9

4.3 Configuration for MCU Wake-Up (Optional) ..... 10

**5 AT Command Format** ..... **11**

5.1 Command Format ..... 11

5.2 Basic Command Format ..... 11

5.3 Extended Command Format ..... 11

5.4 Response Format ..... 12

**6 Basic Function Commands** ..... **14**

**7 Network Function Commands** ..... **28**

**8 Wi-Fi Function Commands** ..... **36**

8.1 Wi-Fi Function Commands for WPA3 ..... 57

8.2 Wi-Fi Function Commands for WPA Enterprise ..... 61

8.2.1 WPA-Enterprise Connection Example ..... 63

**9 Advanced Function Commands** ..... **64**

9.1 MQTT Commands ..... 64

9.1.1 MQTT Client Connection Example ..... 78

9.1.2 MQTT TLS Connection Example ..... 79

9.1.3 MQTT Example with DPM ..... 80

9.1.4 MQTT Example: Changing Subscription Topic when Running ..... 82

9.1.5 MQTT Example: Reading Subscription Topic when Running ..... 83

9.1.6 MQTT Example: Using CleanSession=0 ..... 84

9.2 HTTP-Client Commands ..... 92

9.2.1 HTTP-Client Connection Example ..... 95

9.3 HTTP-Server Commands ..... 96

9.3.1 HTTP/HTTPS-Server Start Example ..... 97

9.4 WebSocket-Client Commands ..... 97

9.5 OTA Commands ..... 98

9.5.1 OTA Download Example ..... 105

9.5.2 OTA Download Progress Example ..... 105

9.5.3 OTA Renew Example ..... 105

9.5.4 MCU FW Transport Example ..... 105

9.5.5 SFLASH User Area Address Setting Example ..... 106

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9.5.6 SFLASH READ/COPY/ERASE Example ..... 106
9.6 Zeroconf Commands ..... 106
9.6.1 Zeroconf Example ..... 108
9.7 Provision Commands ..... 109
9.8 Bluetooth® LE Commands ..... 110
**10 Transfer Function Commands ..... 112**
10.1 Socket Commands ..... 112
10.1.1 Data Transfer Commands ..... 116
10.1.2 Data Transfer with DPM ..... 118
10.2 Secure Socket Commands ..... 120
**11 RF Test Function Commands ..... 126**
**12 System and Peripheral Function Commands ..... 132**
12.1 SPI Commands ..... 132
12.2 OTP Commands ..... 133
12.3 XTAL Commands ..... 136
12.4 Flash Dump Commands ..... 136
12.5 GPIO Commands ..... 137
12.6 LED Commands ..... 139
12.7 PWM Commands ..... 140
12.8 ADC Commands ..... 141
12.9 I2C Commands ..... 142
12.10 Sleep Commands ..... 143
12.11 CALWL Commands ..... 144
**13 Examples ..... 145**
13.1 Data Transfer Test ..... 145
13.1.1 TCP Server Socket Test ..... 146
13.1.2 TCP Client Socket Test ..... 147
13.1.3 UDP Socket Test ..... 148
**Appendix A License Information ..... 149**
**Appendix B HTTP API Return Values ..... 150**
B.1 Return value as defined by NetX Duo HTTP ..... 150
B.2 Return value as defined by LWIP HTTP ..... 150
**Appendix C User UART Configuration ..... 152**
C.1 How to Run AT-CMD on UART2 ..... 152
C.2 User UART Configuration ..... 152
C.3 Use Case ..... 153
**Appendix D Example Sequence to Initiate AT-CMD through SPI or SDIO ..... 154**
D.1 How to Run AT-CMD on SPI ..... 154
D.2 How to Run AT-CMD on SDIO ..... 154
D.3 Example Sequence for SDIO interface ..... 155
**Appendix E DA16200 (DA16600) Cipher Suites ..... 156**
**Appendix F Reason Code For Wi-Fi Connection Failure or Disconnection ..... 158**
**Appendix G Fast-reconnect on SLEEP mode 2 ..... 160**
G.1 Technical overview ..... 160

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**Appendix H Bluetooth® LE Coexistence Feature ..... 161**

**Appendix I Wi-Fi Passive-Scan ..... 162**

I.1 Passive-Scan with specified Channel and Scan-time Limit ..... 162

I.2 Passive-Scan Result ..... 162

I.3 Passive-Scan Stop ..... 162

I.4 Passive Scan Sequence ..... 163

**Revision History ..... 164**

# Figures

Figure 1: AT Command Development Environment ..... 8
Figure 2: Check COM Ports on Device Manager ..... 9
Figure 3: Initial Setup to Get Started with “AT” Command ..... 10
Figure 4: GPIO Wake-Up ..... 10
Figure 5: Example Sequence to Initiate MQTT Protocol with DPM ..... 80
Figure 6: Procedure to Send MQTT Messages ..... 81
Figure 7: Procedure to Process MQTT Messages ..... 82
Figure 8: Broker console - CleanSession=1 connection ..... 84
Figure 9: Broker console - CleanSession=0 connection ..... 84
Figure 10: IO Ninja – TCP Client Socket Setting ..... 146
Figure 11: IO Ninja – TCP Server Socket Setting ..... 147
Figure 12: IO Ninja – UDP Socket Setting ..... 148
Figure 13: Example Sequence to Initiate AT-CMD through SDIO Interface ..... 155
Figure 14: Passive Scan Report ..... 163

# Tables

Table 1: Error Response Codes ..... 12
Table 2: Basic Function Command List ..... 14
Table 3: Initiation Response List ..... 27
Table 4: Network Function Command List ..... 28
Table 5: Certificate Command ..... 34
Table 6: Wi-Fi Function Command List ..... 36
Table 7: Wi-Fi Function Response List ..... 56
Table 8: List of WPA3-Relevant Wi-Fi Function Commands ..... 57
Table 9: WPA-Enterprise Wi-Fi Function Commands ..... 62
Table 10: WPA-Enterprise Network Function Command ..... 63
Table 11: MQTT Command List ..... 64
Table 12: MQTT Optional Configuration Commands ..... 74
Table 13: MQTT Response List ..... 76
Table 14 CleanSession and QoS matrix in message Rx ..... 86
Table 15 CleanSession and QoS matrix in message Tx ..... 87
Table 16: HTTP-Client Command List ..... 92
Table 17: HTTP-Client Response List ..... 94
Table 18: HTTP-Server Command List ..... 96
Table 19: WebSocket-Client Command List ..... 97
Table 20: WebSocket-Client Response List ..... 98
Table 21: OTA Command List ..... 98
Table 22: OTA Response List ..... 103
Table 23: OTA Response Code List ..... 104
Table 24: Zeroconf Command List ..... 106
Table 25: Provision Command List ..... 109
Table 26: atcmd_provision_stat Enumeration ..... 110

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Table 27: Bluetooth® LE Command List ..... 110
Table 28: Socket Command List ..... 112
Table 29: Socket Connection Response List ..... 114
Table 30: Data Transmission Command ..... 116
Table 31: Data Reception Responses ..... 117
Table 32: Secure Socket Command List ..... 120
Table 33: RF Test Command List ..... 126
Table 34: RF Test Examples ..... 131
Table 35: SPI Command List ..... 132
Table 36: OTP Command List ..... 133
Table 37: OTP Memory Address for Writing MAC Address ..... 135
Table 38: Size of Memory by XTAL Offset ..... 135
Table 39: XTAL Command List ..... 136
Table 40: Flash Dump Command List ..... 136
Table 41: GPIO Command List ..... 137
Table 42: LED Command List ..... 139
Table 43: PWM Command List ..... 140
Table 44: ADC Command List ..... 141
Table 45: I2C Command List ..... 142
Table 46: Sleep Command List ..... 143
Table 47: CALWL Command List ..... 144

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# 1 Terms and Definitions

| | |
|---|---|
| AP | Access Point |
| ASCII | American Standard Code for Information Interchange |
| AT | Attention |
| CA | Certificate Authority |
| CID | Client ID |
| CMD | Command |
| COM | Communication Port |
| CRC | Cyclic Redundancy Check |
| CW | Continuous Wave |
| DHCP | Dynamic Host Configuration Protocol |
| DPM | Dynamic Power Management |
| FW | Firmware |
| GPIO | General Purpose Input Output |
| HTTP | Hypertext Transfer Protocol |
| ICMP | Internet Control Message Protocol |
| IEEE | Institute of Electrical and Electronics Engineers |
| IP | Internet Protocol |
| LMAC | Low MAC |
| MAC | Medium Access Control |
| MCU | Micro Controller Unit |
| MQTT | Message Queuing Telemetry Transport |
| NVRAM | Non-Volatile RAM |
| OTA | Over the Air |
| OTP | One-Time Programmable memory |
| PBC | Push Button Connection |
| PC | Personal Computer |
| PER | Packet Error Rate |
| QoS | Quality of Service |
| RTC | Real-Time Clock |
| RTOS | Real-Time Operating System |
| RTS | Request to Send |
| RX | Receive |
| SDK | Software Development Kit |
| SLIB | System Library |
| SNTP | Simple Network Time Protocol |
| SPI | Serial Peripheral Interface |
| SSID | Service Set Identifier |
| STA | Station |
| TCP | Transport Control Protocol |
| TIM | Traffic Indicator Map |
| TLS | Transport Layer Security |
| TX | Transmit |
| UART | Universal Asynchronous Receiver-Transmitter |
| UDP | User Datagram Protocol |
| USB | Universal Serial Bus |


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| | |
|---|---|
| URL | Universal Resource Locator |
| WEP | Wired Equivalent Privacy |
| WLAN | Wireless Local Area Network |
| WMM | Wi-Fi Multimedia |
| WPA | Wi-Fi Protected Access |
| WPS | Wi-Fi Protected Setup |
| XTAL | Crystal |

# 2 References

[1] DA16200 EVK User Manual, Renesas Electronics
[2] AN-WI-002, DA16200 DA16600 Bluetooth® LE Coexistence, Renesas Electronics


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# 3 Introduction

Configuration and control of the DA16200 (DA16600) is provided through an ASCII based command string called “AT Command”. “AT Command” is a standard that was originally defined by Hayes Microcomputer for controlling smart modems and is widely used in many products.

AT is an abbreviation of “Attention”, which means to take note of or fix one's sight upon something. An example of an AT Command is “ATZ” which instructs a modem to become initialized and return to a state with no command input.

An AT Command has a very simple structure consisting of a prefix “AT” concatenated with a command string. This is a very convenient method for sending a series of commands over a serial interface such as a UART. Commands may consist of capital letters, lowercase letters, spaces and some special characters.

# 4 AT Command Development Environment Configuration

## 4.1 How to Connect the DA16200 (DA16600) Board

This section describes the installation procedure for the drivers, the configuration of the serial port, and all necessary steps to set up and check the connection with the PC.

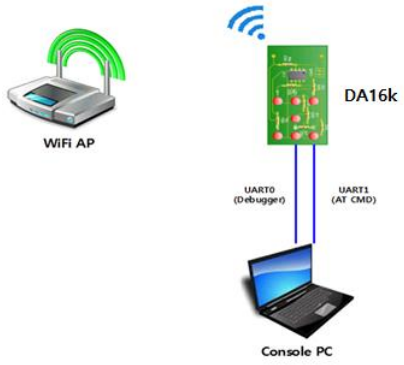



**Figure 1: AT Command Development Environment**

On first connection to a host PC with Microsoft Windows as operating system, the system will detect several devices and will automatically install all necessary drivers. If the driver is not automatically installed, then get the driver from the following URL:
http://www.ftdichip.com/Drivers/CDM/CDM21224_Setup.zip.

There are two virtual COM ports created by the Windows driver. The first COM port (lower number, COM69 in Figure 2) provides a UART interface for debugging or firmware download between the PC

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and the DA161200. The second (higher number, COM70 in Figure 2) is used for AT COMMAND as shown in Figure 2.

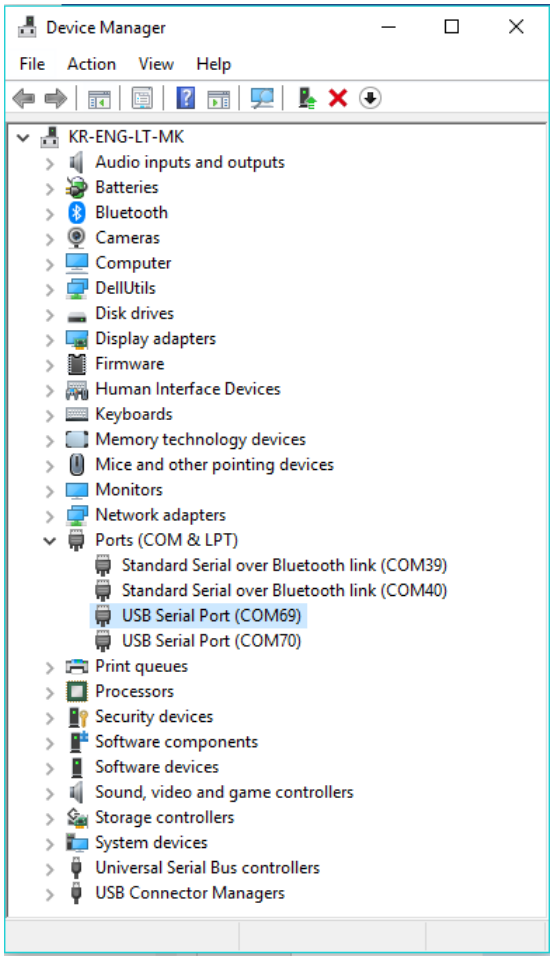



**Figure 2: Check COM Ports on Device Manager**

## 4.2 Configure the Serial Port for UART

On a Windows Host, the utility **Tera Term** is used to connect to Ref. [1].

**Tera Term** is a free terminal emulator (communication program) that supports multiple communication including serial port connections.

1. Download Tera Term from https://ttssh2.osdn.jp/.
2. Run the teraterm-x.yy.exe.
3. Follow the installation wizard.

To make sure that the communication between the DA16200 (DA16600) EVK and the host PC is established correctly, check the UART connection between the two nodes. Do the following steps:

1. Use a USB cable to connect the DA16200 (DA16600) EVK to the PC.
2. Make sure that the PC discovered the two serial ports in Windows Device Manager as shown in Figure 2. The higher COM port number is connected to UART1.
3. Open **Tera Term** from the Windows Start menu.
4. In the **Tera Term: New connection** Renesas Electronics:
   a. Select **Serial**.
   b. Select the COM Port to use.
   c. Click **OK**.


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5. Select **Setup** > **Serial Port** and configure the UART port with the parameters as shown in Figure 3. Select the higher COM port number as discovered in step 2.

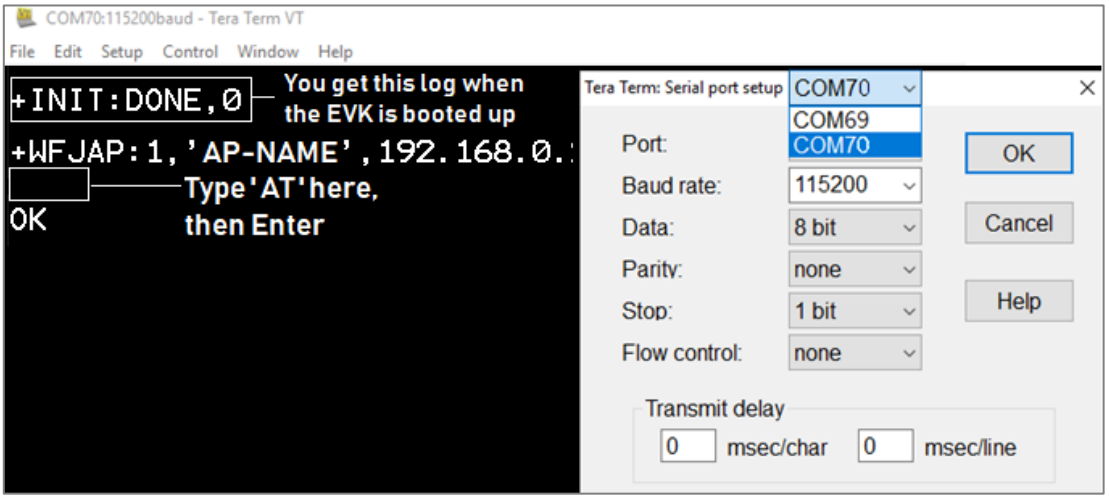



**Figure 3: Initial Setup to Get Started with "AT" Command**

## 4.3 Configuration for MCU Wake-Up (Optional)

Depending on the application scenarios, both MCU and DA16200 (DA16600) may want to be in the SLEEP state and MCU wants to be woken up (by DA16200 (DA16600)) when DA16200 (DA16600) wakes up from DPM Sleep. This can be achieved with the MCU wake-up feature of the DA16200 (DA16600).

To use the MCU wake-up feature, connect pin GPIO_11 of the DA16200 (DA16600) to the wake-up pin on the MCU. Then, when the DA16200 (DA16600) wakes up, GPIO_11 becomes an Output and is set to High (Active High) to trigger the wake-up of the MCU. The wake-up PIN of MCU should be configured to detect the rising edge of GPIO_11 for wake-up.

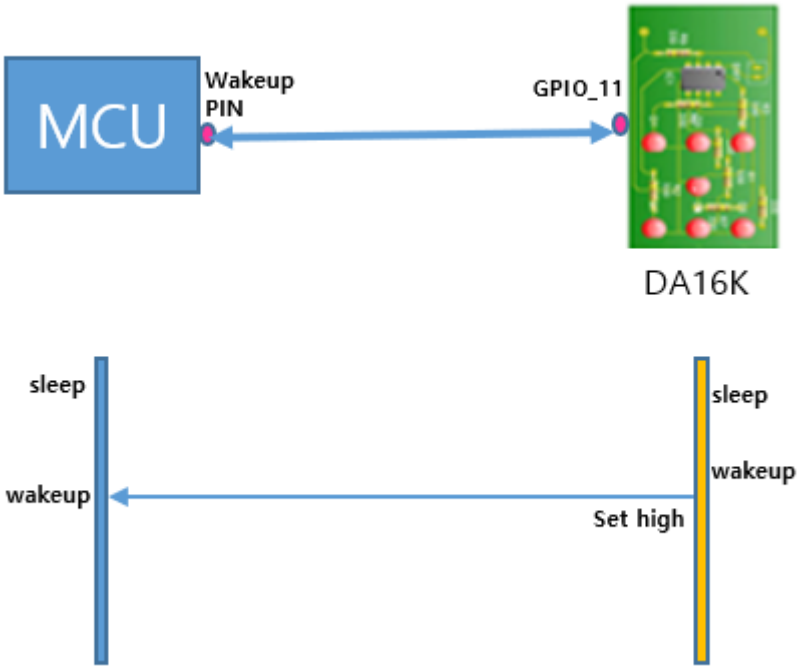



**Figure 4: GPIO Wake-Up**


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# 5 AT Command Format

## 5.1 Command Format

- <CMD>[=|<space>][[']<PARAM_1>[']],[[']<PARAM_2>[']],…
  - [=|<space>] : delimiter between a command <CMD> and its parameters <PARAM>. Either '=' or <space> can be used.
  - comma (,) : delimiter between parameters.
  - enclosing single quote (') : delimiter for a string parameter where single quote (') or comma (,) character is included in the string parameter.

## 5.2 Basic Command Format

- Write CMD
  Basic command execution

  ATXX

  For example: `ATZ`

  `OK`

- Read CMD
  Get the parameter values of the command
  ATXX=?

  For example: `ATQ=?`

  Display result on

  `OK`

## 5.3 Extended Command Format

- Write CMD
  Extended command execution
  AT+XXX=<param1>,<param2>,<param3>,<param4>…<paramN>

  For example: `AT+NWIP=0,172.16.0.100,255.255.255.0,172.16.0.1`

  `OK`

If the SSID contains a comma or single quote, the SSID must be enclosed in single quotes.

For example:

```
SSID = MY,SSID'CS
sec = 4
idx = 2
Password = N12345678
```

Is encoded as:

```
AT+WFJAP='MY,SSID'CS',4,2,N12345678
OK
```

**NOTE**

The use of a single-quote followed by a comma in a parameter is prohibited.
ie. `AT+WFJAP='MY,SSID',CS',4,2,N12345678` is invalid

- Read CMD
  Get the parameter values of the command

  AT+XXX=?

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For example: AT+NWIP=?

```
+ANIP:172.16.0.17,255.255.255.0,172.16.0.1

OK
```

**NOTE**

Not all commands support the AT+XXX=? query function such as AT+RESTART, ATF, etc...
Check the command table for the valid operation of each command.

## 5.4 Response Format

- Start-up response
  The AT command response when DA16200 (DA16600) is reset
  <CR><LF>+INIT:DONE,<mode><CR><LF>
  The AT command response when DA16200 (DA16600) wakes up from DPM sleep
- Basic response
  Basic response gives the command result and is accompanied by a carriage return and a line feed
  <CR><LF>+INIT:WAKEUP,<type><CR><LF>
- Normal response
  <CR><LF>OK<CR><LF>
- Error response
  <CR><LF>ERROR:<error code><CR><LF>
- Extended response
  Extended response gives the command setting values and is followed by a basic response
  <CR><LF>+XXX:[value1],[value2],…
  <CR><LF>OK<CR><LF>

**NOTE**

When an MCU (AT-CMD Host) waits for a response of a command (for those commands that give extended response as well) to take the next action, it should wait for both *normal* response (**OK**/**ERROR**) and *extended* response (also known as **Operation Result**).

- Error response codes

**Table 1: Error Response Codes**

| Code | Description |
|---|---|
| -1 | Unknown command |
| -2 | Insufficient parameter |
| -3 | Too many parameters |
| -4 | Wrong parameter value |
| -5 | Unsupported function |
| -6 | Not connected to an AP or Communication peer |
| -7 | No result |
| -8 | Response buffer overflow |
| -9 | Function is not configured |
| -10 | Command timeout |
| -11 | NVRAM write failure |
| -12 | Retention memory write failure |


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| | |
|---|---|
| -99 | Unknown error |


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# 6 Basic Function Commands

**Table 2: Basic Function Command List**

| Command | Parameters | Description |
|---|---|---|
| ? | (none) | Show AT command usage. |
| | *Example*<br>`?`<br>`AT Commands:`<br>`  ?`<br>`  - No example for ?`<br>`  HELP=<command>`<br>`  - Print help message.`<br>`  AT`<br>`  - Attention command`<br>`  AT+`<br>`  - List available commands`<br>`  ATZ`<br>`  - AT command initialize`<br>`  ATF`<br>`  - Restore to Factory mode (NVRAM clean)`<br>`  ATE`<br>`  - Command echo`<br>`  ATQ`<br>`  - Result Codes On/Off`<br>`  AT+RESTART`<br>`  - System Restart`<br>`--- Middle omission ---`<br>`  AT+TRSAVE`<br>`  - Save current status of all session`<br>`=== User AT-CMD ======`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| help | <cmd_name> | AT command name to query for usage info |
| | (none) | Same with the “?” command. |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>HELP<br>AT Commands:<br>?<br>- No example for ?<br>HELP=<command><br>- Print help message.<br>AT<br>- Attention command<br>AT+<br>- List available commands<br>ATZ<br>- AT command initialize<br>ATF<br>- Restore to Factory mode (NVRAM clean)<br>ATE<br>- Command echo<br>ATQ<br>- Result Codes On/Off<br>AT+RESTART<br>- System Restart<br>…<br>--- *Middle omission* ---<br><br>OK<br><br>HELP=ATE<br>ATE<br>- Command echo<br>OK<br><br>*Note*<br>● Enabled by default in the SDK | |
| AT+ | (none) | Show AT command list. |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>AT+<br>AT<br>AT+<br>ATZ<br>ATF<br>ATE<br>ATQ<br>AT+RESTART<br><br>--- *Middle omission* ---<br><br>AT+TRSAVE<br>OK<br>*Note*<br>● Enabled by default in the SDK | |
| ATZ | (none) | Initialize AT command. |
| | *Example*<br>ATZ<br><br>Display result on<br>Echo off<br>OK<br>*Note*<br>● Enabled by default in the SDK | |
| ATF | (none) | DA16200 (DA16600) factory reset.<br>Response: "+INIT:DONE,0" |
| | *Example*<br>ATF<br><br>+INIT:DONE,0<br>*Note*<br>● Enabled by default in the SDK<br>● All NVRAM parameters including Wi-Fi profile (Soft-AP or STA) settings are removed, DUT restarts, and “+INIT:DONE,0” will be received | |
| ATE | (none) | ECHO on/off. |
| | ? | Shows Echo status - on/off |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`ATE`<br><br>`Echo on`<br>`OK`<br><br>`ATE`<br><br>`Echo off`<br>`OK`<br><br>`ATE=?`<br><br>`Echo on`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| ATQ | (none) | Set displaying result code on/off. |
| | ? | Shows displaying result code on/off state |
| | *Example*<br>`ATQ`<br>`Display result off`<br><br>`ATQ=?`<br>`Display result on`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| ATB | <baudrate><br>[[,<databits>]<br>[,<parity>]<br>[,<stopbits>]<br>[,<flow control>]] | Set UART parameters (the main purpose is to change baud rate)<br><baudrate>: 9600/19200/38400/57600/115200/230400/460800/921600<br><databits>: [optional], 5/6/7/8 (Default)<br><parity>: [optional], n (None, Default)/e (Even)/o (Odd)<br><stopbits>: [optional], 1 (Default)/2<br><flow control>: [optional], 0 (Default)/1 |
| | ? | Shows the current baudrate |
| | *Example*<br>`ATB=230400`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● If __USER_UART_CONFIG__ is enabled in SDK (See Appendix A), this command will be disabled. | |
| AT+RESTART | (none) | System restart. |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+RESTART`<br>`OK`<br><br>`+INIT:DONE,0`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+RESET | (none) | System reset. Go to the Boot mode ([MROM] prompt). |
| | *Example*<br>`AT+RESET`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● Once the system goes to MROM mode, AT command is not available, therefore, MCU needs to force POR booting or enter 'boot' command via UART0 console | |
| AT+CHIPNAME | (none) | Get chip name, DA16200 or DA16600. |
| | *Example*<br>`AT+CHIPNAME`<br>`+CHIPNAME:DA16200`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+VER | (none) | Get version info.<br>For SDK V2.x.x.x<br>Response: +VER:<ramlib version>,<ptim version>,<main version><br>For SDK V3.x.x.x<br>Response: +VER:<main version> |
| | *Example*<br>`AT+VER`<br>`+VER:RTOS-GEN01-01-14648-000000,SLIB-GEN01-01-14645-000000`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+SDKVER | (none) | Get the SDK Version info.<br>Response:<br>+SDKVER:<major >.<minor >.<revision>.<eng_number><br><major>: SDK major number<br><minor>: SDK minor number<br><revision>: SDK Revision number<br><eng_number>: SDK engineering number |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+SDKVER`<br>`+SDKVER:2.4.2.0`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+TIME | <date>,<time> | Set the current time.<br><date>: yyyy-mm-dd<br><time>: hh:mm:ss<br>Response: OK or ERROR |
| | ? | Get the current time.<br>Response: +TIME:<yyyy-mm-dd> <hh:mm:ss> |
| | *Example*<br>`AT+TIME=2021-07-15,16:14:30`<br>`OK`<br><br>`AT+TIME=?`<br>`+TIME:2021-07-15,16:14:32`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+RLT | (none) | Get system running time.<br>Response: +RLT:<days>,<hh:mm.ss> |
| | *Example*<br>`AT+RLT`<br>`+RLT:0,01:06.18`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+TZONE | <sec> | GMT Time zone setting (-43200 ~ 43200).<br><sec>: Time zone setting parameter.<br>Response: OK or ERROR |
| | ? | Get GMT Time zone parameter.<br>Response: +TZONE:<sec> |


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Example<br>AT+TZONE=?<br>+TZONE:0<br>OK<br><br>AT+TZONE=32400<br>OK<br><br>AT+TZONE=?<br>+TZONE:32400<br>OK<br>Note<br>● Enabled by default in the SDK<br>● The <sec> parameter must be a multiple of 60 seconds. If the value for <sec> is not a multiple of 60 seconds then the remainder is discarded.</td></tr>
<tr><td rowspan="2">AT+DEFAP</td><td>(none)</td><td>All profiles in NVRAM are removed and set up in Soft-AP mode with the default configuration. To initialize the Soft-AP interface, the system will reboot automatically.<br>Response: OK or ERROR (reboot)</td></tr>
<tr><td colspan="2">Example<br>AT+DEFAP<br>OK<br><br>+INIT:DONE,1<br>Note<br>● Enabled by default in the SDK<br>● Default configuration:<br>○ SSID: DA16200 (DA16600)_ XXXXXX (for example, 9FFCF3: the last three hexadecimal values of the board’s MAC address)<br>○ Authentication: WPA2/CCMP<br>○ IP address: 10.0.0.1<br>○ Netmask: 255.255.255.0<br>○ Gateway: 10.0.0.1<br>○ PSK: 12345678<br>○ DHCP server started.<br>– DHCP range: 10.0.0.2 ~ 10.0.0.11<br>– DHCP DNS: 8.8.8.8<br>● To query the configuration status, AT+WFSAP and/or AT+NWDHR can be used</td></tr>
<tr><td rowspan="2">AT+BIDX</td><td><idx></td><td>Set Boot index.<br><idx>: Boot index (0 or 1).<br>Response: OK or ERROR</td></tr>
<tr><td>?</td><td>Get the current Boot index.<br>Response: +BIDX:<0|1></td></tr>
</table>

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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Example<br>AT+BIDX=?<br>+BIDX:0<br>OK<br><br>AT+BIDX=1<br>OK<br><br>AT+BIDX=?<br>+BIDX:1<br>OK<br>Note<br>● Enabled by default in the SDK<br>● System restart requires to reflect boot index. “AT+RESTART” command can be used to restart the system</td></tr>
<tr><td rowspan="3">AT+DPM</td><td><dpm><br>[,<nvm_only>]</td><td>Set DPM on/off. To make DPM mode (On/Off) take effect, the system restart is needed.<br><dpm>: 0 (Off), 1 (On).<br><nvm_only>: 1 (write dpm mode to nvram only, and not reboot), 0 or not specified (change dpm mode and reboot)<br>Response: OK or ERROR</td></tr>
<tr><td>?</td><td>Get the current DPM setting.<br>Response: +DPM:<0|1></td></tr>
<tr><td colspan="2">Prerequisite<br>Only available in station mode.<br>Example<br>AT+DPM=?<br>+DPM:0<br>OK<br><br>AT+DPM=1 ; DPM enables and system restarts automatically.<br>OK<br><br>+INIT:DONE,0<br><br>AT+DPM=1,1 ; DPM enable without system restarts.<br>OK<br><br>AT+DPM=?<br>+DPM:1<br>OK<br>Note<br>● Enabled by default in the SDK<br>● DPM configuration is stored in NVRAM<br>● DA16200 (DA16600) is restarted if the “<nvram_only>” parameter is zero or not specified and AT command response is OK<br>○ +INIT:DONE,0 message is sent when DA16200 (DA16600) boots up</td></tr>
</table>


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| Command | Parameters | Description |
|---|---|---|
| | ○ If the usage of the AT command is not valid, then DA16200 (DA16600) sends an **ERROR** message without restarting<br>● If the “nvram_only” parameter is “1”, then restart the system manually using “AT+RESTART”<br>● When DA16200 (DA16600) reboots, DA16200 (DA16600) tries to connect to the AP if the Wi-Fi connection information is available in the NVRAM<br>○ +WFJAP:0,<reason> or +WFJAP:1,'<SSID>',<IP Address> as result of Wi-Fi connection<br>○ If Wi-Fi connection fails during boot-up due to some unexpected condition (for example, AP is offline, temporary communication issue with AP, wrong password is stored, and so on), +WFJAP:x may not be sent immediately and takes some time, in which case, wait until +WFJAP:x is received. When a timeout occurs, depending on the application use case, either cancel the connection trial (AT+WFQAP) or retry the connection with the right info (AT+WFJAP/AT+WFJAPA)<br>● If MQTT is configured, DA16200 (DA16600) tries to connect to the MQTT broker after a Wi-Fi connection is established. The Operation Result – +NWMQCL:0 or +NWMQCL:1 – is sent over UART1 as a result<br>● DA16200 (DA16600) operates DPM if it is set to 1 (**TRUE**)<br>○ If Wi-Fi connection is NOT established in DPM mode, DA16200 (DA16600) enters an abnormal DPM operation<br>○ ** Abnormal DPM operation:<br>While DA16200 (DA16600) operates in DPM sleep, DA16200 (DA16600) executes an Abnormal DPM operation if DA16200 (DA16600) is in a “disconnected” state with the specified AP for some reason<br>Abnormal DPM works like this:<br>DA16200 (DA16600) enters sleep with a predefined RTC timer (This is called Abnormal DPM RTC Timer) configured if the AP connection fails. If DA16200 (DA16600) is woken up by the Abnormal DPM RTC timer, DA16200 (DA16600) tries to connect to the specified AP within a predefined period and sleeps again for a predefined time. The DA16200 library provides default predefined values for Abnormal DPM, but users can modify the relevant parameters based on their application use case scenarios<br>○ If the Wi-Fi connection is established but MQTT connection is NOT established (if MQTT is enabled), DA16200 (DA16600) tries to connect to the MQTT broker several times and enters DPM Sleep based on MQTT’s abnormal DPM operation | |
| AT+DPMKA | <period> | Set DPM keepalive period.<br><period>: Keepalive period (millisecond, 0 ~ 600000).<br>Response: OK or ERROR |
| | ? | Get DPM keepalive period.<br>Response: +DPMKA=<millisecond> |
| | (none) | |


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Example<br>AT+DPMKA<br>+DPMKA:30000<br>OK<br><br>AT+DPMKA=5000<br>OK<br><br>AT+DPMKA=?<br>+DPMKA:5000<br>OK<br>Note<br>● Enabled by default in the SDK<br>● The configuration is stored in NVRAM<br>● System restart is required to take effect</td></tr>
<tr><td rowspan="3">AT+DPMTIMWU</td><td><count></td><td>Set DPM TIM wake-up count.<br><count>: TIM wake-up count (1 ~ 65535).<br>Response: OK or ERROR</td></tr>
<tr><td>?<br>(none)</td><td>Get DPM TIM wake-up count.<br>Response: +DPMTIMWU=<count></td></tr>
<tr><td colspan="2">Example<br>AT+DPMTIMWU<br>+DPMTIMWU:10<br>OK<br><br>AT+DPMTIMWU=20<br>OK<br><br>AT+DPMTIMWU=?<br>+DPMTIMWU:20<br>OK<br>Note<br>● Enabled by default in the SDK<br>● The configuration is stored in NVRAM<br>● System restart is required to take effect</td></tr>
<tr><td rowspan="2">AT+DPMUSERWU</td><td><time></td><td>Set DPM user wake-up time.<br><time>: User wake-up period (second, 0 ~ 86400).<br>Response: OK or ERROR</td></tr>
<tr><td>?<br>(none)</td><td>Get DPM user wake-up time.<br>Response: + DPMUSERWU =<second></td></tr>
</table>


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+DPMUSERWU`<br>`+DPMUSERWU:0`<br>`OK`<br><br>`AT+DPMUSERWU=300`<br>`OK`<br><br>`AT+DPMUSERWU=?`<br>`+DPMUSERWU:300`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● The configuration is stored in NVRAM<br>● System restart is required to take effect | |
| AT+CLRDPMSLPEXT | (none) | Set the user application not to enter DPM sleep.<br>Response: OK or ERROR |
| | *Prerequisite*<br>It works in DPM mode.<br>*Example*<br>`AT+CLRDPMSLPEXT`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● A host should execute this command within 200ms after waking up the DA16200 (DA16600) through the external wake-up pin, otherwise, DA16200 (DA16600) will go into DPM Sleep | |
| AT+SETDPMSLPEXT | (none) | Set the user application ready to enter DPM sleep.<br>Response: OK or ERROR |
| | *Prerequisite*<br>It works in DPM mode.<br>*Example*<br>`AT+SETDPMSLPEXT`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● If DA16200 (DA16600) is woken up by an external wake-up signal and the “AT+CLRDPMSLPEXT” command is executed, this command should be issued once every job you wanted to do is done. If this command is not run after the job is done, DA16200 (DA16600) will not enter DPM Sleep | |
| AT+SETSLEEP2EXT | <period>,<retain_dpm_memory> | Enter DPM Sleep 2 mode for the period specified.<br><period>: wake-up timeout, in second(s). Max. period: 3600<br><retain_dpm_memory>: 1 (retain), 0 (not retain)<br>Response: OK or ERROR |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+SETSLEEP2EXT=10,1`<br>`OK`<br>`+INIT:DONE,0`<br>*Note*<br>● Enabled by default in the SDK<br>● DA16200 (DA16600) can be woken up by RTC_WAKE_UP while in sleep by AT+SETSLEEP2EXT<br>● DA16200 (DA16600) sends "+INIT:DONE:0" when it wakes up<br>● A value of 0 for the <period> parameter sets the DPM to wakeup only when an RTC_WAKE_UP event occurs. | |
| AT+SETSLEEP1EXT | *Deprecated*. | |
| | <retain_ dpm_memory> | Enter DPM Sleep 2 mode.<br><retain_dpm_memory>: 1 (retain), 0 (not retain)<br>Response: OK or ERROR |
| | *Example*<br>`AT+SETSLEEP1EXT=1`<br>`OK`<br>`+INIT:DONE,0`<br>*Note*<br>● This command is the same as AT+SETSLEEP2EXT with the period set to 0.<br>● It is recommended to use the AT+SETSLEEP2EXT command instead of this one.<br>● Enabled by default in the SDK<br>● DA16200 (DA16600) can only be woken up by RTC_WAKE_UP or GPIO which has been assigned as a wake-up source<br>● DA16200 (DA16600) sends "+INIT:DONE:0" once it wakes up | |
| AT+GETFASTCONN | (none) | Get the Wi-Fi Fast-reconnection mode status value<br>Response: +GETFASTCONN:<0\|1> |
| | *Example*<br>`AT+GETFASTCONN`<br>`+GETFASTCONN:0`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● See Appendix G for "Wi-Fi Fast-reconnect" for the DA16200/DA16600 | |
| AT+SETFASTCONN | <flag> | Enable/Disable the Wi-Fi Fast-reconnection mode<br><mode>: 0 (Disable), 1 (Enable)<br>Response: OK or ERROR |
| | *Example*<br>`AT+SETFASTCONN=1`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK v3.2.3.0 or later.<br>● Refer Appendix G for "Wi-Fi Fast-reconnect" for the DA16200/DA16600 | |


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| Command | Parameters | Description |
|---|---|---|
| AT+MCUWUDONE | (none) | Notify that the MCU wakes up completely. After this command is received, DA16200 (DA16600) starts to send messages to the MCU (that is, MCU should send this command immediately after executing “External wakeup”.)<br>Response: OK or ERROR |
| | *Example*<br>`AT+MCUWUDONE`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● When DA16200 (DA16600) receives the command, it starts to send messages to the MCU<br>● MCU should send this command immediately when it receives a notification like “+INIT:WAKEUP,UC”<br>○ If the “__DPM_TEST_WITHOUT_MCU__” is defined, then MCU does not need to send this command which means it is assumed that MCU is always ready to read a message(response) from DA16200 (DA16600) | |
| AT+HOSTINITDONE | (none) | Notifies the DA16200 that the MCU has completed initialization. (In the case of the SDIO interface, the MCU must send this command immediately after initialization.) The DA16200 returns its initialization status as a response. Refer to Table 3.<br>Response: +INIT:DONE,<mode><br>or +INIT:WAKEUP,<type>) |
| | *Example*<br>`AT+HOSTINITDONE`<br>`+INIT:DONE,0`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+DPMABNWFCCNT | <count> | Set Wi-Fi Connection Retry counts until System enters DPM Abnormal Sleep.<br><count>: 0 (This feature not used. DPM Abnormal Sleep scheme is followed), 1 to 6 (Wi-Fi Connection Retry count)<br>Response: OK or ERROR |
| | ? | Get the current DPM Abnormal Wi-Fi Connection Retry counts set<br>Response: +DPMABNWFCCNT:<count> |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>*; If Wi-Fi connection trials are not successful two times in a row, the system goes to DPM Abnormal sleep.*<br><br>`AT+DPMABNWFCCNT=2`<br>`OK`<br><br>`AT+DPMABNWFCCNT=?`<br>`+DPMABNWFCCNT:2`<br>`OK`<br>*Note*<br>● Disabled by default in the SDK<br>● If __WF_CONN_RETRY_CNT_ABN_DPM__ is enabled in the SDK (config_generic_sdk.h), this command will be enabled<br>● The configuration is stored in NVRAM<br>● If the cause of the Wi-Fi connection failure is “Wrong password” input, and if the application wants to cancel the auto-reconnect trial right away, __WIFI_CONN_RETRY_STOP_AT_WK_CONN_FAIL__ should be defined in sys_common_features.h | |

**Table 3: Initiation Response List**

| Response | Parameters | Description |
|---|---|---|
| +INIT | DONE,<mode> | DA16200 (DA16600) booting is complete.<br><mode>: 0 (STA), 1 (Soft-AP)<br>For example: +INIT:DONE,0 |
| | WAKEUP,<type> | DA16200 (DA16600) wake-up is complete from DPM SLEEP state.<br><type> wake-up type<br>● UC: Unicast packet received<br>● NOBCN: No beacon from the connected AP<br>● DEAUTH: Disconnected from the connected AP<br>● EXT: External wakeup<br>● RTC: By a timer registered<br>For example: +INIT:WAKEUP,UC |


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# 7 Network Function Commands

**Table 4: Network Function Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+NWIP | <iface>,<ip_addr>,<br><netmask>,<gw> | Set the IP address.<br><iface>: WLAN interface. 0 (WLAN0, STA), 1 (WLAN1, Soft-AP)<br><ip_addr>: IP Address.<br><netmask>: Subnet mask.<br><gw>: Gateway.<br>Response: OK or ERROR |
| | ? | Get the IP address of the current WLAN interface.<br>Response: +NWIP: <iface>,<ip_addr>,<netmask>,<gw> |
| | (none) | |
| | *Example*<br>`AT+NWIP=0,192.168.0.100,255.255.255.0,192.168.0.1`<br>`OK`<br><br>`AT+NWIP`<br>`+NWIP:0,192.168.0.100,255.255.255.0,192.168.0.1`<br>`OK`<br><br>`At+NWIP=?`<br>`+NWIP:0,192.168.0.100,255.255.255.0,192.168.0.1`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● In Soft-AP mode, if the IP configuration is changed while the DHCP Server is running, then the DHCP Server must be restarted using the AT+RESTART or AT+NWDHS=0 > AT+NWDHS=1 command. | |
| AT+NWDNS | <dns_ip> | Set the DNS server IP address of STA interface.<br><dns_ip>: DNS server IP address.<br>Response: OK or ERROR |
| | ? | Get the DNS server IP address of STA interface.<br>Response: +NWDNS:<dns_ip> |
| | (none) | |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+NWDNS=8.8.8.8`<br>`OK`<br><br>`AT+NWDNS`<br>`+NWDNS:8.8.8.8`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● If AT+NWDNS=? is run under DHCP mode, it returns the DNS IP address from DHCP provision data regardless of whatever DNS IP address you set with AT+NWDNS=<dns_ip>. ERROR:-7 (“No result” or “Not configured”) can be returned if there’s no DHCP provision data existing.<br>● If AT+NWDNS=? is run under Static IP mode, it returns the DNS IP address from AT+NWDNS=<dns_ip> that you run previously or default one.<br>● If AT+NWDNS=<dns_ip> is run under DHCP mode, and to take effect in Static IP mode, system restart is required | |
| AT+NWDNS2 | <dns_ip> | Set the 2nd DNS server IP address of STA interface.<br><dns_ip>: DNS server IP address.<br>Response: OK or ERROR |
| | ? | Get the 2nd DNS server IP address of STA interface.<br>Response: +NWDNS2:<dns_ip> |
| | (none) | |
| | *Example*<br>`AT+NWDNS2=8.8.8.8`<br>`OK`<br><br>`AT+NWDNS2`<br>`+NWDNS2:8.8.8.8`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+NWHOST | <name> | Get the host IP address by name.<br><name>: Domain name.<br>Response: +NWHOST:<ip> |
| | *Example*<br>`at+nwhost=www. Renesas Electronics-semiconductor.com`<br>`+NWHOST:54.192.175.64`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+NWPING | <iface>,<dst_ip>,<br><count> | Ping test.<br><iface>: WLAN interface. 0 (WLAN0), 1 (WLAN1).<br><dst_ip>: Target IP address.<br><count>: The number of ICMP message transmissions.<br>Response: +NWPING:<sent_count>,<recv_count>,<br><avg_time>,<min_time>,<max_time> |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+NWPING=0,192.168.0.1,4`<br>`+NWPING:4,4,0,0,0`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+NWDHC | <dhcpc> | Start/Stop the DHCP client.<br><dhcpc>: 0 (stop), 1 (start).<br>Response: OK or ERROR |
| | ? | Get the DHCP client status.<br>Response: +NWDHC:<dhcpc> |
| | (none) | |
| | *Prerequisite*<br>This command is valid only after DA16200 (DA16600) is connected to an AP.<br>*Example*<br>`AT+NWDHC=1`<br>`OK`<br><br>`AT+NWDHC`<br>`+NWDHC:1`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+NWDHR | <start_ip>,<end_ip> | Set an IP address range of the DHCP server.<br><start_ip>: Starting IP address assigned by the DHCP server.<br><end_ip>: Ending IP address assigned by the DHCP server.<br>Response: OK or ERROR |
| | ? | Get an IP address range of the DHCP server.<br>Response: +NWDHR:<start_ip>,<end_ip> |
| | (none) | |
| | *Prerequisite*<br>*Use this command in Soft-AP mode.*<br>*Example*<br>`AT+NWDHR=10.0.0.2,10.0.0.11`<br>`OK`<br><br>`AT+NWDHR`<br>`+NWDHR:10.0.0.2,10.0.0.11`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● DHCP Server restart (AT+RESTART or AT+NWDHS=0 > AT+NWDHS=0) is required for this command to take effect. | |
| AT+NWDHLT | <lease_time> | Set an IP lease time (in seconds) of the DHCP server.<br><lease_time>: IP lease time (from 60 sec to 86400 sec)<br>Response: OK or ERROR |
| | ? | Get an IP lease time of the DHCP server.<br>Response: +NWDHLT:<lease_time> |
| | (none) | |


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Prerequisite<br>Use this command in Soft-AP mode.<br>Example<br><pre>AT+NWDHLT=1800
OK

AT+NWDHLT
+NWDHLT:1800
OK</pre>Note<br>● Enabled by default in the SDK<br>● DHCP Server restart (AT+RESTART or AT+NWDHS=0 > AT+NWDHS=0) is required for this command to take effect.</td></tr>
<tr><td rowspan="4">AT+NWDHS</td><td><dhcpd></td><td>Start/Stop DHCP server.<br><dhcpd>: 0 (stop), 1 (start).<br>Response: OK or ERROR</td></tr>
<tr><td><dhcpd>,<br><start_ip>,<end_ip>,<br><lease_time></td><td>Start the DHCP server with options.<br><dhcpd>: 1 (start).<br><start_ip>: Starting IP address for the DHCP client.<br><end_ip>: Ending IP address for the DHCP client.<br><lease_time>: IP lease time (optional, in second, default is 1800).<br>Response: OK or ERROR</td></tr>
<tr><td>?<br>(none)</td><td>Get the DHCP client status.<br>Response: +NWDHS:<dhcpd></td></tr>
<tr><td colspan="2">Prerequisite<br>Use this command in Soft-AP mode.<br>Example<br><pre>AT+NWDHS=1
OK

AT+NWDHS=1,10.0.0.2,10.0.0.10,1800
OK

AT+NWDHS
+NWDHS:1,10.0.0.2,10.0.0.10,1800
OK</pre>Note<br>● Enabled by default in the SDK</td></tr>
<tr><td>AT+NWDHIP</td><td>(none)</td><td>Show the information of the DHCP Client(s) connected.<br>Response: OK or ERROR<br>In case of OK, the following response comes first before OK<br>+NWDHIP:<mac_addr_1>,<ip_addr_1>;<mac_addr_2>,<ip_addr_2>;.....</td></tr>
</table>


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>// One DHCP Client is in connected state.<br>`AT+NWDHIP`<br>`+NWDHIP:80:35:c1:79:c1:da,10.0.0.2`<br>`OK`<br><br>// Two DHCP Clients are in connected state.<br>`AT+NWDHIP`<br>`+NWDHIP:80:35:c1:79:c1:da,10.0.0.2;b4:f1:da:b4:27:11,10.0.0.3`<br>`OK`<br><br>// No DHCP Client exists.<br>`AT+NWDHIP`<br>`ERROR:-7` // "NO RESULT" meaning no client's connected<br>*Note*<br>● Enabled by default in the SDK<br>● Use this command when DHCP Server is running | |
| AT+NWSNS<br>AT+NWSNS1<br>AT+NWSNS2 | <server_ip> | Set the SNTP server IP address/domain name.<br><server_ip>: SNTP server IP address/domain name.<br>Response: OK or ERROR |
| | ? | Get the SNTP server IP address.<br>Response: +NWSNS:<sntp> |
| | (none) | |
| | *Example*<br>`AT+NWSNS=8.8.8.8`<br>`OK`<br><br>`AT+NWSNS`<br>`+NWSNS:8.8.8.8`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● Up to three SNTP servers can be specified by users; an SNTP server is contacted in round robin manner if DA16200 (DA16600) fails to synchronize the system time with a server<br>● If not specified, default SNTP servers are tried | |
| AT+NWSNUP | <period> | Set the SNTP client update period (in seconds).<br><period>: SNTP client update period (from 60 sec to 129600 sec)<br>Response: OK or ERROR |
| | ? | Get the SNTP client update period.<br>Response: +NWSNUP:<period> |
| | (none) | |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+NWSNUP=86400`<br>`OK`<br><br>`AT+NWSNUP`<br>`+NWSNUP:86400`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+NWSNTP<br>AT+NWSNTP1<br>AT+NWSNTP2 | <sntp> | Start/Stop the SNTP Client.<br><sntp>: 0 (stop), 1 (start).<br>Response: OK or ERROR |
| | <sntp>,<br><server_ip>,<br><period> | Start the SNTP Client with options.<br><sntp>: 1 (start).<br><server_ip>: SNTP server IP address (or domain).<br><period>: SNTP client update period (optional, second, default is 86400).<br>Response: OK or ERROR |
| | ?<br>(none) | Get the SNTP status.<br>Response: +NWSNTP:<sntp> |
| | *Example*<br>`AT+NWSNTP=0`<br>`OK`<br><br>`AT+NWSNTP=1,pool.ntp.org,86400`<br>`OK`<br><br>`AT+NWSNTP`<br>`+NWSNTP:1,pool.ntp.org,86400`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● If <sntp> is 1, SNTP Client is started immediately and tries to do time sync with the server specified. <sntp>=1 also enables “auto start” of the SNTP Client if DA16200 (DA16600) reboots. <sntp>=0 removes “auto start” flag from NVRAM, so DA16200 (DA16600) does not try to sync time when rebooted | |
| AT+NWCCRT | (none) | Check if certificates exist.<br>There are two sets of certificates:<br>● Set #1: for MQTT<br>Root CA (bit 2)/Cert (bit 1)/Key (bit 0)<br>● Set #2: for HTTPS client for OTA<br>Root CA (bit 5)/Cert (bit 4)/Key (bit 3)<br>For example: if DA16200 (DA16600) has the Root CA and Cert in Set #1, the return value is 6.<br>Response: +VER:<cert> |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+NWCCRT`<br>`+NWCCRT:6 ; MQTT`<br>`OK`<br><br>`AT+NWCCRT`<br>`+NWCCRT:56 ; HTTPS`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+NWDCRT | (none) | Delete all TLS certificates including private key.<br>Response: OK or ERROR |
| | *Example*<br>`AT+NWDCRT`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |

**Table 5: Certificate Command**

| Escape Sequence | Parameters | Description |
|---|---|---|
| <ESC>C | <cert_id>,<content><ETX> | Store certificate or private key.<br><ESC>C: To enter certificate input mode, type in <ESC>(0x1B) and C keys together<br><cert_id>: Certificate ID.<br>There are two sets of certificates:<br>● Set #1: for MQTT<br>0 (Root CA)/1 (Client Certificate)/2 (Private Key)<br>● Set #2: for HTTPS client for OTA<br>3 (Root CA)/4 (Client Certificate)/5 (Private Key)<br><content>: Certificate data. Copy and paste cert ascii text. Max length is 2048.<br><ETX>: Indication of the end of content (Ctrl+C, 0x03).<br>Response: OK or ERROR<br>For example:<br><ESC>C1,----- BEGIN CERTIFICATE -----<br>MIIodknvfano923nf/<br>...<ETX> |
| | *Example*<br>`<ESC>C0,Root CA<ETX>`<br>`OK`<br>`<ESC>C1,Client CA<ETX>`<br>`OK`<br>`<ESC>C2,Provate Key<ETX>`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |


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# 8 Wi-Fi Function Commands

**Table 6: Wi-Fi Function Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+WFMODE | <mode> | Set the Wi-Fi mode.<br><mode>: 0 (STA), 1 (Soft-AP).<br>Response: OK or ERROR |
| | ? | Get the current Wi-Fi mode.<br>Response: +WFMODE:<mode> |
| | (none) | |
| | *Example*<br>`AT+WFMODE=0` ; Set Station mode<br>`OK`<br><br>`AT+WFMODE=1` ; Set Soft-AP mode<br>`OK`<br><br>`AT+WFMODE` ; Get current Wi-Fi mode<br>`+WFMODE:1`<br>`OK`<br><br>`AT+WFMODE=?` ; Get current Wi-Fi mode<br>`+WFMODE:1`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● Wi-Fi mode is stored in NVRAM<br>● To take effect, a system restart is required | |
| AT+WFMAC | <mac> | Write a user MAC address in the NVRAM.<br>Response: OK or ERROR |
| | ? | Get the current MAC address of the activated WLAN interface.<br>Response: +WFMAC:<mac> |
| | (none) | |


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Prerequisite<br>The last digit should be an even number to be a valid MAC address.<br>Example<br><pre>AT+WFMAC=EC:9F:0D:9F:FA:64
OK

AT+WFMAC=?
+WFMAC:EC:9F:0D:9F:FA:64
OK

AT+WFMAC
+WFMAC:EC:9F:0D:9F:FA:64
OK

AT+WFMAC=?                    ; In Soft-AP mode
+WFMAC:EC:9F:0D:9F:FA:65
OK</pre>Note<br>● Enabled by default in the SDK<br>● A user MAC address is stored in NVRAM and a system restart is required to take effect<br>● DA16200 (DA16600) provides three types of the MAC address and the priority is in the following order: Spoofing MAC address, User MAC address, OTP MAC address<br>● When you read the MAC address in Soft-AP mode, it becomes the MAC address you wrote + 1</td></tr>
<tr><td rowspan="2">AT+WFSPF</td><td><mac></td><td>Write the spoofing MAC address in the NVRAM.<br>Response: OK or ERROR</td></tr>
<tr><td colspan="2">Example<br><pre>AT+WFSPF=EC:9F:0D:90:00:48
OK

AT+WFMAC=?
+WFMAC:EC:9F:0D:90:00:48
OK</pre>Note<br>● Enabled by default in the SDK<br>● Either odd or even number last digit of MAC address is accepted. Use this command only in STA mode<br>● A spoofing MAC address is stored in NVRAM and a system restart is required to take effect<br>● DA16200 (DA16600) provides three types of the MAC address and the priority is in the following order: Spoofing MAC address, User MAC address, OTP MAC address<br>● The AT+WFMAC=? command can be used to read back the spoofing MAC address as this command does not support query</td></tr>
<tr><td>AT+WFOTP</td><td><mac></td><td>Write the MAC address in the OTP memory.<br>Response: OK or ERROR<br>The MAC address written in the OTP is used as WLAN0 MAC address and MAC address + 1 will be used as WLAN1 MAC address.</td></tr>
</table>

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| Command | Parameters | Description |
|---|---|---|
| | *Prerequisite*<br>The last digit should be an even number to be a valid MAC address.<br>*Example*<br>`AT+WFOTP=EC:9F:0D:90:00:48`<br>`OK`<br><br>`AT+WFMAC=?`<br>`+WFMAC:EC:9F:0D:90:00:48`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● An OTP MAC address is stored in OTP and system restart is required to take effect<br>● An old MAC address in the OTP will be invalidated if it exists<br>● There are four MAC address slots available in OTP. It is possible to write the OTP MAC address four times in total at the production<br>● DA16200 (DA16600) provides three types of the MAC address and the priority is in the following order: Spoofing MAC address, User MAC address, OTP MAC address<br>● The AT+WFMAC=? command can be used to read back the OTP MAC address as this command does not support query<br>● When you read the MAC address in Soft-AP mode, it becomes the MAC address you wrote + 1 | |
| AT+WFSTAT | (none) | Get Wi-Fi configuration.<br>Response: +WFSTAT:<Wi-Fi interface><var>… |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+WFSTAT`<br><br>`+WFSTAT:sta0`<br>`mac_address= ec:9f:0d:9f:fa:64`<br>`wpa_state=DISCONNECTED`<br>`disconnect_reason=0`<br><br>`OK`<br><br>`AT+WFSTAT`<br><br>`+WFSTAT:softap1`<br>`mac_address=ec:9f:0d:9f:fa:65`<br>`wpa_state=DISCONNECTED`<br>`disconnect_reason=0`<br><br>`OK`<br><br>`AT+WFSTAT`<br><br>`+WFSTAT:sta0`<br>`mac_address=ec:9f:0d:9f:fa:64`<br>`bssid=70:5d:cc:32:15:32`<br>`ssid=`*`MY_AP_SSID`*<br>`id=0`<br>`mode=STATION`<br>`key_mgmt=WPA2-PSK`<br>`pairwise_cipher=CCMP`<br>`group_cipher=CCMP`<br>`channel=3`<br>`wpa_state=COMPLETED`<br><br>`OK`<br>*Note*<br>• Enabled by default in the SDK<br>• A Response can be different depending on the current DA16200 (DA16600) status/mode | |
| AT+WFPBC | (none) | Run the WPS PBC method.<br>Response: OK or ERROR |


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Prerequisite<br>A router should support WPS and PBC.<br>Example<br>AT+WFPBC<br><br>OK<br>+WFJAP:1,'MY_APS_SSID',192.168.0.3<br>Note<br>● Enabled by default in the SDK<br>● A WPS button can be pressed after issuing the command<br>● The existing connection, if any, will be lost when this command is run</td></tr>
<tr><td rowspan="3">AT+WFPIN</td><td><pin><br>(none)</td><td>Run the WPS PIN method.<br><pin>: PIN (eight digits).<br>(none): Generate a random PIN.<br>Response: +WFPIN:<pin><br>OK or ERROR</td></tr>
<tr><td>?</td><td>Get the current PIN.<br>Response: +WFPIN:<pin></td></tr>
<tr><td colspan="2">Prerequisite<br>An AP should support WPS PIN.<br>Example<br>AT+WFPIN=13557799<br>+WFPIN:13557799<br>OK<br><br>AT+WFPIN<br>+WFPIN:36269112 ; Generate random number.<br>OK<br><br>AT+WFPIN=?<br>+WFPIN:36269112<br>OK<br>Note<br>● Enabled by default in the SDK</td></tr>
<tr><td>AT+WFCWPS</td><td>(none)</td><td>Cancel WPS (both PBC and PIN).<br>Response: OK or ERROR</td></tr>
<tr><td colspan="2">Prerequisite<br>WPS should be in progress.<br>Example<br>AT+WFCWPS<br>OK<br>Note<br>● Enabled by default in the SDK<br>● An error will be returned if WPS is not in progress</td></tr>
</table>


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td rowspan="4">AT+WFCC</td><td><code></td><td>Set a country code.<br><code>: Country code (defined by ISO 3166-1 alpha-2 standard). Such as KR, US, JP, CH, and so on.<br>Response: OK or ERROR</td></tr>
<tr><td>?</td><td rowspan="2">Get the current country code.<br>Response: AT+WFCC=<code></td></tr>
<tr><td>(none)</td></tr>
<tr><td colspan="2">Example<br>AT+WFCC=KR<br>OK<br><br>AT+WFCC<br>+WFCC:KR<br>OK<br><br>AT+WFCC=?<br>+WFCC:KR<br>OK<br>Note<br>● Enabled by default in the SDK<br>● A country code is stored in the NVRAM<br>● A country code consists of two characters<br>● If a country is invalid, DA16200 (DA16600) returns an error code that is -4</td></tr>
<tr><td rowspan="2">AT+WFRSSI</td><td>(none)</td><td>Get the current RSSI value.<br>Response:<br>+RSSI: -34</td></tr>
<tr><td colspan="2">Prerequisite<br>Connection with an AP should be established.<br>Example<br>AT+WFRSSI<br>+RSSI:-25<br>OK<br>….<br>(if there’s no connection to an AP)<br>AT+WFRSSI<br>+RSSI:NOT_CONN<br><br>ERROR:-7<br>Note<br>● Enabled by default in the SDK<br>● DA16200 (DA16600) will respond “+RSSI:NOT_CONN” with error(-7) if the connection is not established</td></tr>
<tr><td>AT+WFSCAN</td><td>(none)</td><td>Scan APs.<br>Response: +WFSCAN:<bssid><\t><frequency><\t><signal strength><\t><flag><\t><ssid><LF>…</td></tr>
</table>


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+WFSCAN`<br><br>`+WFSCAN:70:5d:cc:32:15:32 2422 -30 [WPA2-PSK-CCMP][WPS][ESS] IPTIME_N604BLACK_ERIC`<br>`b4:a9:4f:62:39:46 2422 -32 [WPA-PSK-CCMP+TKIP][WPA2-PSK-CCMP+TKIP][WPS][ESS] SK_WiFiGIGA3943`<br><br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● An SSID can be missed in case of hidden AP | |
| AT+WFPSCAN | <channel time limit>, <ch>..<ch> | Get the passive scan result for the given parameters.<br><channel time limit>: Channel scan time limit (should be more than 30000 microsecond).<br><ch>: Carrier frequency (from 0 to14)<br>Response: BSSID Wi-Fi_Channel RSSI SSID Security Type |
| | *Prerequisite*<br>Only available in station mode.<br><br>*Example*<br>`AT+WFPSCAN=120000,1,3,5`<br><br>`70:5d:cc:8b:49:8e 2412 -47 Gen_Port_*.5_AP [WPA2-PSK-CCMP][WPS][ESS]`<br>`72:5d:cc:c0:9a:c4 2412 -47 IPTIME_A3004NS-M_Bell [WPS][ESS]`<br>…<br>`+PSCAN:TIMEOUT`<br><br>*Note:*<br>● Enabled by default in the SDK v3.2.3.0 or higher.<br>● Multiple parameters can be typed in <ch> as example. (‘0’ means all channel) | |
| AT+WFPCDTMIN | <bssid>,<br><min_threshold> | Set the passive scan minimum RSSI threshold condition.<br><bssid>: BSSID<br><min_threshold>: minimum threshold (from -10 to -100)<br>Response: OK or ERROR |
| | ? | Get the current condition. |
| | (none) | |


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Prerequisite<br>Only available in station mode.<br><br>Example<br><pre>AT+WFPCDTMIN=72:5d:cc:d0:82:bc,-80
OK

AT+WFPCDTMIN
+WFPCDTMIN: 72:5d:cc:d0:82:bc, -80
OK</pre>Note<br>● Enabled by default in the SDK v3.2.3.0 or higher.</td></tr>
<tr><td rowspan="4">AT+WFPCDTMAX</td><td><bssid>,<br><max_threshold></td><td>Set the passive scan maximum RSSI threshold condition.<br><bssid>: BSSID<br><max_threshold>: maximum threshold (from -10 to -100)<br>Response: OK or ERROR</td></tr>
<tr><td>?</td><td rowspan="2">Get the current condition.</td></tr>
<tr><td>(none)</td></tr>
<tr><td colspan="2">Prerequisite<br>Only available in station mode.<br><br>Example<br><pre>AT+WFPCDTMAX=72:5d:cc:c0:82:bc,-20
OK

AT+WFPCDTMAX
+WFPCDTMAX: 72:5d:cc:c0:82:bc, -20
OK</pre>Note<br>● Enabled by default in the SDK v3.2.3.0 or higher.</td></tr>
<tr><td rowspan="2">AT+WFPSTOP</td><td>(none)</td><td>Stop passive scan.<br>Response: OK or ERROR</td></tr>
<tr><td colspan="2">Prerequisite<br>Only available in station mode.<br><br>Example<br><pre>AT+WFPSTOP
OK</pre>Note<br>● Enabled by default in the SDK v3.2.3.0 or higher.</td></tr>
<tr><td rowspan="2">AT+WFJAP</td><td><ssid>,<sec>[,<hidden>]<br>(sec=0)</td><td rowspan="2">Connect to an AP.<br><ssid>: AP SSID.<br><sec>: Security protocol. 0 (OPEN), 1 (WEP), 2 (WPA), 3 (WPA2), 4 (WPA+WPA2).<br><idx>: Key index for WEP. 0~3<br><enc>: Encryption. 0 (TKIP), 1 (AES), 2 (TKIP+AES).<br><key>: Passphrase. 8 ~ 63 characters allowed.</td></tr>
<tr><td><ssid>,<sec>,<br><idx>,<key>[,<hidden>]<br>(sec=1)</td></tr>
</table>


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| Command | Parameters | Description |
|---|---|---|
| | <ssid>,<sec>,<br><enc>,<key>[,<hidden>]<br>(sec=2\|3\|4) | <hidden>: 1 (<ssid> is hidden), 0 or [not specified] (<ssid> is NOT hidden)<br>Response: OK or ERROR<br><br>Operation Results:<br>+WFJAP:<OPS_RESULT>[,'<SSID>','<IP_ADDRESS>']<br>+WFJAP:<OPS_RESULT>,<REASON>,[<REASON_CODE>]<br><OPS_RESULT> : 1 (SUCCESS), 0 (FAILED)<br>If <OPS_RESULT> : 1<br><SSID>: The SSID will be surrounded by single quotation mark<br><IP_ADDRESS>: Assigned IP address and format is xxx.xxx.xxx.xxx<br>If <OPS_RESULT> : 0<br><REASON> : well-known reason in text<br><REASON_CODE> : if < REASON > is OTHER, this shows the reason code<br>For an explanation of <REASON> or <REASON_CODE>, see Table 7 |
| | ?<br>(none) | Get the AP provisioning information.<br><br>Operation Results:<br>If provisioning data available:<br>+WFJAP:'<SSID>',<sec>,<enc>,'<Passphrase>'<br>If provisioning data is not available:<br>ERROR:-7 (No result) |
| | *Example*<br>`AT+WFJAP=MY_AP_SSID,0` *; Open security*<br>`OK`<br>`+WFJAP:1,'MY_AP_SSID',192.168.43.32`<br><br>`AT+WFJAP=MY_AP_SSID,0,1` ; *Open security + hidden SSID*<br>`OK`<br>`+WFJAP:1,'MY_AP_SSID',192.168.43.32`<br><br>`AT+WFJAP=MY_AP_SSID,1,0,12345` *; WEP security*<br>`OK`<br>`+WFJAP:1,'MY_AP_SSID',192.168.0.7`<br><br>`AT+WFJAP=MY_AP_SSID,1,0,12345,1` *; WEP + hidden AP*<br>`OK`<br>`+WFJAP:1,'MY_AP_SSID',192.168.0.7`<br><br>`AT+WFJAP=MY_AP_SSID,4,2,N12345678` ; WPA2 security<br>`OK`<br>`+WFJAP:1,'MY_AP_SSID',192.168.0.7`<br><br>`AT+WFJAP=MY_AP_SSID,4,2,N12345678,1` *; WPA2 + hidden AP*<br>`OK`<br>`+WFJAP:1,'MY_AP_SSID',192.168.0.7` | |


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| Command | Parameters | Description |
|---|---|---|
| | | AT+WFJAP=?<br>+WFJAP:'MY_AP_SSID',4,2,'N12345678'<br>OK<br>*Note*<br>● Enabled by default in the SDK<br>● The host should wait for both command response **OK** or **ERROR** and **Operation Result ;** wait for OK, and +WFJAP:1,'<SSID>',<IP Address> for successful connection<br>● Depending on the network condition, it may take more time to get an Operation Result due to internal connection re-trials<br>● No system reboot happens after running this command<br>● The AP configuration parameters (AP Profile) are stored in NVRAM |
| AT+WFJAPA | <ssid>[,<key>][,<hidden>] | Connect to an AP.<br>If <key> exists, security protocol is WPA+WPA2 and encryption is TKIP+AES.<br>if <key> is omitted, security protocol is OPEN.<br><hidden>: 1 (<ssid> is hidden), 0 or [not specified] (<ssid> is NOT hidden)<br>if <hidden> is omitted, <ssid> is not hidden.<br><ssid>: AP SSID.<br><key>: Passphrase. 8 ~ 63 characters are allowed<br>Response: OK or ERROR<br><br>Operation Results:<br>+WFJAP:<OPS_RESULT>[,'<SSID>','<IP_ADDRESS>']<br>+WFJAP:<OPS_RESULT>,<REASON>,[<REASON_CODE>]<br><OPS_RESULT> : 1 (SUCCESS), 0 (FAILED)<br>If <OPS_RESULT> : 1<br><SSID>: The SSID will be surrounded by single quotation mark<br><IP_ADDRESS>: Assigned IP address and format is xxx.xxx.xxx.xxx<br>If <OPS_RESULT> : 0<br><REASON> : well-known reason in text<br><REASON_CODE> : if < REASON > is OTHER, this shows the reason code.<br>For an explanation of <REASON> or <REASON_CODE>, see Table 7 |
| | ? | Get the AP provisioning information (SSID and Passphrase only)<br>Operation Results:<br>If provisioning data is available:<br>+WFJAPA:'<SSID>','<Passphrase>'<br>If provisioning data is not available:<br>ERROR:-7 (No result) |
| | (none) | |


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Example<br>
<pre>AT+WFJAPA=MY_AP_SSID                ; Open security
OK
+WFJAP:1,'MY_AP_SSID',192.168.43.32

AT+WFJAPA=MY_AP_SSID,1              ; Open security + hidden SSID
OK
+WFJAP:1,'MY_AP_SSID',192.168.43.32

AT+WFJAPA=MY_AP_SSID,N12345678      ; WPA2 security
OK
+WFJAP:1,'MY_AP_SSID',192.168.43.32

AT+WFJAPA=MY_AP_SSID,N12345678,1    ; WPA2 security + hidden AP
OK
+WFJAP:1,'MY_AP_SSID',192.168.43.32

AT+WFJAPA=?
+WFJAPA:'MY_AP_SSID','N12345678'
OK</pre>
Note<br>
● Enabled by default in the SDK<br>
● The host should wait for both command response OK or ERROR and Operation Result ; wait for OK, and +WFJAP:1,'<SSID>',<IP Address> for successful connection<br>
● Depending on the network condition, it may take more time to get an Operation Result due to internal connection re-trials<br>
● No system reboot happens after running this command<br>
● The AP configuration parameters (AP Profile) are stored in NVRAM</td></tr>
<tr><td rowspan="2">AT+WFCAP</td><td>(none)</td><td>Connect to an AP with the current WLAN0 interface configuration.<br>Response: OK or ERROR</td></tr>
<tr><td colspan="2">Prerequisite<br>AP profile parameters should exist in NVRAM.<br>Example<br>
<pre>AT+WFCAP
OK
+WFJAP:1,'MY_AP_SSID',192.168.0.7

AT+WFCAP
ERROR:-9                ; No AP profile found

AT+WFCAP
ERROR:-7                ; Already connected</pre>
Note<br>
● Enabled by default in the SDK<br>
● An AP profile can be stored in NVRAM by issuing the “AT+WFJAPA” or “AT+WFJAP” command<br>
● If there is no AP profile in NVRAM, DA16200 (DA16600) returns an error (-9)<br>
● If DA16200 (DA16600) is already in connection with an AP, it returns an error (-7)</td></tr>
</table>

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| Command | Parameters | Description |
|---|---|---|
| AT+WFQAP | (none) | Disconnect from the currently associated AP.<br>Response: OK or ERROR |
| | *Prerequisite*<br>DA16200 (DA16600) is in connection status.<br>*Example*<br>`AT+ WFQAP`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● No error returns if it has already been disconnected from an AP | |
| AT+WFSTA | (none) | Check Wi-Fi connection.<br>Response: +WFSTA:<status><br><status> 1 (Connected), 0 (disconnected) |
| | *Prerequisite*<br>DA16200 (DA16600) is running in station mode.<br>*Example*<br>`AT+WFSTA`<br>`+WFSTA:0`<br>`OK`<br><br>`AT+WFSTA`<br>`+WFSTA:1`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● If DA16200 (DA16600) runs the command in Soft-AP mode, it returns an error (-7) | |
| AT+WFROAP | <roam> | Operate the STA roaming.<br><roam>: 1 (run), 0 (stop).<br>Response: OK or ERROR |
| | ? | Get the roaming status.<br>Response: +WFROAP:<roam> |
| | (none) | |


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Prerequisite<br>DA16200 (DA16600) is running in station mode.<br>Example<br><pre>AT+WFROAP=1
OK

AT+WFROAP=0
OK

AT+WFROAP=?
+WFROAP:1
OK</pre>Note<br>● Enabled by default in SDK<br>● This command enables “simple” roaming. The roaming configuration consists of one parameter called the roaming threshold (set to AT+WFROTH, -65 by). Assume that the DA16200 (DA16600) is connected to an AP, and there are other APs that have the same SSID and security settings around the DA16200 (DA16600). As the DA16200 (DA16600) moves , if the RSSI value of the currently connected AP is lower than the specified threshold, it will try to connect to an AP with a higher RSSI (same SSID and security). If the condition is met, the DA16200 (DA16600) will silently switch to the new AP without a disconnection event.<br>● The auto roaming start flag is stored in NVRAM when <roam> is set to, and roaming operation is enabled when the flag is set upon system reboot.. If <roam> is 0, the roaming flag is removed from NVRAM and roaming is disabled.<br>● "Simple" roaming is not supported in DPM mode. So, setting DPM mode with command "AT+DPM=1" disables "The auto roaming start flag".</td></tr>
<tr><td rowspan="3">AT+WFROTH</td><td><rssi></td><td>Set the STA roaming threshold.<br><rssi>: Roaming threshold value (from 0 to -95 dBm).<br>Response: OK or ERROR</td></tr>
<tr><td>?</td><td rowspan="2">Get the STA roaming threshold.<br>Response: +WFROTH:<rssi></td></tr>
<tr><td>(none)</td></tr>
<tr><td colspan="2">Prerequisite<br>DA16200 (DA16600) is running in station mode.<br>Example<br><pre>AT+WFROTH=-55
OK
AT+WFROTH=?

+WFROTH:-55
OK</pre>Note<br>● Enabled by default in the SDK<br>● This command writes roaming threshold in NVRAM<br>● When AT+WFROAP=1 is run, the roaming is enabled with the new threshold</td></tr>
<tr><td>AT+WFDIS</td><td><disabled></td><td>Set the Wi-Fi STA profile unused. If set to 1, DA16200 (DA16600) will not start to connect to the configured AP when rebooting.<br><disabled >: 1 (Unused), 0 (Used).<br>Response: OK or ERROR</td></tr>
</table>


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td rowspan="3"></td><td>?</td><td rowspan="2">Get the status of the Wi-Fi profile.<br>Response: +WFDIS:<disabled></td></tr>
<tr><td>(none)</td></tr>
<tr><td colspan="2">Example<br>AT+WFDIS=1<br>OK<br><br>AT+WFDIS=?<br>+WFDIS:1<br>OK<br>Note<br>● Enabled by default in the SDK<br>● The “unused” flag is stored in the NVRAM<br>● The flag affects DA16200 (DA16600) during boot-up procedure which means system restart is required</td></tr>
<tr><td rowspan="5">AT+WFSAP</td><td><ssid>,<sec>,<br><ch>,<code><br>(sec=0)</td><td rowspan="2">Set up Soft-AP interface.<br><ssid>: AP SSID.<br><sec>: Security protocol. 0 (OPEN), 2 (WPA), 3 (WPA2), 4 (WPA+WPA2).<br><enc>: Encryption. 0 (TKIP), 1 (AES), 2 (TKIP+AES).<br><key>: Passphrase. 8 ~ 63 characters are allowed.<br><ch>: Operating channel (optional). Default is 1 or uses the current channel if Soft-AP is operating.<br><code>: Country code (optional). If exists, <ch> is essential.<br>Response: OK or ERROR</td></tr>
<tr><td><ssid>,<sec>,<br><enc>,<key>,<br><ch>,<code><br>(sec=2|3|4)</td></tr>
<tr><td>?</td><td rowspan="2">Get the Soft-AP interface configuration.<br>Response:<br>+WFSAP:'<ssid>',<auth>,<enc>,'<key>',<ch>,<code><br>Operation Result:<br>+WFSAP:<ssid> is printed on success</td></tr>
<tr><td>(none)</td></tr>
<tr><td colspan="2">Example<br>AT+WFSAP=DA16200_MY_SSID,0,1,KR<br>+WFSAP:DA16200_MY_SSID<br>OK<br><br>AT+WFSAP=?<br>+WFSAP:'DA16200_MY_SSID',0,1,KR<br>OK<br><br>AT+WFSAP=DA16200_MY_SSID,3,1,12345678,1,KR<br>+WFSAP:DA16200_MY_SSID<br>OK<br><br>AT+WFSAP=?<br>+WFSAP:'DA16200_MY_SSID',3,1,'12345678',1,KR<br>OK<br><br>AT+WFSAP='DA16200,MY_SSID',3,1,'12345678',1,KR<br>+WFSAP:DA16200,MY_SSID</td></tr>
</table>


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">OK<br><br>AT+WFSAP=?<br><br>+WFSAP:'DA16200,MY_SSID',3,1,'12345678',1,KR<br>OK<br>Note<br>● Enabled by default in the SDK<br>● The Soft-AP configuration parameters are stored in NVRAM<br>● If the command is issued in station mode, a reboot is required to start as Soft-AP mode. (If the command is issued in Soft-AP mode, then no system restart is required)<br>● The ‘,’(comma) is included in the SSID string, enclose the SSID with a single quotation mark</td></tr>
<tr><td>AT+WFOAP</td><td>(none)</td><td>Operate Soft-AP interface.<br>Response: OK or ERROR</td></tr>
<tr><td></td><td colspan="2">Prerequisite<br>A Soft-AP profile should be stored in NVRAM.<br>Example<br>AT+WFOAP<br>OK<br>Note<br>● Enabled by default in the SDK<br>● Run this command in Soft-AP mode<br>● If there is no profile in NVRAM, it returns an error (-99)<br>● DA16200 (DA16600) returns an error (-99) if it already operates as Soft-AP</td></tr>
<tr><td>AT+WFTAP</td><td>(none)</td><td>Stop the Soft-AP interface.<br>Response: OK or ERROR</td></tr>
<tr><td></td><td colspan="2">Prerequisite<br>Run this command in Soft-AP mode.<br>Example<br>AT+WFTAP<br>OK<br>Note<br>● Enabled by default in the SDK<br>● This command is valid while DA16200 (DA16600) is running in Soft-AP mode</td></tr>
<tr><td colspan="3">Additional Note for AT+WFSAP, AT+WFOAP, AT+WFTAP:<br>Use case example:<br>For example, in STA mode, in factory reset state<br>…<br>AT+WFSAP=DA16200_OPEN,0 // set up Soft-AP<br>AT+RESTART // reboot to start in the configured Soft-AP mode<br>…<br>DUT starts as Soft-AP<br>….<br>AT+WFTAP // stop Soft-AP if you want<br>AT+WFOAP // start Soft-AP if you want</td></tr>
<tr><td>AT+WFRAP</td><td>(none)</td><td>Restart the Soft-AP interface.<br>Response: OK or ERROR</td></tr>
</table>

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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Prerequisite<br>A profile for Soft-AP should be stored in NVRAM.<br>Example<br>AT+WFRAP<br><br>OK<br>Note<br>● Enabled by default in the SDK<br>● This command is valid in Soft-AP mode<br>● If it runs in station mode, DA16200 (DA16600) returns an error (-99)</td></tr>
<tr><td rowspan="2">AT+WFLCST</td><td>(none)</td><td>Get connected station information.<br>Response: +WFLCST:<mac><LF><flags><LF><var>…</td></tr>
<tr><td colspan="2">Example<br>AT+WFLCST<br><br>+WFLCST:a6:f2:7c:d4:53:1c<br>flags=[AUTH][ASSOC][AUTHORIZED][SHORT_PREAMBLE][WMM][MAYBE_WPS][HT]<br>aid=1<br>capability=0x421<br>listen_interval=10<br>wifi_mode=802.11n<br>timeout_next=NULLFUNC POLL<br>rx_packets=290<br>tx_packets=4<br>rx_bytes=29625<br>tx_bytes=10658<br>inact_cnt=0<br>connected_time=20<br>sta_count=1<br><br>OK<br><br>AT+WFLCST<br>+WFLCST:NOT_FOUND<br>OK<br>Note<br>● Enabled by default in the SDK<br>● If there is no station connected, then DA16200 (DA16600) returns “+WFLCST:NOT_FOUND”</td></tr>
<tr><td rowspan="3">AT+WFAPWM</td><td><mode></td><td>Set IEEE 802.11 Wi-Fi mode of Soft-AP interface.<br><mode>: 0 (B/G/N), 1 (G/N), 2 (B/G), 3 (N), 4 (G), 5 (B)<br>Response: OK or ERROR</td></tr>
<tr><td>?</td><td rowspan="2">Get IEEE 802.11 Wi-Fi mode of Soft-AP interface.<br>Response: +WFAPWM:<mode></td></tr>
<tr><td>(none)</td></tr>
</table>


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Example<br>AT+WFAPWM=0<br>OK<br><br>AT+WFAPWM=1<br>OK<br><br>AT+WFAPWM=?<br>+WFAPWM:1<br>OK<br>Note<br>● Enabled by default in the SDK<br>● The configuration is stored in NVRAM<br>● System restart is required to take effect</td></tr>
<tr><td rowspan="4">AT+WFAPCH</td><td><ch></td><td><ch>: Operating channel (from 0 to 14, 0 is auto).<br>Response: OK or ERROR</td></tr>
<tr><td>?</td><td rowspan="2">Get the operating channel number for the Soft-AP interface.<br>Response: +WFAPCH:<ch></td></tr>
<tr><td>(none)</td></tr>
<tr><td colspan="2">Example<br>AT+WFAPCH=5<br>OK<br><br>AT+WFAPCH=?<br>+WFAPCH:5<br>OK<br>Note<br>● Enabled by default in the SDK<br>● The configuration is stored in NVRAM<br>● System restart is required to take effect</td></tr>
<tr><td rowspan="4">AT+WFAPBI</td><td><interval></td><td>Set the AP beacon interval.<br><interval>: Beacon interval (ms).<br>Response: OK or ERROR</td></tr>
<tr><td>?</td><td rowspan="2">Get the AP beacon interval.<br>Response: +WFAPBI:<interval></td></tr>
<tr><td>(none)</td></tr>
<tr><td colspan="2">Example<br>AT+WFAPBI=200<br>OK<br><br>AT+WFAPBI=?<br>+WFAPBI:200<br>OK<br>Note<br>● Enabled by default in the SDK<br>● The configuration is stored in NVRAM<br>● System restart is required to take effect</td></tr>
</table>


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| Command | Parameters | Description |
|---|---|---|
| AT+WFAPUI | <timeout> | Set station disconnection timeout in Soft-AP mode.<br><timeout>: Disconnection timeout (sec).<br>(from 30 to 86400, step = 10)<br>Response: OK or ERROR |
| | ? | Get station disconnection timeout in Soft-AP mode.<br>Response: +WFAPUI:<timeout> |
| | (none) | |
| | *Example*<br>`AT+WFAPUI=60`<br>`OK`<br><br>`AT+WFAPUI=?`<br>`+WFAPUI:60`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● Within the specified time, if an STA does not send any frame, Soft-AP sends a NULL frame after the timeout is expired to check STA's inactivity. If no ACK is received from the STA, Soft-AP removes the STA<br>● The configuration is stored in NVRAM<br>● System restart is required to take effect | |
| AT+WFAPRT | <threshold> | Set the AP RTS threshold (octets).<br><threshold>: RTS threshold (from 1 to 2347).<br>Response: OK or ERROR |
| | ? | Get the AP RTS threshold.<br>Response: +WFAPRT:<threshold> |
| | (none) | |
| | *Example*<br>`AT+WFAPRT=2100`<br>`OK`<br><br>`AT+WFAPRT=?`<br>`+WFAPRT:2100`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● If a frame that is bigger than the RTS threshold specified is to be sent, RTS/CTS frames are sent first to avoid collision in the air. By default, the RTS threshold is 2347<br>● The configuration is stored in NVRAM<br>● System restart is required to take effect | |
| AT+WFAPDE | <mac> | Send de-authentication frame to the connected station.<br><mac>: MAC address of the connected station.<br>Response: OK or ERROR |


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Prerequisite<br>There should be a station connected.<br>Example<br><pre>AT+WFAPDE=E6:0D:E5:A5:5D:B3
+WFDST:e6:0d:e5:a5:5d:b3
OK</pre>Note<br>● Enabled by default in the SDK<br>● Use this command in Soft-AP mode<br>● You can check the MAC address of an STA that you want to send deauthentication frame by using the command “AT+WFLCST”<br>● If the operation is not successful (for example, a wrong MAC address is specified), the operation result (+WFDST:<mac_addr>) does not come</td></tr>
<tr><td rowspan="2">AT+WFAPDI</td><td><mac></td><td>Send disassociation frame to the connected station.<br><mac>: MAC address of the connected station.<br>Response: OK or ERROR</td></tr>
<tr><td colspan="2">Prerequisite<br>There should be a station connected.<br>Example<br><pre>AT+WFAPDI=E6:0D:E5:A5:5D:B3
+WFDST:e6:0d:e5:a5:5d:b3
OK</pre>Note<br>● Enabled by default in the SDK<br>● Use this command in Soft-AP mode<br>● You can check the MAC address of an STA that you want to send disassociation frame by using the command “AT+WFLCST”<br>● If the operation is not successful (for example, a wrong MAC address is specified), the operation result (+WFDST:<mac_addr>) does not come</td></tr>
<tr><td rowspan="4">AT+WFWMM</td><td><wmm></td><td>Set WMM on/off.<br><wmm>: 0 (off), 1 (on).<br>Response: OK or ERROR</td></tr>
<tr><td>?</td><td rowspan="2">Get the WMM status.<br>Response: +WFWMM:<wmm></td></tr>
<tr><td>(none)</td></tr>
<tr><td colspan="2">Prerequisite<br>Run this command in Soft-AP mode.<br>Example<br><pre>AT+WFWMM=1
OK

AT+WFWMM=?
+WFWMM:1
OK</pre>Note<br>● Enabled by default in the SDK<br>● WMM is enabled by default. If WMM is enabled, Beacon/Probe Rsp/Assoc frames will have a WMM info element. WMM enables QoS on the AC category<br>● The configuration is stored in NVRAM</td></tr>
</table>


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| Command | Parameters | Description |
|---|---|---|
| AT+WFWMP | <wmmps> | Set WMM-PS (WMM Power Save) on/off.<br><wmmps>: 0 (off), 1 (on).<br>Response: OK or ERROR |
| | ? | Get the WMM-PS status.<br>Response: +WFWMP:<wmmps> |
| | *Prerequisite*<br>Run this command in Soft-AP mode.<br>*Example*<br>`AT+WFWMM=0`<br>`OK`<br><br>`AT+WFWMM=?`<br>`+WFWMM:0`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● By default, WMM-PS is disabled. If WMM-PS is enabled, Beacon/Probe Rsp/Assoc Rsp frames sent from Soft-AP will have a U-APSD flag set. For WMM and WMM-PS to properly work, the STA should also have WMM and WMM-PS certified | |


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**Table 7: Wi-Fi Function Response List**

| Response | Parameters | Description |
|---|---|---|
| +WFJAP | <result>,<ssid>,<ip><br><br><result>,<br><well-known-reason><br>[,<reason_code>] | The result of AP connection in STA mode. (The result of AT+WFJAP or AT+WFJAPA or AT+WFCAP.)<br><result>: 0 (failed), 1 (succeeded).<br>For <result>:1<br><ssid>: SSID of the AP when succeeded.<br><ip>: IP address of the station when succeeded.<br>For <result>:0<br><well-known-reason> : connection trial failure reason in text format, TIMEOUT / WRONGPWD / ACCESSLIMIT / OTHER<br>TIMEOUT: connection attempt failed after continuous connection attempts.<br>WRONGPWD: WPA 4-Way Handshake failed, Pre-shared key (password) may be incorrect<br>ACCESSLIMIT: disconnected because the authorized access number limit has been reached.<br>OTHER: other reason<br><reason_code>: if <well-known-reason> is OTHER, this field shows which reason caused the connection trial failure<br>See Appendix F<br>For example:<br>+WFJAP:0,TIMEOUT<br>+WFJAP:1,'ap_test',192.168.0.10 // The Wi-Fi connection is established, and the assigned IP address is 192.168.0.10. |
| +WFDAP | <reserved>,<br><well-known-reason><br>[,<reason_code>] | Disconnected from the AP.<br><reserved>: 0<br><well-known-reason>: disconnection reason in text format,<br>AUTH_NOT_VALID / DEAUTH / INACTIVITY / APBUSY / OTHER<br>AUTH_NOT_VALID: Previous authentication no longer valid<br>DEAUTH: De-authenticated as STA is leaving<br>INACTIVITY: Disassociated due to inactivity<br>APBUSY: Disassociated because AP is unable to handle all currently associated STAs<br><reason_code> : if <well-known-reason> is OTHER, this field shows which reason caused the disconnection.<br>See Appendix F<br>For example:<br>+WFDAP:0,INACTIVITY<br>+WFDAP:0,DEAUTH<br>+WFDAP:0,OTHER,8 → WLAN_REASON_CLASS2_FRAME_FROM_NONAUTH_STA (8) |
| +WFCST | <mac> | A Wi-Fi station connected in Soft-AP mode.<br><mac>: MAC address of the connected station. |
| +WFDST | <mac> | A Wi-Fi station disconnected in Soft-AP mode.<br><mac>: MAC address of the disconnected station. |


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## 8.1 Wi-Fi Function Commands for WPA3

Customer/User can configure DA16200 (DA16600) as WPA3 STA or WPA3 Soft-AP with a special DA16200 (DA16600) SDK where the WPA3 feature is enabled. By default, WPA3 is not enabled in DA16200 (DA16600) Generic SDK. Contact Renesas Electronics if you need this specific SDK.

Syntax of all the Wi-Fi function commands is the same as described in Table 8 apart from the following commands where you need to specify WPA3 specific parameters.

**Table 8: List of WPA3-Relevant Wi-Fi Function Commands**

<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td rowspan="5">AT+WFJAP</td><td><ssid>,<sec>[,<hidden>]<br>(sec=0|5)</td><td rowspan="3">Connect to an AP.<br><ssid>: AP SSID.<br><sec>: Security protocol. 0 (OPEN), 1 (WEP), 2 (WPA), 3 (WPA2), 4 (WPA+WPA2) , 5 (WPA3_OWE), 6 (WPA3_SAE), 7 (WPA2+WPA3_SAE)<br><idx>: Key index for WEP. 0~3<br><enc>: Encryption. 0 (TKIP), 1 (AES), 2 (TKIP+AES).<br><key>: Passphrase. 8 ~ 63 characters are allowed.<br><hidden>: 1 (<ssid> is hidden), 0 or [not specified] (<ssid> is NOT hidden)<br>Response: OK or ERROR<br><br>Operation Results:<br>+WFJAP:<OPS_RESULT>[,'<SSID>','<IP_ADDRESS>']<br>+WFJAP:<OPS_RESULT>,<REASON>,[<REASON_CODE>]<br><OPS_RESULT> : 1 (SUCCESS), 0 (FAILED)<br>If <OPS_RESULT> : 1<br><SSID>: The SSID will be surrounded by single quotation mark<br><IP_ADDRESS>: Assigned IP address and format is xxx.xxx.xxx.xxx<br>If <OPS_RESULT> : 0<br><REASON> : well-known reason in text<br><REASON_CODE> : if < REASON > is OTHER, this shows the reason code<br>For explanation on <REASON> or <REASON_CODE>, see Table 7</td></tr>
<tr><td><ssid>,<sec>,<br><idx>,<key>[,<hidden>]<br>(sec=1)</td></tr>
<tr><td><ssid>,<sec>,<br><enc>,<key>[,<hidden>]<br>(sec=2|3|4|6|7)</td></tr>
<tr><td>?</td><td rowspan="2">Get the AP provisioning information<br>Operation Results:<br>If provisioning data available:<br>+WFJAP:'<SSID>',<sec>,<enc>,'<Passphrase>'<br>If provisioning data is not available:<br>ERROR:-7 (No result)</td></tr>
<tr><td>(none)</td></tr>
<tr><td colspan="2">Example<br><br>AT+WFJAP=MY_AP_SSID,5 ; OWE<br>OK<br>+WFJAP:1,'MY_AP_SSID',192.168.43.32<br><br>AT+WFJAP=MY_AP_SSID,5,1 ; OWE + hidden SSID<br>OK<br>+WFJAP:1,'MY_AP_SSID',192.168.43.32</td></tr>
</table>


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| Command | Parameters | Description |
|---|---|---|
| | | |

```
AT+WFJAP=MY_AP_SSID,6,1,12345678          ; WPA3_SAE
OK
+WFJAP:1,'MY_AP_SSID',192.168.0.7

AT+WFJAP=MY_AP_SSID,6,1,12345678,1        ; WPA3_SAE + hidden AP
OK
+WFJAP:1,'MY_AP_SSID',192.168.0.7

AT+WFJAP=MY_AP_SSID,7,1,12345678          ; WPA2+WPA3_SAE
OK
+WFJAP:1,'MY_AP_SSID',192.168.0.7

AT+WFJAP=?
+WFJAP:'MY_AP_SSID',6,1,'N12345678'
OK
```

*Note*

- Disabled by default in the SDK. To use this command, use WPA3 SDK or Contact Renesas Electronics
- The host should wait for both command response **OK** or **ERROR** and **Operation Result ;** wait for OK, and +WFJAP:1,'<SSID>',<IP Address> for successful connection
- Depending on the network condition, it may take more time to get an Operation Result due to internal connection re-trials
- No system reboot happens after running this command
- The AP configuration parameters (AP Profile) are stored in NVRAM


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td rowspan="3">AT+WFJAPA</td><td><is_wpa3>,<ssid>[,<key>]<br>[,<hidden>]</td><td>Connect to an AP.<br>For connection to a WPA3 (OWE, SAE, or WPA2+WPA3) AP, give <is_wpa3> 1, otherwise, give 0.<br>If <key> exists, then security protocol is WPA+WPA2 and encryption is TKIP+AES.<br>if <key> is omitted, then security protocol is OPEN (is_wpa3=0) or OWE (is_wpa3=1).<br><hidden>: 1 (<ssid> is hidden), 0 or [not specified] (<ssid> is NOT hidden)<br>if <hidden> is omitted, <ssid> is not hidden.<br><ssid>: AP SSID.<br><key>: Passphrase. 8 ~ 63 characters allowed.<br>Response: OK or ERROR<br><br>Operation Results:<br>+WFJAP:<OPS_RESULT>[,'<SSID>','<IP_ADDRESS>']<br>+WFJAP:<OPS_RESULT>,<REASON>,[<REASON_CODE>]<br><OPS_RESULT> : 1 (SUCCESS), 0 (FAILED)<br>If <OPS_RESULT> : 1<br><SSID>: The SSID will be surrounded by single quotation mark<br><IP_ADDRESS>: Assigned IP address and format is xxx.xxx.xxx.xxx<br>If <OPS_RESULT> : 0<br><REASON> : well-known reason in text<br><REASON_CODE> : if < REASON > is OTHER, this shows the reason code<br>For explanation on <REASON> or <REASON_CODE>, see Table 7</td></tr>
<tr><td>?<br>(none)</td><td>Get the AP provisioning information (SSID and Passphrase only)<br>Operation Results:<br>If provisioning data is available:<br>+WFJAPA:'<SSID>','<Passphrase>'<br>If provisioning data is not available:<br>ERROR:-7 (No result)</td></tr>
<tr><td colspan="2">Example<br><br>AT+WFJAPA=1,MY_AP_SSID ; OWE security<br>OK<br>+WFJAP:1,'MY_AP_SSID',192.168.43.32<br><br>AT+WFJAPA=1,MY_AP_SSID,1 ; OWE security + hidden SSID<br>OK<br>+WFJAP:1,'MY_AP_SSID',192.168.43.32<br><br>AT+WFJAPA=1,MY_AP_SSID,N12345678 ; WPA3_SAE security<br>OK<br>+WFJAP:1,'MY_AP_SSID',192.168.43.32<br><br>AT+WFJAPA=1,MY_AP_SSID,N12345678,1 ; WPA3_SAE security + hidden AP</td></tr>
</table>


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| Command | Parameters | Description |
|---|---|---|
| | OK<br><br>+WFJAP:1,'MY_AP_SSID',192.168.43.32<br><br>AT+WFJAPA=?<br>+WFJAPA:'MY_AP_SSID','N12345678'<br>OK<br>*Note*<br>● Disabled by default in the SDK. To use this command, use WPA3 SDK or Contact Renesas Electronics<br>● The host should wait for both command response **OK** or **ERROR** and **Operation Result ;** wait for OK, and +WFJAP:1,'<SSID>',<IP Address> for successful connection<br>● Depending on the network condition, it may take more time to get an Operation Result due to internal connection re-trials<br>● No system reboot happens after running this command<br>● The AP configuration parameters (AP Profile) are stored in NVRAM | |
| AT+WFSAP | <ssid>,<sec>,<br><ch>,<code><br>(sec=0\|5) | Set up the Soft-AP interface.<br><ssid>: AP SSID.<br><sec>: Security protocol. 0 (OPEN), 2 (WPA), 3 (WPA2), 4 (WPA+WPA2), WPA3_OWE (5), WPA3_SAE (6), WPA2+WPA3 (7)<br><enc>: Encryption. 0 (TKIP), 1 (AES), 2 (TKIP+AES)<br><key>: Passphrase. 8 ~ 63 characters are allowed.<br><ch>: Operating channel (optional). Default is 1 or uses the current channel if Soft-AP is operating<br><code>: Country code (optional). If exists, <ch> is essential.<br>Response: OK or ERROR |
| | <ssid>,<sec>,<br><enc>,<key>,<br><ch>,<code><br>(sec=2\|3\|4\|6\|7) | |
| | ? | Get the Soft-AP interface configuration.<br>Response:<br>+WFSAP:'<ssid>',<auth>,<enc>,'<key>',<ch>,<code><br>Operation Result:<br>+WFSAP:<ssid> is printed on success |
| | (none) | |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+WFSAP=DA16200_MY_SSID,5,1,KR` ; OWE<br>`+WFSAP:DA16200_MY_SSID`<br>`OK`<br><br>`AT+WFSAP=?`<br>`+WFSAP:'DA16200_MY_SSID',5,1,KR`<br>`OK`<br><br>`AT+WFSAP=DA16200_MY_SSID,6,1,12345678,1,KR` ; WPA3_SAE<br>`+WFSAP:DA16200_MY_SSID`<br>`OK`<br><br>`AT+WFSAP=?`<br>`+WFSAP:'DA16200_MY_SSID',6,1,'12345678',1,KR`<br>`OK`<br><br>`AT+WFSAP='DA16200,MY_SSID',7,1,12345678,1,KR` ; WPA2+WPA3<br>`+WFSAP:DA16200,MY_SSID`<br>`OK`<br><br>`AT+WFSAP=?`<br>`+WFSAP:'DA16200,MY_SSID',7,1,'12345678',1,KR`<br>`OK`<br>*Note*<br>● Disabled by default in the SDK. To use this command, use the DA16200 SDK for WPA3 or contact to Renesas.<br>● The Soft-AP configuration parameters are stored in NVRAM<br>● If the command is issued in station mode, a reboot is required to start as Soft-AP mode. (If the command is issued in Soft-AP mode, then no system restart is required)<br>● The ‘,’(comma) is included in the SSID string, enclose the SSID with a single quotation mark | |

## 8.2 Wi-Fi Function Commands for WPA Enterprise

AT-CMD of DA16200 provides Wi-Fi commands that can be used as STA in WPA-Enterprise environment. To connect to the WPA-Enterprise AP, the DA16200 need to have profile information for the WPA-Enterprise AP and user account information.


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**Table 9: WPA-Enterprise Wi-Fi Function Commands**

| Command | Parameters | Description |
|---|---|---|
| AT+WFENTAP | <ssid>,<auth>,<enc>,<br><phase1>[,<phase2>] | Create Enterprise profile to NVRAM.<br><ssid>: Enterprise AP SSID.<br><auth>: Authentication mode for WAP-Enterprise. 8 (WPA-EAP), 9 (WPA2-EAP), 10 (WPA/WPA2-EAP).<br><enc>: Encryption Type. 0 (TKIP), 1 (AES), 2 (TKIP+AES).<br><phase1>: Phase #1 EAP type. 0 (Mixed), 1 (PEAP0), 2 (PEAP1), 3 (FAST), 4 (TTLS), 5 (TLS)<br><pahse2>: Phase #2 EAP type. 0 (Mixed), 1 (MSCHAPV2), 2 (GTC)<br><br>Response: OK or ERROR<br>Operation Results: +WFENTAP:<SSID> |
| | ?<br>(none) | Get the WPA-Enterprise configuration.<br>Response:<br>+WFENTAP:<ssid>,<auth>,<enc>,<phase1>,<pahse2> |
| | *Example*<br>`AT+WFENTAP=MY_AP_SSID,10,2,0` *; Phase#1 Mixed mode*<br>`OK`<br>`+WFJAP:'MY_AP_SSID'`<br><br>`AT+WFENTAP=MY_AP_SSID,10,2,0,0` *; Phase#1, #2 Mixed mode*<br>`OK`<br>`+WFJAP:'MY_AP_SSID'`<br><br>`AT+WFENTAP=MY_AP_SSID`<br>`OK`<br>`+WFJAP:'MY_AP_SSID', 10,2,0,0`<br>*Note*<br>● Enabled by default in the SDK v3.2.3.0 or higher.<br>● WPA-Enterprise profile is stored in NVRAM.<br>● “EAP-FAST” type for <phase_1> does not support in SDK 3.2.3.0 and earlier SDK.<br>● Need user account to connect to WPA-Enterprise AP. See “AT+WFENTLI” command.<br>● To take effect, a system restart is required. | |
| AT+WFENTLI | <id>,<pw> | Set User-ID / Password for WPA-Enterprise user account.<br><id>: Login-ID for WPA-Enterprise user account<br><pw>: Login-Password for WPA-Enterprise user account<br><br>Response: OK or ERROR |
| | ?<br>(none) | Get current saved User-ID / Password for WPA-Enterprise user account.<br><br>Response: OK or ERROR<br>Operation Result: +WFENTLI:<id>,<pwd> |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+WFENTLI='USER_ACCOUNT_ID','USER_ACCOUNT_PWD'`<br>`OK`<br><br>`AT+WFENTLI`<br>`+WFENTLI='USER_ACCOUNT_ID','USER_ACCOUNT_PWD'`<br><br>`AT+WFENTLI=?`<br>`+WFENTLI='USER_ACCOUNT_ID','USER_ACCOUNT_PWD'`<br>*Note*<br>● Enabled by default in the SDK<br>● User account ID and PASSWORD are stored in NVRAM.<br>● To take effect, a system restart is required. | |

**Table 10: WPA-Enterprise Network Function Command**

| Command | Parameters | Description |
|---|---|---|
| AT+NWTLSV | <ver> | Set the minimum accepted TLS protocol version when Phase#1 EAP type is TTLS or TLS of WPA Enterprise. The maximum accepted TLS protocol version is TLSv1.2. For example, If TLSv1.0 is setup, the version of TLS session can be TLSv1.0, TLSv1.1 or TLSv1.2.<br><ver>: Enterprise AP SSID. 0 <TLSv1.0>, 1 <TLSv1.1>, 2 <TLSv1.2><br><br>Response: OK or ERROR |
| | ? | Get current saved TLS version.<br>Response:<br>+NWTLSV:<tls_version> |
| | (none) | |
| | *Example :*<br>`AT+NWTLSV=NEW_TLS_VER`<br>`OK`<br><br>`AT+NWTLSV`<br>`+NWTLSV:2`<br>`OK`<br><br>`AT+NWTLSV=?`<br>`+NWTLSV:2`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK v3.2.3.0 or higher.<br>● TLS Version number is stored as internal value.<br>● Default minimum accepted TLS protocol version is TLSv1.2.<br>● To take effect, a system restart is required. | |

### 8.2.1 WPA-Enterprise Connection Example

Create WPA-Enterprise profile and restart the DA16200/DA16600 to start Wi-Fi connection :


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All cases, WPA-Enterprise user account information is needed.

```
Case #1, "Mixed" mode for EAP-type.
In this case, Encryption-type is configured as "Mixed" mdoe.
EAP-type and Encryption type are selected automatically.
AT+WFENTAP='WPA-Ent-AP-SSID',10,2,0
AT+WFENTLI='WPA-Ent_User_ID','WPA-Ent_PWD'
AT+RESTART

Case #2, "Mixed" mode for EAP-type and Encryption type.
EAP-type and Encryption type are selected automatically.
AT+WFENTAP='WPA-Ent-AP-SSID',10,2,0,0
AT+WFENTLI='WPA-Ent_User_ID','WPA-Ent_PWD'
AT+RESTART

Case #3, in case of PEAP0 and MSCHAPV2 for WPA-Enterprise.
AT+WFENTAP='WPA-Ent-AP-SSID',10,2,1,1
AT+WFENTLI='WPA-Ent_User_ID','WPA-Ent_PWD'
AT+RESTART

Case #4, "Mixed" mode for EAP-type and set with TLS v1.0.
In this case, Encryption-type is configured as "Mixed" mdoe.
EAP-type and Encryption type are selected automatically.
AT+NWTLSV=1
AT+WFENTAP='WPA-Ent-AP-SSID',10,2,0
AT+WFENTLI='WPA-Ent_User_ID','WPA-Ent_PWD'
AT+RESTART
```

# 9 Advanced Function Commands

## 9.1 MQTT Commands

**Table 11: MQTT Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+NWMQCL | <mqtt_client> | Enable/disable the MQTT client.<br><mqtt_client>: 0 (disable), 1 (enable)<br>Response: OK or ERROR |
| | ? | Get the MQTT client status.<br>Response: +NWMQCL:<mqtt_client> |
| | (none) | |


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| Command | Parameters | Description |
|---|---|---|
| | *Prerequisite*<br>DA16200 (DA16600) should be connected to a network. If not, this command will not work.<br>*Example*<br>`AT+NWMQCL=1`<br>`OK`<br><br>`AT+NWMQCL`<br>`+NWMQCL:1`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● If the system restarts, then the MQTT client is not started automatically as this command is just to start/stop the MQTT client<br>● Setting MQTT configuration parameters such as MQTT broker IP, port number, subscriber topic, and so on are required to be done before issuing this command | |
| AT+NWMQMSG | <msg>,<topic> | Publish an MQTT message.<br><msg>: Message to be published (see note).<br><topic>: MQTT topic (optional).<br>Response: OK or ERROR<br>Operation Results:<br>Send Success: +NWMQMSGSND:1<br>Send Failure: +NWMQMSGSND:0,<err_code> |


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| Command | Parameters | Description |
|---|---|---|
| | *Prerequisite*<br>MQTT client should be enabled. (+NWMQCL:1)<br>*Example*<br>`AT+NWMQMSG=Hello world !!!`<br>`OK`<br>`+NWMQMSGSND:1`<br><br>`AT+NWMQMSG='{"car":"red", "type":"bus"}'`<br>`OK`<br>`+NWMQMSGSND:1`<br><br>`AT+NWMQMSG=Hello`<br>`OK`<br>`+NWMQMSGSND:0,-6` // send failed due to mqtt is not in connected state<br><br>*Note*<br>● Enabled by default in the SDK v3.2.3.0 or higher.<br>● If a single quotation is used in a message, surrounded by double quotation marks<br>● In the default AT command image, the maximum total combined string length allowed for <msg> + <topic> should be less than or equal to 2063 bytes. So, if you want to send a max length <msg> (2048 bytes long) with an explicit <topic> specified, the <topic> length should be 15 bytes long or less. If you want to send a message with the maximum length topic allowed (which is 64), you can send a 1999 byte <msg> in maximum.<br>● About Operation Results (+NWMQMSGSND:1 or +NWMQMSGSND:0,<err_code>) :<br>- This command is not included in SDK v2.4.2.0. Later versions of SDK V2.4 will include this command, but it will be disabled by default in those versions.<br>- Depending on network condition, a message publishing transaction (Qos 0, 1, 2) may take some time if network condition is not good.<br>- A new async response +NWMQMSGSND:1 or +NWMQMSGSND:0 that comes after “OK” indicates either completion of a publish transaction or failure.<br>- In CleanSession=1 mode, if mqtt is disconnected, the host can immediately get +NWMQMSGSND:0, but in CleanSession=0 mode, +NWMQMSGSND:0 is not sent but instead the transaction resumes when mqtt client re-connects. See MQTT Example: Using CleanSession=0 | |
| AT+NWMQBR | <host_name>,<port> | Set the host name (or IP address) and the port number of the MQTT Broker.<br><host_name>: Broker’s domain name, or IP address.<br><port>: Broker’s port number.<br>Response: OK or ERROR |
| | ?<br>(none) | Get the host name or IP address and the port number of the MQTT Broker.<br>Response: +NWMQBR:<host_name>,<port> |


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Example<br>AT+NWMQBR=192.168.0.65,1884<br>OK<br><br>AT+NWMQBR=?<br>+NWMQBR:192.168.0.65,1884<br>OK<br>Note<br>● Enabled by default in the SDK<br>● The broker host name (or IP address) and port configured are stored in the NVRAM<br>● MQTT restart is required to take the new configuration effect</td></tr>
<tr><td rowspan="4">AT+NWMQQOS</td><td><qos></td><td>Set the MQTT QoS level.<br><qos>: 0 (at most once), 1 (at least once), 2 (exactly once)<br>Response: OK or ERROR</td></tr>
<tr><td>?</td><td rowspan="2">Get the MQTT QoS level.<br>Response: +NWMQQOS:<qos></td></tr>
<tr><td>(none)</td></tr>
<tr><td colspan="2">Example<br>AT+NWMQQOS=1<br>OK<br><br>AT+NWMQQOS<br>+NWMQQOS:1<br>OK<br>Note<br>● Enabled by default in the SDK<br>● MQTT restart is required to take the new configuration effect</td></tr>
<tr><td rowspan="4">AT+NWMQTLS</td><td><tls></td><td>Enable/disable the MQTT TLS function.<br><tls>: 1 (enable), 0 (disable)<br>Response: OK or ERROR</td></tr>
<tr><td>?</td><td rowspan="2">Get MQTT TLS status.<br>Response: +NWMQTLS:<tls></td></tr>
<tr><td>(none)</td></tr>
<tr><td colspan="2">Prerequisite<br>Certificate should be stored before enabling the TLS function. Refer to Table 5.<br>Example<br>AT+NWMQTLS=1<br>OK<br><br>AT+NWMQTLS<br>+NWMQQOS:1<br>OK<br>Note<br>● Enabled by default in the SDK<br>● MQTT restart is required to take the new configuration effect</td></tr>
</table>


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td rowspan="4">AT+NWMQCS</td><td><clean_session></td><td>Set clean session mode<br><clean_session>: 1(session cleared), 0(session retained)<br>Response: OK or ERROR</td></tr>
<tr><td>?</td><td rowspan="2">Get clean session status.<br>Response: +NWMQCS:<clean_session></td></tr>
<tr><td>(none)</td></tr>
<tr><td colspan="2">Example<br><pre>AT+NWMQCS=1
OK

AT+NWMQCS=?
+NWMQCS:1
OK</pre>Note<br>● Enabled by default in SDK v3.2.3.0.<br>● This command is not included in SDK v2.4.2.0. The next v2.4 SDK will include this command (but this command will be disabled by default in SDK v2.4.x.x)<br>● To disable / enable this feature, use __MQTT_CLEAN_SESSION_MODE_SUPPORT__ in config_generic_sdk.h<br>● If __MQTT_CLEAN_SESSION_MODE_SUPPORT__ is not defined, mqtt client always connects to a mqtt broker in CleanSession=1 mode.<br>● MQTT re-connection is required to take this new configuration effect<br>● See MQTT Example: Using CleanSession=0</td></tr>
<tr><td rowspan="4">AT+NWMQTS</td><td><num>,<topic#1>,<br><topic#2>,<br>…</td><td>Set the topic(s) of the MQTT subscriber.<br><num>: Number of topics.<br><topic#n>: MQTT subscriber topic(s). Max topic length = 64<br>Response: OK or ERROR</td></tr>
<tr><td>?</td><td rowspan="2">Get the MQTT subscriber topic(s).<br>Response: +NWMQTS:<num>,<topic#1>,<topic#2>,…</td></tr>
<tr><td>(none)</td></tr>
<tr><td colspan="2">Example<br><pre>AT+NWMQTS=?
ERROR:-7

AT+NWMQTS=1,da16k_sub
OK

AT+NWMQTS=?
+NWMQTS:1,”da16k_sub”
OK</pre>Note<br>● Enabled by default in the SDK<br>● “ERROR:-7” returns when there’s no subscriber topic set<br>● After this command is run, the previously configured subscriber topic(s) is(are) cleared and set to the new one(s)<br>● MQTT restart is required to take the new configuration effect</td></tr>
<tr><td>AT+NWMQATS</td><td><topic></td><td>Add the specified topic to MQTT configuration</td></tr>
</table>


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>AT+NWMQATS=ABCD<br>OK<br><br>AT+NWMQTS=?<br>+NWMQTS:1,"ABCD"<br>OK<br>*Note*<br>● Enabled by default in the SDK<br>● Query command (AT+NWMQATS=?) not supported. If AT+NWMQATS=? Is run, "?" is added as a topic | |
| AT+NWMQDTS | <topic> | Delete the specified topic from MQTT configuration |
| | *Example*<br>AT+NWMQTS=?<br>+NWMQTS:2,"ABCD",”EFGH”<br>OK<br><br>AT+NWMQDTS=ABCD<br>OK<br><br>AT+NWMQTS=?<br>+NWMQTS:1,”EFGH”<br>OK<br>*Note*<br>● Enabled by default in the SDK | |
| AT+NWMQUTS | <topic> | Unsubscribe from the specified topic |
| | *Example*<br>AT+NWMQUTS=ABCD<br>OK<br>*Note*<br>● Enabled by default in the SDK (available from SDK v3.2.3.0 or higher. For SDK v2.4, the next release will include this command)<br>● This command should be run while the MQTT client is in a connected state with Broker. | |
| AT+NWMQTP | <topic> | Set the topic(s) of the MQTT publisher.<br><topic>: MQTT publisher topic.<br>Response: OK or ERROR |
| | ? | Get the MQTT publisher topic.<br>Response: +NWMQTP:<topic> |
| | (none) | |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+NWMQTP=?`<br>`ERROR:-7`<br><br>`AT+NWMQTP=da16k_pub`<br>`OK`<br><br>`AT+NWMQTP=?`<br>`+NWMQTP:da16k_pub`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● "ERROR:-7" means ERR_NO_RESULT<br>● MQTT restart is required to take the new configuration effect | |
| AT+NWMQV311 | <use_v311> | Use MQTT protocol v3.1.1. Default is v3.1<br><use_v311>: 1 (v3.1.1) / 0 (v3.1) |
| | ? | Shows the MQTT protocol version currently set |
| | *Example*<br>`AT+NWMQV311=?`<br>`+NWMQV311:0`<br>`OK`<br><br>`AT+NWMQV311=1`<br>`OK`<br><br>`AT+NWMQV311=?`<br>`+NWMQV311:1`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● MQTT restart is required to take the new configuration effect | |
| AT+NWMQPING | <period> | Set MQTT ping period.<br><period>: Ping period (second).<br>Response: OK or ERROR |
| | ? | Get the current MQTT ping period.<br>Response: +NWMQPING:<period> |
| | (none) | |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+NWMQPING=?`<br>`+NWMQPING:600`<br>`OK`<br><br>`AT+NWMQPING=300`<br>`OK`<br><br>`AT+NWMQPING`<br>`+NWMQPING:300`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● MQTT restart is required to take the new configuration effect | |
| AT+NWMQCID | <client_id> | Set the MQTT Client ID.<br><client_id>: Client ID.<br>Response: OK or ERROR |
| | ? | Get the current MQTT Client ID.<br>Response: +NWMQCID:<client_id> |
| | (none) | |
| | *Example*<br>`AT+NWMQCID=?`<br>`ERROR:-7`<br><br>`AT+NWMQCID=client-1`<br>`OK`<br><br>`AT+NWMQCID`<br>`+NWMQCID:client-1`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● “ERROR:-7” means ERR_NO_RESULT<br>● MQTT restart is required to take the new configuration effect | |
| AT+NWMQLI | <name>,<pw> | MQTT login information.<br><name>: ID.<br><pw>: Password.<br>Response: OK or ERROR |
| | ? | Get the MQTT login information.<br>Response: +NWMQLI:<name>,<pw> |
| | (none) | |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+NWMQLI=?`<br>`ERROR:-7`<br><br>`AT+NWMQLI=da16k_user,12345678`<br>`OK`<br><br>`AT+NWMQLI`<br>`+NWMQLI:da16k_user,12345678`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● "ERROR:-7" means ERR_NO_RESULT<br>● MQTT restart is required to take the new configuration effect | |
| AT+NWMQAUTO | <auto> | Enable/Disable auto-start of MQTT Client at reboot.<br><auto>: 1 (Enable), 0 (Disable) |
| | ?<br>(none) | Get the MQTT Client's auto start configuration status.<br>Response: +NWMQAUTO:<auto> |
| | *Example*<br>`AT+NWMQAUTO=?`<br>`+NWMQAUTO:0`<br>`OK`<br><br>`AT+NWMQAUTO=1`<br>`OK`<br><br>`AT+NWMQAUTO`<br>`+NWMQAUTO:1`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● Default is 0 (disable)<br>● MQTT restart is required to take the new configuration effect | |
| AT+NWMQWILL | <topic>,<msg>,<qos> | Set MQTT Will message.<br><topic>: Will topic.<br><msg>: Will message.<br><qos>: Will QoS. 0 (at most once), 1 (at least once), 2 (exactly once).<br>Response: OK or ERROR |
| | ?<br>(none) | Get the MQTT Will message.<br>Response: +NWMQWILL:<topic>,<msg>,<qos> |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>AT+NWMQWILL=?<br>ERROR:-7<br><br>AT+NWMQWILL=da16k_will,bye,0<br>OK<br><br>AT+NWMQWILL<br>+NWMQWILL:da16k_will,bye,0<br>OK<br>*Note*<br>● Enabled by default in the SDK<br>● "ERROR:-7" means ERR_NO_RESULT<br>● MQTT restart is required to take the new configuration effect | |
| AT+NWMQDEL | (none) | Reset the MQTT configurations.<br>Response: OK or ERROR |
| | *Prerequisite*<br>MQTT client should be disabled. (+NWMQCL:0)<br>*Example*<br>AT+NWMQDEL<br>OK<br>*Note*<br>● This command will reset all MQTT configurations<br>● If the MQTT client is running, run this command after the MQTT client is disabled by AT+NWMQCL=0 | |
| AT+NWMQTT | <host_name>,<port>,<br><sub_topic>,<br><pub_topic>,<br><qos>,<tls>,<br><username>,<br><password> | Run the MQTT Client with options. After entering this command, system will reboot automatically. At reboot, DA16200 (DA16600) tries to connect to the MQTT broker after the Wi-Fi connection is successfully established.<br><host_name>: Broker's domain name or IP address,<br><port>: Broker's port number.<br><sub_topic>: MQTT subscriber topic.<br><pub_topic>: MQTT publisher topic.<br><qos>: MQTT QoS level.<br><tls>: Enable/disable MQTT TLS. 1 (enable), 0 (disable).<br><username>: Login ID (optional).<br><password>: Login password (optional).<br>Response: OK or ERROR |


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| Command | Parameters | Description |
|---|---|---|
| | *Prerequisite*<br>MQTT client should be disabled (+NWMQCL:0) before running this command<br>*Example*<br>`AT+NWMQTT=192.168.0.65,1884,da16k_sub,da16k_pub,0,0`<br>`;` *Below are logs after DA16200 reboot*<br>`+INIT:DONE,0`<br>`+WFJAP:1,'test_ap_ssid',192.168.0.88`<br>`+NWMQCL:1`<br>*Note*<br>● Enabled by default in the SDK<br>● After the system reboot, operation result is sent, see "+NWMQCL" response | |

The table below shows optional MQTT configuration commands for the MQTT brokers that require TLS ALPN, SNI, or Cipher Suite info from MQTT Client at the connection stage. These commands are enabled by default in SDK v3.2.3.0 or higher, but disabled in SDK v2.x.x.x.

To enable these commands in SDK v2.x.x.x, build SDK with `__MQTT_TLS_OPTIONAL_CONFIG__` defined in config_generic_sdk.h.

**Table 12: MQTT Optional Configuration Commands**

| Command | Parameters | Description |
|---|---|---|
| AT+NWMQALPN | <num>,<br><alpn#1>,<br><alpn#2>,<br><alpn#3> | Set the TLS ALPN protocol name for MQTT.<br><num>: Number of ALPNs. Maximum number of ALPN is three<br><alpn#n>: TLS ALPN protocol name. Maximum length of each ALPN protocol name is 24<br>Response: OK or ERROR |
| | ? | Get the TLS ALPN(s) that have been set.<br>Response:<br>+NWMQALPN:<num>,<alpn#1>,<alpn#2>,<alpn#3> |


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Prerequisite<br>MQTT client should be disabled (+NWMQCL:0) before running this command<br>Example<br>AT+NWMQALPN=?<br>ERROR:-7<br><br>AT+NWMQALPN=2,alpn-protrol-name-an,alpn-protocol-name-ax<br>OK<br><br>AT+NWMQALPN<br>+NWMQALPN:2,"alpn-protrol-name-an","alpn-protocol-name-ax"<br>OK<br>Note<br>● Disabled by default in SDK v2.x.x.x.<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __MQTT_TLS_OPTIONAL_CONFIG__ is enabled in the SDK, this command will be enabled<br>● “ERROR:-7” means ERR_NO_RESULT</td></tr>
<tr><td rowspan="3">AT+NWMQSNI</td><td><sni></td><td>Set TLS SNI for MQTT.<br><sni>: Server Name Indication. Maximum length of SNI is 64<br>Response: OK or ERROR</td></tr>
<tr><td>?</td><td>Get the TLS SNI that has been set.<br>Response: +NWMQSNI:<sni></td></tr>
<tr><td colspan="2">Prerequisite<br>MQTT client should be disabled (+NWMQCL:0) before running this command<br>Example<br>AT+NWMQSNI=?<br>ERROR:-7<br><br>AT+NWMQSNI=a38a9rhiu3roqb-ats.myserver.com<br>OK<br><br>AT+NWMQSNI<br>+NWMQSNI: a38a9rhiu3roqb-ats.myserver.com<br>OK<br>Note<br>● Disabled by default in SDK v2.x.x.x.<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __MQTT_TLS_OPTIONAL_CONFIG__ is enabled in the SDK, this command will be enabled<br>● “ERROR:-7” means ERR_NO_RESULT</td></tr>
<tr><td rowspan="2">AT+NWMQCSUIT</td><td><cipher suite 1>,<br><cipher suite 2>,<br>…</td><td>Set TLS Cipher suites.<br><cipher suite>: A hex decimal value of cipher suite. See Appendix E. Maximum number of cipher suites is 17.<br>Response: OK or ERROR</td></tr>
<tr><td>?</td><td>Get TLS cipher suites that have been set<br>Response: +NWMQSNI:<number of cipher suites>,<cipher suite 1>,<cipher suite 2>,…</td></tr>
</table>

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| Command | Parameters | Description |
|---|---|---|
| | *Prerequisite*<br>MQTT client should be disabled (+NWMQCL:0) before running this command<br>*Example*<br>AT+NWMQCSUIT=?<br>ERROR:-7<br><br>AT+NWMQCSUIT=c024,c023,c00a,c009,c00d,c032<br>OK<br><br>AT+NWMQCSUIT<br>+NWMQCSUIT:6,c024,c023,c00a,c009,c00d,c032<br>OK<br>*Note*<br>● Disabled by default in SDK v2.x.x.x.<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __MQTT_TLS_OPTIONAL_CONFIG__ is enabled in the SDK, this command will be enabled<br>● "ERROR:-7" means ERR_NO_RESULT<br>● The hex pre-fix "0x" should be removed when one is typed<br>● DA16200 (DA16600) does not support all the cipher suites due to memory limitation. Contact Renesas Electronics if you want to use other cipher suites that are not specified in Appendix E | |

**Table 13: MQTT Response List**

| Response | Parameters | Description |
|---|---|---|
| +NWMQCL | <result><br>[,TOO_LONG_MSG_RX] | The result of the MQTT client connection.<br><result>: 0 (disconnected), 1 (connected)<br>For example:<br>+NWMQCL:1<br>If MQTT connection to the MQTT broker is successfully established.<br>+NWMQCL:0<br>If the MQTT connection is NOT successfully established.<br>+NWMQCL:0,TOO_LONG_MSG_RX<br>Can be sent when the MQTT is disconnected by unsupported message length only if the current mqtt connection is with clean_session=0 and qos greater than or equal to 1. See Note section for detail. |
| | *Prerequisite*<br>Sent only when MQTT Client is run<br>*Example*<br>; When MQTT connection to the MQTT broker is successfully established<br>+NWMQCL:1<br><br>; When MQTT broker is down,<br>+NWMQCL:0<br>*Note*<br>● DA16200 (DA16600) Debug Console log (UART0) | |


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| Response | Parameters | Description |
|---|---|---|
| | ○ When MQTT connection to the MQTT broker is successfully established, DA16200 (DA16600) sends “+NWMQCL:1” message to MCU (or AT command console). At this moment, you can see the following log:<br>*>>> MQTT Client connection OK*<br>○ When the MQTT broker is down, DA16200 (DA16600) sends “+NWMQCL:0” message to MCU (or AT command console) after retrying connection. In console log, you can see the following logs:<br>*Failed to receive pkt. (0x38)*<br>*Failed to read pkt(0x7880)*<br>*MQTT Client disconnected ... (state=6)*<br>*[SUB] REQ mqtt_restart (count=1)*<br>*Connecting FAIL (0x38)*<br>*Unable to connect (The connection was refused.)*<br>*…*<br>*[SUB] REQ mqtt_restart (count=5)*<br>*Connecting FAIL (0x38)*<br>*Unable to connect (The connection was refused.)*<br>*[SUB] MAX Retry (Retry Cnt=6).*<br>● Until you get this **Operation Result**, it may take more time if the DA16200 (DA16600) connection retrial happens depending on your test network condition.<br>● This message is also sent after AT+NWMQTT is run or if any MQTT configuration command is run and then the system is restarted.<br>○ DA16200 (DA16600) restarts if the AT command format is **OK**.<br>– +INIT:DONE,0 message is sent as DA16200 (DA16600) boots up.<br>– If usage of the AT command is not valid, DA16200 (DA16600) sends an **ERROR** message without restarting.<br>○ DA16200 (DA16600) tries to connect to the AP after the reboot.<br>– +WFJAP:0,<reason> or +WFJAP:1,'<SSID>',<IP Address> as result of the Wi-Fi connection.<br>– If the Wi-Fi connection information such as SSID or key is NOT stored correctly in the DA16200 (DA16600) NVRAM, +WFJAP:x response is NOT sent and the MQTT connection is NOT attempted as well. Because the MQTT connection needs a successful Wi-Fi connection first.<br>○ DA16200 (DA16600) tries to connect to the MQTT broker after the Wi-Fi connection is established. The MQTT broker information is stored in NVRAM. Connection result – +NWMQCL:0 or +NWMQCL:1 – is sent over UART1 as a result.<br>● +NWMQCL:0 is sent when the Wi-Fi Link goes down due to the following conditions<br>○ Sometimes, ATCMD host can get an unsolicited +NWMQCL:0 message when DPM Wakeup under poor signal condition with the AP connected. An example is when Wi-Fi connection with AP becomes unstable – such as when DPM Keepalive fails, Beacon loss detected, etc. In these cases, as STA, DA16200 / DA16600 tries to get “Wi-Fi link” down and up to reconnect with AP (while doing this, Wi-Fi is disconnected and then re-connected). MQTT client, when this kind of situation is detected, will try to re-connect to Broker after forcing disconnection. When MQTT disconnection happens, +NWMQCL:0 is sent to ATCMD host. On receipt of this unsolicited message, ATCMD host should wait for +NWMQCL:1.<br>● +NWMQCL:0,TOO_LONG_MSG_RX<br>○ If the current mqtt_client connection with Broker is configured with **clean_session=0** and **qos is greater than or equal to 1**, +NWMQCL:0,TOO_LONG_MSG_RX can be sent to ATCMD host (exceptional case). This message indicates that mqtt_client is disconnected by receiving a message that exceeds the message length limit (2048) of da16x mqtt client. What ATCMD host should do, in this situation, is that configuring clean_session to 1 and connect to Broker to delete the message, and then disconnect from Broker and | |

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| Response | Parameters | Description |
|---|---|---|
| | reconnect with clean_session=0 again to start over (a kind of Recovery). According to MQTT Spec, Broker keeps sending a message that has not been ACKed by the client. In this case, the long message (valid for Broker, but not valid for da16x mqtt client) can repeatedly be sent to da16x mqtt client unless connection with clean_session=1 is made from da16x mqtt client.<br>○ This response message is enabled by default in SDK v3.2.3.0.<br>○ This response message is not included in SDK v2.4.2.0. The next v2.4 SDK will include this command (but this command will be disabled by default in SDK v2.4.x.x)<br>○ Example recovery flow on receipt of +NWMQCL:0,TOO_LONG_MSG_RX<br>`...`<br>`// mqtt client is disconnected by receiving a message with unsupported length`<br>**`+NWMQCL:0,TOO_LONG_MSG_RX`**<br>`AT+NWMQCS=1 // set clean_session=1`<br>`OK`<br>`// connect to Broker (to clear the invalid long message)`<br>`AT+NWMQCL=1`<br>`OK`<br><br>`+NWMQCL:1`<br>`AT+NWMQCL=0 // disconnect from Broker`<br>`+NWMQCL:0`<br><br>`OK`<br>`AT+NWMQCS=0 // set clean_session=0`<br>`OK`<br>`AT+NWMQCL=1 // connect to Broker with clean_session=0`<br>`OK`<br><br>`+NWMQCL:1` | |
| +NWMQMSG | <msg>,<topic>,<length> | Received the MQTT message.<br><msg>: Message data.<br><topic>: Received topic.<br><length>: Message length. |
| | *Prerequisite*<br>MQTT client is in a connected state with the broker. (+NWMQCL:1)<br>MQTT publisher is ready to send a message.<br>*Example*<br>; When DA16200 (DA16600) receives a message from the MQTT publisher, the following message will be sent from DA16200 (DA16600) to AT command console:<br><br>+NWMQMSG:Hello world!!!!,da16k_sub,15<br>*Note*<br><msg> : Hello world!!!!<br><topic> : da16k_sub<br><length> : 15 (length of “Hello world!!!!”) | |

### 9.1.1 MQTT Client Connection Example

Configure the parameters and start the MQTT Client (After Wi-Fi Connection):


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```
AT+NWMQBR=172.16.0.1,1884
AT+NWMQTS=1,da16k_sub
AT+NWMQTP=da16k_pub
AT+NWMQAUTO=1 (Optional, if DPM mode is used, setting this parameter is needed)
AT+NWMQCL=1
```

If the connection is successful, the following is shown:

```
+NWMQCL:1
```

If DA16K receives a PUBLISH from a broker, the following is shown:

```
+NWMQMSG:Hello World,da16k,11
```

DA16K can send a PUBLISH to a broker. Type the following command:

```
AT+NWMQMSG='Hello I'm DA16K'
```

### 9.1.2 MQTT TLS Connection Example

```
Configure the MQTT parameters:AT+NWMQBR=172.16.0.1,8883
AT+NWMQTS=1,da16k_sub
AT+NWMQTP=da16k_pub
AT+NWMQTLS=1
AT+NWMQAUTO=1 (Optional, if DPM mode is used, setting this parameter is needed)
```

To check the validity of a certificate, the DA16K should set the exact current time:

```
AT+TIME=yyyy-mm-dd,hh:mm:ss
```

And store the certificate and private key if needed. (<ESC>C in Section 7)

After all settings are made, start the client:

```
AT+NWMQCL=1
```


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### 9.1.3 MQTT Example with DPM

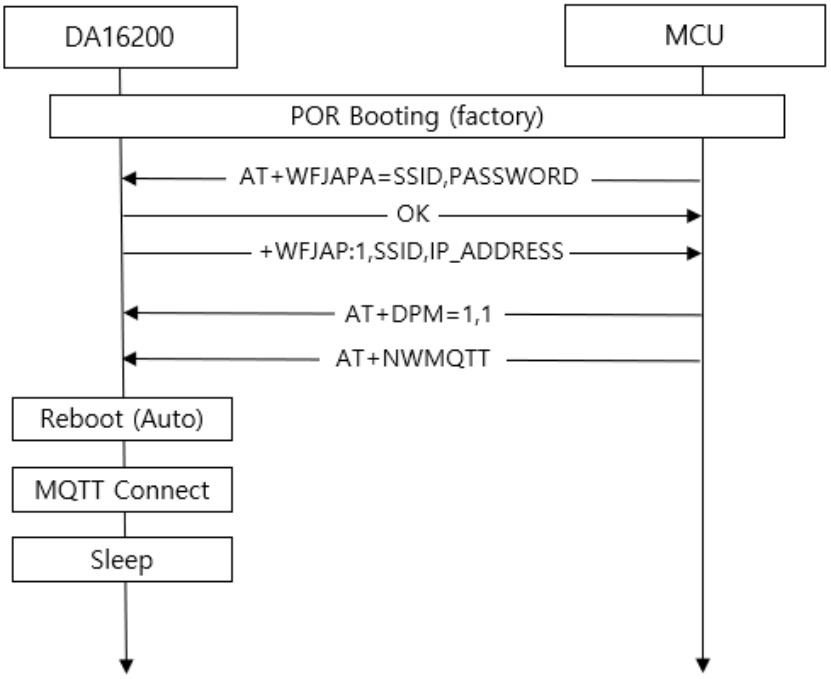



**Figure 5: Example Sequence to Initiate MQTT Protocol with DPM**

Figure 5 is an example sequence to initiate the MQTT protocol with DPM in the DA16200 (DA16600).

In the normal BOOT state, connect to an AP (AT+WFJAPA) and change the DA16200 (DA16600) run mode to DPM mode (AT+DPM=1,1 ← optional parameter '1' means writing "dpm mode" to NVRAM and does not reboot. To make DPM mode take effect, a reboot is required).

To configure the MQTT connection information, enter command AT+CLRDPMSLPEXT and type the following as an example:

```
AT+NWMQTT=test.mosquitto.org,1883,sub_topic,pub_topic,0,0
```


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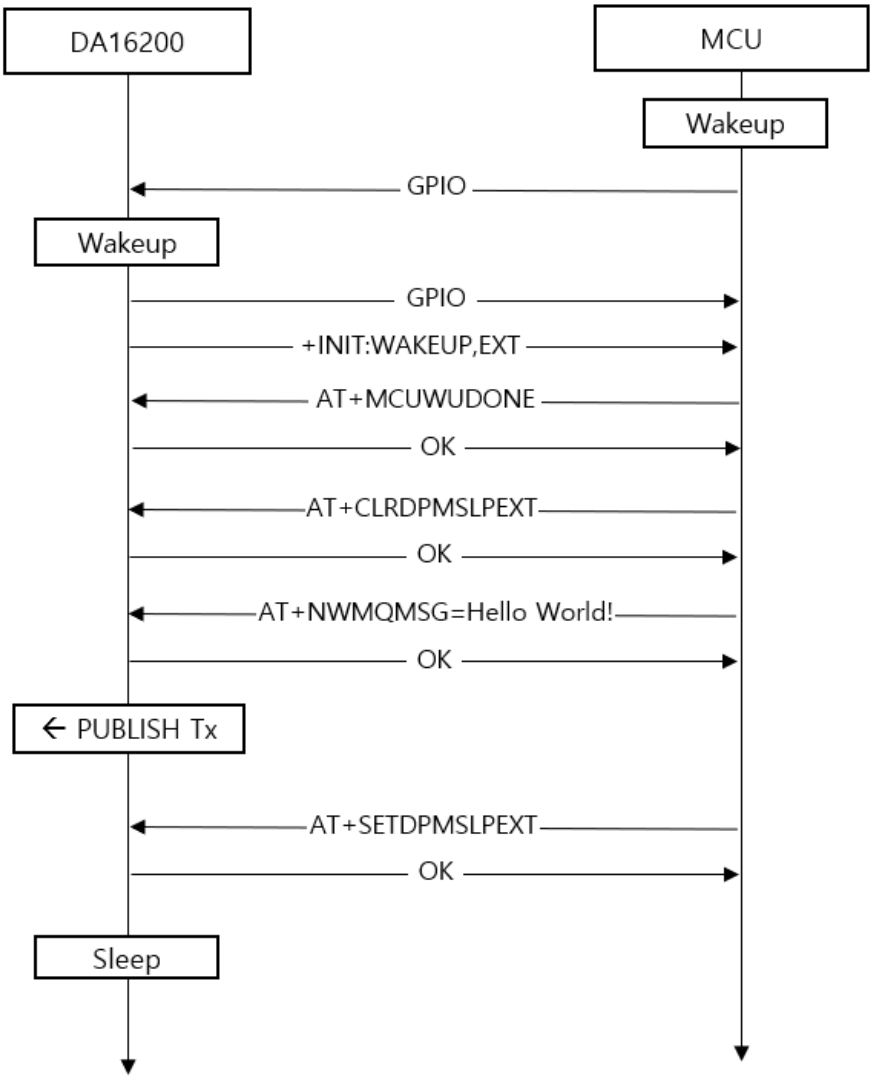



**Figure 6: Procedure to Send MQTT Messages**

Figure 6 shows the procedure to send an MQTT message in Sleep mode.

When MCU wakes up the DA16200 (DA16600), the response +INIT:WAKEUP,EXT is sent. The MCU sends the command AT+MCUWUDONE to inform that MCU is ready to operate. To prevent that the DA16200 (DA16600) enters DPM Sleep mode, MCU should send command AT+CLRDPMSLPEXT before an MQTT PUBLISH is sent. To make the DA16200 (DA16600) enter DPM Sleep mode again, send a PUBLISH with command AT+NWMQMSG, and then enter command AT+SETDPMSLPEXT.


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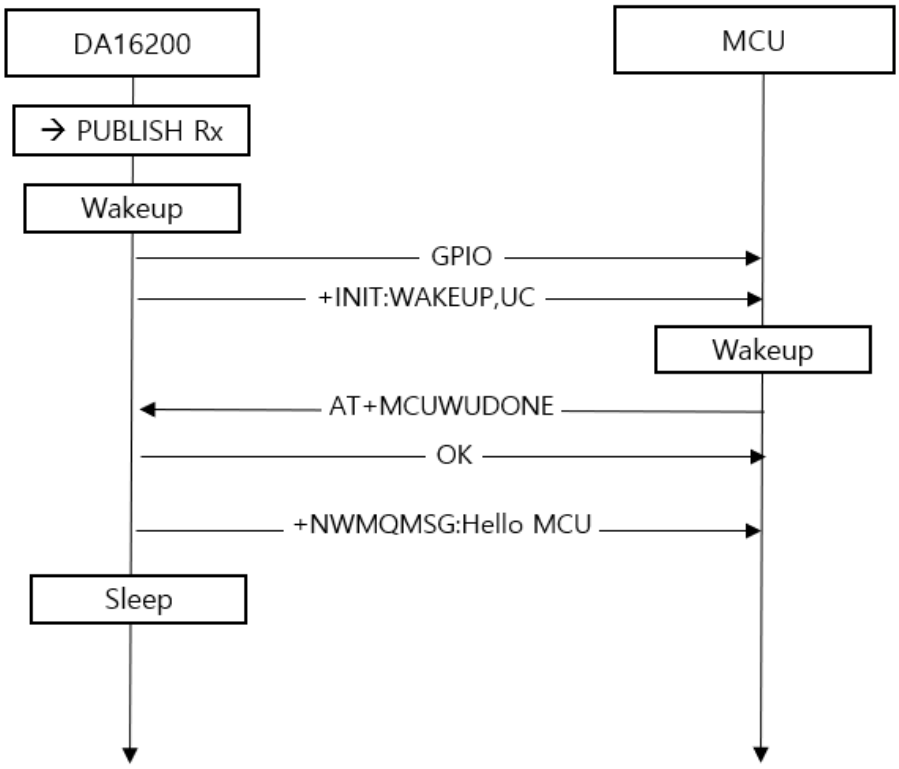



**Figure 7: Procedure to Process MQTT Messages**

Figure 7 shows the procedure to process an MQTT message received while in Sleep mode.

When the DA16200 (DA16600) wakes up by a PUBLISH message from an MQTT broker, the response +INIT:WAKEUP,UC is sent. The MCU sends the AT+MCUWUDONE to inform that it is ready to operate. Next, the DA16200 (DA16600) sends the received PUBLISH to the MCU and enters DPM Sleep mode again.

### 9.1.4 MQTT Example: Changing Subscription Topic when Running

Let's assume that the Wi-Fi/MQTT connection is configured properly and DPM is set to 1 (TRUE).

Below is the recommended sequence. Note that the double quotation marks are used.


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```
A. Trigger RTC_WAKE_UP Event (by MCU)
B. Wait for "+INIT:WAKEUP,EXT" Response. Send AT+MCUWUDONE, and wait for “OK”
C. Run "AT+CLRDPMSLPEXT" command
D. Wait for "OK" response
E. loop running "AT+NWMQCL=?"
     E.1 if responses are "+NWMQCL:0" and "OK"
     E.2      then, goto E. to run "AT+NWMQCL=?" command
     E.3 else if responses are "+NWMQCL:1" and "OK"
     E.4      then, goto next, F.
     E.5 else if response is "ERROR:x"
     E.6      then, Run "AT+SETDPMSLPEXT"
     E.7            Wait for "OK" response
     E.8            return
F. Run "AT+NWMQCL=0"
G. Wait for "+NWMQCL:0" and "OK" response
H. Run "AT+NWMQTS=<New MQTT Subscription Topic>"
I. Wait for "OK" response
J. Run "AT+RESTART"
K. Wait for "+INIT:DONE,0" response
L. Wait for "+WFJAP:1,'<SSID>',<IP ADDRESS>"
M. Wait for "+NWMQCL:1" response
```

### 9.1.5 MQTT Example: Reading Subscription Topic when Running

Let's assume that the Wi-Fi/MQTT connection is configured properly and DPM is set to 1 (TRUE).

The reading of the MQTT publishing topic would be similar.

Below is the recommended sequence. Note that the double quotation marks are used.

1. Trigger RTC_WAKE_UP Event

Wait for "+INIT:WAKEUP,EXT" Response. Send AT+MCUWUDONE, and wait for "OK"

2. Run "AT+CLRDPMSLPEXT" command
3. Wait for "OK" response
4. Run "AT+NWMQTS=?"
5. Wait for "+NWMQTS:<MQTT Subscription Topic>" and "OK" response

   Note that you need to consider that you can get the ERROR response if that format of the command may have some error.

6. Run "AT+SETDPMSLPEXT"
7. Wait for "OK" response

Let's assume that the Wi-Fi/MQTT connection is configured properly and DPM is set to 1 (TRUE).

The reading of the MQTT publishing topic would be similar.

Below is the recommended sequence. Note that the double quotation marks are used.

1. Trigger RTC_WAKE_UP Event
2. Wait for "+INIT:WAKEUP,EXT" Response. Send AT+MCUWUDONE, and wait for “OK”
3. Run "AT+CLRDPMSLPEXT" command
4. Wait for "OK" response
5. Run "AT+NWMQTS=?"
6. Wait for "+NWMQTS:<MQTT Subscription Topic>" and "OK" response
7. Run "AT+SETDPMSLPEXT"


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### 9.1.6 MQTT Example: Using CleanSession=0

#### 9.1.6.1 CleanSession=0 Mode

When an MQTT Client (Mqttc onward) establishes connection with an MQTT Broker (Broker onward), there are two types of session; CleanSession=1 and CleanSession=0.

**CleanSession=1**: default session type. when Broker gets a connect request from an Mqttc that tries to connect with an option “CleanSession=1” (which is default config on DA16x), Broker treats the connection as a “new” session. So if there’s any existing session associated with the same client_id found, Broker clears that previous session and creats a new one with the client_id.

**CleanSession=0**: when Broker gets a connect request from an Mqttc that tries to connect with an option “CleanSession=0”, Broker tries to find a session (session data) with the same client_id first. If it finds one, it keeps using that session for the new Mqttc.

While Mqttc is in operation with Broker, there may be times when the TCP connection gets unstable and disconnected (e.g. mqtt ping failed) which may cause some messages that had been published to Broker at that specific disconnected time may not be delivered to a subscriber. If new messages (with QoS > 0) are published to Broker and for sessions that have been configured in “CleanSession=0”, Broker retains and re-send them when the Mqttc is re-connected. Mqttc (if CleanSession=0 is enabled) also should retain the state of the unfinished / unacked messages until reconnection.

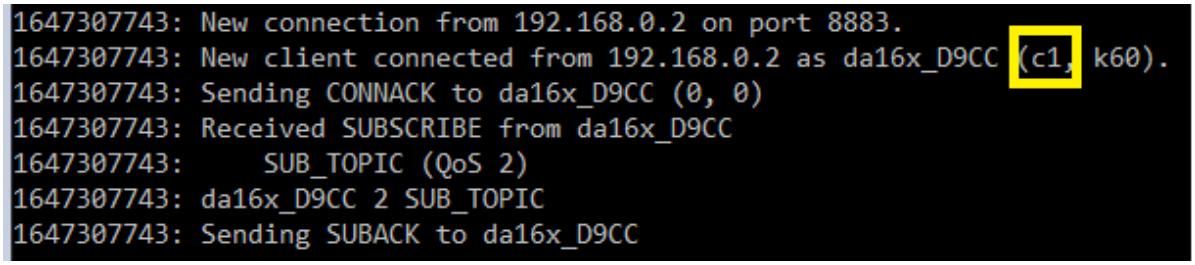



**Figure 8: Broker console - CleanSession=1 connection**

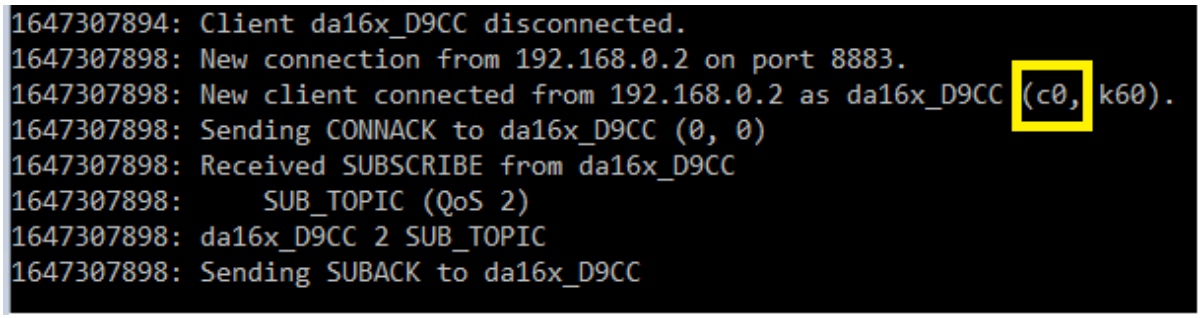



**Figure 9: Broker console - CleanSession=0 connection**

Even with CleanSession=0 connection, Broker does not maintain session data if Mqttc is disconnected in the following cases.

- If a new message is published with QoS 0 after Mqttc is disconnected
- If Mqttc’s connection QoS is 0.

DA16x supports CleanSession=0 mode in the following way

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- "CleanSession=0 feature" is enabled by default in SDK v3.2.3.0 or higher, disabled by default in SDK v2.4.x.x. For verfiication of this feature in SDK v2.4.x.x, "CleanSession=0 feature" should be enabled in config_generic_sdk.h (__MQTT_CLEAN_SESSION_MODE_SUPPORT__)
- If customer application decides that **QoS 1 or Qos 2 and CleanSession=0** is used in their application, the message (payload) size (both Tx and Rx) should be pre-decided (because there's limitation in dpm user pool size). By default 100 bytes are defined.

  #define MQTT_MSG_TBL_PRESVD_MAX_PLAYLOAD_LEN 100

  Depending on application's expected maximum payload size while operation, other value can be defined.

  DPM User Pool has limited amount ( about 8K in total ) in the system.

  Check available "free" DPM User Pool size first (by using the console command "dpm user_pool"), and then calculate the max payload length and message number for your application if needed.

  The default configuraiton (payload_len: 100, max_count: 10)

  allocates about 1.9KB of dpm user pool (Check mq_msg_tbl_presvd_t for detail).

  Search the following compiler options in config_generic_sdk.h.

  ```
  //max payload length of a preserved message
  #define MQTT_MSG_TBL_PRESVD_MAX_PLAYLOAD_LEN100
  // max number of preserved messages
  #define MQTT_MSG_TBL_PRESVD_MAX_MSG_CNT                10
  ```

- Supported command to set CleanSession mode : AT+NWMQCS=<1|0>

### CleanSession and QoS Matrix Table for PUBLISH Rx

| Subscriber | | | Unacked Message Delivery | QoS | Publisher |
|---|---|---|---|---|---|
| Case | Clean Session | QoS | (After MQTT Reconnection) | (Effective Actual) | Message's QoS |
| 1 | 1 | 0 | X | **0** | 0 |
| 2 | 1 | 1 | X | **0** | 0 |
| 3 | 1 | 2 | X | **0** | 0 |
| 4 | 1 | 0 | X | **0** | 1 |
| 5 | 1 | 1 | X | **1** | 1 |
| 6 | 1 | 2 | X | **1** | 1 |
| 7 | 1 | 0 | X | **0** | 2 |


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| | | | | | |
|---|---|---|---|---|---|
| 8 | 1 | 1 | X | 1 | 2 |
| 9 | 1 | 2 | X | 2 | 2 |
| 10 | 0 | 0 | X | 0 | 0 |
| 11 | 0 | 1 | X | 0 | 0 |
| 12 | 0 | 2 | X | 0 | 0 |
| 13 | 0 | 0 | X | 0 | 1 |
| 14 | 0 | 1 | O | 1 | 1 |
| 15 | 0 | 2 | O | 1 | 1 |
| 16 | 0 | 0 | X | 0 | 2 |
| 17 | 0 | 1 | O | 1 | 2 |
| 18 | 0 | 2 | O | 2 | 2 |

**Table 14 CleanSession and QoS matrix in message Rx**

Basically with CleanSession=1, no unacked message delivery happens when MQTT reconnect happens (marked as x)

With CleanSession=0, only case 14, 15, 17, and 18 makes message redelivery happen for messages that had been delivered to Broker while the Mqttc was offline. (marked as O)

### CleanSession and QoS Matrix Table for PUBLISH Tx

| | |
|---|---|
| Expectation 1 | Application should assume “sending message” fails, and wait until mqtt gets re-connected for retry. |
| Behavior 1 | Application does meessage send retry |
| Expectation 2 | Application should assume “sending message” will resume when mqtt gets re-connected |
| Behavior 2 | Application just wait until meessage send retry by mqtt is done |

| Publisher | | | Expectation if MQTT gets disconnected (while QoS 1/2 message is not fully acked or QoS 0 Send is being sent) | Behaviour expected when MQTT Client re-connected |
|---|---|---|---|---|
| Case | Clean Session | QoS | | |
| 1 | 1 | 0 | Expectation 1 | Behaviour 1 |
| 2 | 1 | 1 | Expectation 1 | Behaviour 1 |
| 3 | 1 | 2 | Expectation 1 | Behaviour 1 |


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| | | | | |
|---|---|---|---|---|
| 4 | 0 | 0 | Expectation 1 | Behaviour 1 |
| 5 | 0 | 1 | Expectation 2 | Behaviour 2 |
| 6 | 0 | 2 | Expectation 2 | Behaviour 2 |

**Table 15 CleanSession and QoS matrix in message Tx**

When publishing a message from DA16x, application's expectation and action/behavior may be different if CleanSession=0 and QoS 1 or 2 are used in some specific cases.

In normal network condition, there's no difference in message send behavior between CleanSession=0 and CleanSession=1.

In some abnormal cases where QoS 1/2's ACK message (PUBACK, PUBREC, PUBREL, or PUBCOMP) get lost due to some bad network conditions (which can cause Mqttc re-connection), CleanSession=0 can recover the previous message state and resume the communication with Broker.

However, if CleanSession=1 is used, when Mqttc is disconnected, it can safely re-transmit the message when Mqttc is reconnected. Depending on use cases of applications / host applications, either approach (CleanSession=0 or CleanSession=1 ) can be utilized.

#### 9.1.6.2 How to Connect with CleanSession=0

```
AT+NWMQQOS=2
OK
AT+NWMQCS=0
OK
AT+NWMQCL=1
OK

+NWMQCL:1
```

To activate "**CleanSession=0 support mode**" in DA16x, QoS should be 1 or 2 and CleanSession option should be set to 0.

If either option (CleanSession and QoS) is not set as above, CleanSession=0 support mode is disabled.

#### 9.1.6.3 How to Restart CleanSession=0 test

If you want to do re-test (fresh new test) with CleanSession=0 mode, depending on the previous session type, you may need Broker to "clear the previous session".

The reason is that since an Mqttc connects with CleanSession=0, Broker does not delete the session data until the Mqttc re-connects with CleanSession=1.

Case 1: Previous session is CleanSession=1 and you want to restart a new CleanSession=0 test

```
AT+NWMQCL=0
OK
AT+NWMQCS=0
OK
AT+NWMQCL=1
```


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```
OK

+NWMQCL:1
```

Case 2: Previous session is CleanSession=0 and you want to re-test another CleanSession=0 test run.

```
AT+NWMQCL=0
+NWMQCL:0

OK
AT+NWMQCS=1
OK
AT+NWMQCL=1
OK

+NWMQCL:1
AT+NWMQCL=0
+NWMQCL:0

OK
AT+NWMQCS=0
OK
AT+NWMQCL=1
OK

+NWMQCL:1
```

#### 9.1.6.4 PUBLISH Rx Test Steps

Test steps are as follows under non-DPM and DPM mode.

In non-DPM mode:

- DA16x: connect to Broker
- Publisher: send one or two messages
- DA16x: check if the messages are received.
- DA16x: disconnect from Broker
- Publisher: send one or two messages (let say msg_A)
- DA16x: reconnect to Broker
- DA16x: check if msg_A (sent while DA16x is offline) is received

DPM mode:

- DA16x: connect to Broker. Enter DPM Sleep
- Publisher: send one or two messages
- DA16x: check if the messages are received.
- DA16x: turn off AP. Do not turn on AP, but wait for the mqtt keep alive period (to make sure Broker recognizes the Mqttc disconnection)
- Publisher: send one or two messages (let say msg_A)
- DA16x: turn on AP. Wait until DA16x is connected to AP
- DA16x: reconnected to AP and check if msg_A (sent while DA16x is offline) is received.


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Note:

- mosquitto broker (Broker), mosquitto publisher (Publisher), and DA16x (Subscriber) are used for the test.
- message length from publisher should be less than or equal to 100. If longer messages are sent, they may not be restored properly when mqtt is reconnected.

#### 9.1.6.5 PUBLISH Rx Test Steps - Example 1 (Non-DPM)

Below is the test steps for case 15 (non-DPM mode)

[DA16x] Connect Mqttc with CleanSession=0 and QoS 2

```
AT+NWMQQOS=2
OK
AT+NWMQCS=0
OK
AT+NWMQCL=1
OK

+NWMQCL:1
```

[Other Publisher] Publish messages

```
C:\mosquitto>mosquitto_pub -h 192.168.0.230 -p 8883 --cafile cas.pem --cert
wifiuser.pem --key wifiuser.key --tls-version tlsv1 --insecure -q 2 -t SUB_TOPIC -
m "Hello_q2"

C:\mosquitto>mosquitto_pub -h 192.168.0.230 -p 8883 --cafile cas.pem --cert
wifiuser.pem --key wifiuser.key --tls-version tlsv1 --insecure -q 2 -t SUB_TOPIC -
m "Hello_q2_2"
```

[DA16x] check the messages are successfully received

```
+NWMQMSG:Hello_q2,SUB_TOPIC,8

+NWMQMSG:Hello_q2_2,SUB_TOPIC,10
```

[DA16x] disconnect from Broker

```
AT+NWMQCL=0
+NWMQCL:0

OK
```

[Other Publisher] publish two messages (while DA16x is in disconnected state)

```
C:\mosquitto>mosquitto_pub -h 192.168.0.230 -p 8883 --cafile cas.pem --cert
wifiuser.pem --key wifiuser.key --tls-version tlsv1 --insecure -q 2 -t SUB_TOPIC -
m "Hello_q2_3"

C:\mosquitto>mosquitto_pub -h 192.168.0.230 -p 8883 --cafile cas.pem --cert
wifiuser.pem --key wifiuser.key --tls-version tlsv1 --insecure -q 2 -t SUB_TOPIC -
m "Hello_q2_4"
```


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[DA16x] reconnect to Broker and check if the two messages that had been published while DA16x is in disconnected state are received successfully.

```
AT+NWMQCL=1
OK

+NWMQCL:1

+NWMQMSG:Hello_q2_3,SUB_TOPIC,10

+NWMQMSG:Hello_q2_4,SUB_TOPIC,10
```

#### 9.1.6.6 PUBLISH Rx Test Steps - Example 2 (DPM)

Below is the test steps for case 18 (DPM mode)

(mosquitto broker and mosquitto publisher are used for test)

[DA16x] Connect with CleanSession=0 and QoS 2

```
AT+NWMQQOS=2
OK
AT+NWMQCS=0
OK
AT+RESTART
OK

+INIT:DONE,0

+WFJAP:1,'SYN_TEST_AP',192.168.1.195

+ATPROV=STATUS 0

+NWMQCL:1
```

[Other Publisher] Publish messages

```
C:\mosquitto>mosquitto_pub -h 192.168.0.230 -p 8883 --cafile cas.pem --cert
wifiuser.pem --key wifiuser.key --tls-version tlsv1 --insecure -q 2 -t SUB_TOPIC -
m "Hello_q2_1"

C:\mosquitto>mosquitto_pub -h 192.168.0.230 -p 8883 --cafile cas.pem --cert
wifiuser.pem --key wifiuser.key --tls-version tlsv1 --insecure -q 2 -t SUB_TOPIC -
m "Hello_q2_2"
```

[DA16x] check the messages are successfully received

Note : At wakeup time, ATCMD host should send AT+CLRDPMSLPEXT to get +NWMQMSG after which AT+SETDPMSLPEXT should be sent by ATCMD host to let DA16x enter DPM Sleep.

```
+INIT:WAKEUP,UC

+ATPROV=STATUS 0

+NWMQMSG:Hello_q2_1,SUB_TOPIC,10

+INIT:WAKEUP,UC
```


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```
+ATPROV=STATUS 0

+NWMQMSG:Hello_q2_2,SUB_TOPIC,10
```

[DA16x] Turn OFF AP

```
+INIT:WAKEUP,NOBCN

+ATPROV=STATUS 0

+WFDAP:0,INACTIVITY
...
```

[Broker] make sure Mqttc is disconnected

```
...
1647405247: Socket error on client da16x_D9CC, disconnecting.
...
```

[Other Publisher] publish two messages (while DA16x is in disconnected state)

```
C:\mosquitto>mosquitto_pub -h 192.168.0.230 -p 8883 --cafile cas.pem --cert
wifiuser.pem --key wifiuser.key --tls-version tlsv1 --insecure -q 2 -t SUB_TOPIC -
m "Hello_q2_3"

C:\mosquitto>mosquitto_pub -h 192.168.0.230 -p 8883 --cafile cas.pem --cert
wifiuser.pem --key wifiuser.key --tls-version tlsv1 --insecure -q 2 -t SUB_TOPIC -
m "Hello_q2_4"
```

[DA16x] Turn ON AP

[DA16x] Wait until AP is connected and see whether “hello_qos_3” and “hello_qos_4” are received

```
+INIT:DONE,0

+WFJAP:1,'SYN_TEST_AP',192.168.1.195

+ATPROV=STATUS 0

+NWMQCL:1

+NWMQMSG:Hello_q2_3,SUB_TOPIC,10

+NWMQMSG:Hello_q2_4,SUB_TOPIC,10
```

#### 9.1.6.7 PUBLISH Tx Test Steps

Test steps are as follows:

- DA16x: connect to Broker
- DA16x: send a messages
- DA16x: check if the message send is successful

Note:


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- Message length from DA16x should be less than or equal to 100 bytes for case 5 and 6 configuration. Sending longer messages returns failure. For cases other than case 5 or 6, message length limit is 2048 bytes

#### 9.1.6.8 PUBLISH Tx Test Steps – Example

Below are the test steps for case 6 (non-DPM mode)

```
AT+NWMQQOS=2
OK
AT+NWMQCS=0
OK
AT+NWMQCL=1
OK

+NWMQCL:1
AT+NWMQMSG=hello_q2
OK

+NWMQMSGSND:1
```

## 9.2 HTTP-Client Commands

**Table 16: HTTP-Client Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+NWHTC | <url>,<method>(,<msg>) | Start the HTTP client with options.<br><url>: HTTP server address.<br><method>: GET, POST or PUT.<br><msg>: Request message for POST and PUT methods. |
| | *Prerequisite*<br>Target system should be connected to an AP.<br>HTTP server should exist.<br>*Example 1*<br>`AT+NWHTC=https://httpbin.org/get,get`<br>`OK`<br>*Example 2*<br>`AT+NWHTC=https://httpbin.org/post,post,HTTP-Client POST method sample test!`<br>`OK`<br>*Example 3*<br>`AT+NWHTC=https://httpbin.org/put,put,HTTP-Client POST method sample test!`<br>OK<br>*Note*<br>• Disabled by default in the ThreadX SDK. If __SUPPORT_HTTP_CLIENT__ is enabled in the SDK, this command will be enabled<br>• Enabled by default in the SDK v3.2.3.0 or later. If __SUPPORT_HTTP_CLIENT_FOR_ATCMD__ is enabled in the SDK, this command will be enabled | |


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td rowspan="2">AT+NWHTCSNI</td><td><sni></td><td>Set the server name indication<br><sni>: Server name</td></tr>
<tr><td colspan="2">Example<br>AT+NWHTCSNI=httpbin.org<br>OK<br>Note<br>● Enabled by default in the SDK<br>● It must be set up before connecting to the server</td></tr>
<tr><td rowspan="2">AT+NWHTCALPN</td><td><alpn_number>,<alpn1>,<alpn2>,<alpn3></td><td>Set the application layer protocol negotiation<br><alpn_number>: Number of alps<br><alpn1>: First alpn<br><alpn2>: Second alpn<br><alpn3>: Third alpn</td></tr>
<tr><td colspan="2">Example<br>AT+NWHTCALPN=1,http/1.1<br>OK<br>Note<br>● Enabled by default in the SDK<br>● It must be set up before connecting to the server</td></tr>
<tr><td rowspan="2">AT+NWHTCSNIDEL</td><td>(none)</td><td>Delete the saved SNI</td></tr>
<tr><td colspan="2">Example<br>AT+NWHTCSNIDEL<br>OK<br>Note<br>● Enabled by default in the SDK<br>● It must be set up before connecting to the server</td></tr>
<tr><td rowspan="2">AT+NWHTCALPNDEL</td><td>(none)</td><td>Delete all saved ALPNs</td></tr>
<tr><td colspan="2">Example<br>AT+NWHTCALPNDEL<br>OK<br>Note<br>● Enabled by default in the SDK<br>● It must be set up before connecting to the server</td></tr>
</table>


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**Table 17: HTTP-Client Response List**

| Response | Parameters | Description |
|---|---|---|
| + NWHTCSTATUS | <status> | Returns status along with the received payload according to the requested method.<br><status>: 0x00 is success.<br>See Appendix B.<br>For example: +NWHTCSTATUS:0x00 |


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### 9.2.1 HTTP-Client Connection Example

GET method request:

```
AT+NWHTC=https://httpbin.org/get,get
OK
HTTP/1.1 200 OK
Date: Tue, 07 Dec 2021 01:19:49 GMT
Content-Type: application/json
Content-Length: 457
Connection: keep-alive
Server: gunicorn/19.9.0
Access-Control-Allow-Origin: *
Access-Control-Allow-Credentials: true

{
  "args": {},
  "headers": {
    "Accept": "*/*",
    "Accept-Encoding": "identity",
    "Accept-Language": "ko-KR,Ko;q=0.9,en-US;q=0.8,en;q=0.7",
    "Host": "httpbin.org",
    "User-Agent": "Mozilla/5.0 (windows NT 6.1; Win64; x64) AppleWebkit/537.36
(KHTML, like Gecko) Chrome/64.0.3282.186 Safari/537.36",
    "X-Amzn-Trace-Id": "Root=1-61aeb6b5-67d7324c112a7f1631adcc72"
  },
  "origin": "124.50.108.25",
  "url": "https://httpbin.org/get"
}

+NWHTCSTATUS:0x00
```

POST method request:

```
AT+NWHTC=https://httpbin.org/post,post,HTTP-Client POST method sample test!
OK
HTTP/1.1 200 OK
Date: Tue, 07 Dec 2021 01:25:05 GMT
Content-Type: application/json
Content-Length: 586
Connection: keep-alive
Server: gunicorn/19.9.0
Access-Control-Allow-Origin: *
Access-Control-Allow-Credentials: true

{
  "args": {},
  "data": "HTTP-Client POST method sample test!",
  "files": {},
  "form": {},
  "headers": {
    "Accept": "*/*",
    "Accept-Encoding": "identity",
    "Accept-Language": "ko-KR,Ko;q=0.9,en-US;q=0.8,en;q=0.7",
    "Content-Length": "36",
    "Host": "httpbin.org",
    "User-Agent": "Mozilla/5.0 (windows NT 6.1; Win64; x64) AppleWebkit/537.36
(KHTML, like Gecko) Chrome/64.0.3282.186 Safari/537.36",
    "X-Amzn-Trace-Id": "Root=1-61aeb7f1-341bbb8c3f3d6bc7484370e2"
```


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```
  },
  "json": null,
  "origin": "124.50.108.25",
  "url": "https://httpbin.org/post"
}

+NWHTCSTATUS:0x00
```

PUT method request:

```
AT+NWHTC=https://httpbin.org/put,put,HTTP-Client PUT method sample test!
OK
HTTP/1.1 200 OK
Date: Tue, 07 Dec 2021 02:04:19 GMT
Content-Type: application/json
Content-Length: 584
Connection: keep-alive
Server: gunicorn/19.9.0
Access-Control-Allow-Origin: *
Access-Control-Allow-Credentials: true

{
  "args": {},
  "data": "HTTP-Client PUT method sample test!",
  "files": {},
  "form": {},
  "headers": {
    "Accept": "*/*",
    "Accept-Encoding": "identity",
    "Accept-Language": "ko-KR,Ko;q=0.9,en-US;q=0.8,en;q=0.7",
    "Content-Length": "35",
    "Host": "httpbin.org",
    "User-Agent": "Mozilla/5.0 (windows NT 6.1; Win64; x64) AppleWebkit/537.36
(KHTML, like Gecko) Chrome/64.0.3282.186 Safari/537.36",
    "X-Amzn-Trace-Id": "Root=1-61aec123-4c3c5d390c6b31992bb803be"
  },
  "json": null,
  "origin": "124.50.108.25",
  "url": "https://httpbin.org/put"
}

+NWHTCSTATUS:0x00
```

## 9.3 HTTP-Server Commands

**Table 18: HTTP-Server Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+NWHTS | <flagt> | Start or stop the HTTP server depending on your options.<br><start>: 1 (start), 0 (stop)<br>Response: OK or ERROR. |


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| Command | Parameters | Description |
|---|---|---|
| | *Prerequisite*<br>Target system should be connected to an AP.<br>*Example*<br>`AT+NWHTS=1`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+NWHTSS | <flag> | Start or stop the HTTPS server depending on your options.<br><start>: 1 (start), 0 (stop)<br>Response: OK or ERROR. |
| | *Prerequisite*<br>Target system should be connected to an AP.<br>*Example*<br>`AT+NWHTSS=1`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK. If __SUPPORT_HTTP_SERVER__ is enabled in the SDK, this command will be enabled<br>● Enabled by default in the SDK v3.2.3.0 or later. If __SUPPORT_HTTP_SERVER_FOR_ATCMD__ is enabled in the SDK, this command will be enabled | |

### 9.3.1 HTTP/HTTPS-Server Start Example

HTTP start:

```
AT+NWHTS=1
```

HTTPS start:

```
AT+NWHTSS=1
```

## 9.4 WebSocket-Client Commands

**Table 19: WebSocket-Client Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+NWWSC | <operation>,<uri> (<msg>) | Start the WebSocket client with options.<br><operation> connect, send or disconnect<br><uri>: WebSocket server address.<br><msg>: Request message |
| | *Prerequisite*<br>Target system should be connected to an AP.<br>WebSocket server should exist.<br><br>*Example 1*<br>`AT+NWWSC=connect,ws://192.168.86.182:8080`<br>`+NWWSC:1`<br><br>*Example 2*<br>`AT+NWWSC=send,Send Message Test`<br>`OK`<br><br>*Example 3*<br>`AT+NWWSC=disconnect` | |


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| | |
|---|---|
| | +NWWSC:0<br><br>*Note*<br>● If __SUPPORT_WEBSOCKET_CLIENT__ is enabled in SDK, this command will be enabled |

**Table 20: WebSocket-Client Response List**

| Response | Parameters | Description |
|---|---|---|
| +NWWSC | <status>(,<opcode>,<received msg length>,<received msg>) | Returns status along with the received payload.<br><status><br>0 is disconnected.<br>1 is connected.<br><opcode><br>Continuation Frame : 0x00<br>Text Frame : 0x01<br>Binary Frame : 0x02<br>Close Frame : 0x08<br>Ping Frame : 0x09<br>Pong Frame : 0x0a<br><br>*Example 1*: +NWWSC:1<br>*Example 2*: +NWWSC:0<br>*Example 3*: +NWWSC:1,1,12,Test Message |

## 9.5 OTA Commands

**NOTE**

- In the case of DPM mode enabled:
  If OTA update is in progress (firmware download is in progress), it does not enter DPM Sleep because SFLASH write operation occurs.
  After downloading the firmware, the DA16200 resumes to enter DPM sleep mode.

**Table 21: OTA Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+NWOTADWSTART | <fw_type>,<uri>[,<fw_name>] | Start downloading firmware from an OTA server.<br><*fw_type*>: Set the type of FW to be downloaded.<br><*uri*>: Server URL where a FW exists.<br><*fw_name*>: Optional. It is available if *fw_type* is *mcu_fw*. Maximum input size is eight bytes. MCU_FW will be stored as default if there is no fw_name information. (Only for MCU FW)<br>Response: +NWOTADWSTART:0x00 |


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Prerequisite<br>Target system should be connected to an AP.<br>OTA server should exist.<br>fw_type should be a lowercase character.<br>Example<br>; RTOS download<br>AT+NWOTADWSTART=rtos,https://server/DA16200_RTOS-GEN01-01-1111-000000.img<br>OK<br>+NWOTADWSTART:0x00<br><br>; SLIB download<br>AT+NWOTADWSTART=slib,https://server/DA16200_SLIB-GEN01-01-1111-000000.img<br>OK<br>+NWOTADWSTART:0x00<br><br>; MCU FW download<br>AT+NWOTADWSTART=other_fw,https://server/mcu_firmware.img<br>OK<br>+NWOTADWSTART:0x00<br><br>AT+NWOTADWSTART=other_fw,https://server/mcu_firmware.img,ver01<br>OK<br>+NWOTADWSTART:0x00<br><br>; BLE FW download<br>AT+NWOTADWSTART=ble_fw,https://server/pxr_sr_coex_ext_531_6_0_14_1114_2_ota.img<br>OK<br>+NWOTADWSTART:0x00<br><br>; Cert Key download:<br>AT+NWOTADWSTART=cert_key,https://server/ca.pem<br>OK<br>+NWOTADWSTART:0x00<br>Note<br>● Enabled by default in the SDK<br>● BLE FW download is available on SDK V3.2.3.0 or higher</td></tr>
<tr><td>AT+NWOTARENEW</td><td>(none)</td><td>Reboot with updated FW.</td></tr>
</table>


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Prerequisite<br>RTOS and SLIB or BLE images should be downloaded in advance.<br>Example<br><pre>AT+NWOTARENEW
+NWOTARENEW:0x00</pre>Note<br>● Enabled by default in the SDK<br>● Will reboot automatically after renew is done<br>● BLE image is supported in SDK V3.2.3.0 or higher (not on SDK v2.4.x.x), either RTOS or BLE, or both of them can be done.</td></tr>
<tr><td rowspan="2">AT+NWOTADWPROG</td><td><fw_type></td><td>FW download progress.<br><fw_type>: SLIB/RTOS.<br>Response: +NWOTADWPROG:100<br>(in SDK V3.x.x.x, only RTOS is available)</td></tr>
<tr><td colspan="2">Example<br>; SLIB download progress:<br><pre>AT+NWOTADWPROG=slib
+NWOTADWPROG:100
OK</pre>; RTOS download progress:<br><pre>AT+NWOTADWPROG=rtos
+NWOTADWPROG:100
OK</pre>; MCU FW download progress:<br><pre>AT+NWOTADWPROG=mcu_fw
+NWOTADWPROG:100
OK</pre>; BLE FW download progress:<br><pre>AT+NWOTADWPROG=ble_fw
+NWOTADWPROG:100
OK</pre>; Cert Key download progress:<br><pre>AT+NWOTADWPROG=cert_key
+NWOTADWPROG:100
OK</pre>Note<br>● Enabled by default in the SDK</td></tr>
<tr><td rowspan="2">AT+NWOTADWSTOP</td><td>(none)</td><td>Stop while downloading FW.</td></tr>
<tr><td colspan="2">Example<br><pre>AT+NWOTADWSTOP
OK</pre></td></tr>
<tr><td>AT+NWOTAFWNAME</td><td>(none)</td><td>Read a name in the header of the MCU firmware (Only for MCU FW)</td></tr>
</table>


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>AT+NWOTAFWNAME<br>+NWOTAFWNAME:MCU<br>OK<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __OTA_UPDATE_MCU_FW__ is enabled in the SDK, this command will be enabled | |
| AT+NWOTAFWSIZE | (none) | Read a size in the header of the MCU firmware (Only for MCU FW) |
| | *Example*<br>AT+NWOTAFWSIZE<br>+NWOTAFWSIZE:4128<br>OK<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __OTA_UPDATE_MCU_FW__ is enabled in the SDK, this command will be enabled | |
| AT+NWOTAFWCRC | (none) | Read a CRC in the header of the MCU firmware (Only for MCU FW) |
| | *Example*<br>AT+NWOTAFWCRC<br>+NWOTAFWCRC:5aa8b6c4<br>OK<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __OTA_UPDATE_MCU_FW__ is enabled in the SDK, this command will be enabled | |
| AT+NWOTAREADFW | <read addr>,<br><read_size> | Read the MCU firmware as much as the *read_size* from the *read_addr* and transmit it (Only for MCU FW)<br><*read_addr*>: Hexadecimal without “0x” prefix.<br><*read_size*>: Decimal. |
| | *Example*<br>AT+NWOTAREADFW=1f2000,128<br>DA16FMCUÿ2345678901234561234567890123456123456789012345612345678901234561234567890123456<br>+NWOTAREADFW:COMPLETE<br>OK<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __OTA_UPDATE_MCU_FW__ is enabled in the SDK, this command will be enabled | |
| AT+NWOTATRANSFW | (none) | Transmit an MCU firmware to MCU through UART1. Transmission will be failed if no header (16 bytes) information exist (Only for MCU FW) |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+NWOTATRANSFW`<br>`MCU`<br>`DA16FMCUy234567890123456123456789012345612345678901234561234567890123456123456789012345612345678901234561234567890123456123456789012345612345678901234 .....`<br>`+NWOTATRANSFW:COMPLETE`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __OTA_UPDATE_MCU_FW__ is enabled in the SDK, this command will be enabled | |
| AT+NWOTAERASEFW | (none) | Erase the MCU firmware stored in a serial flash of DA16200 (DA16600). (Only for MCU FW) |
| | *Example*<br>`AT+NWOTAERASEFW`<br>`+NWOTAERASEFW:COMPLETE`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __OTA_UPDATE_MCU_FW__ is enabled in the SDK, this command will be enabled | |
| AT+NWOTASETADDR | <sflash_addr> | Data to be downloaded can be designated as an address within the range of *User Area* and *TLS Certificate Key* in the SFLASH area. The default value is 0x003A_D000. |
| | *Example*<br>`AT+NWOTASETADDR=1f3000`<br>`+NWOTASETADDR:0x00`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+NWOTAGETADDR | <fw_type> | Returns the value set with NWOTASETADDR. |
| | *Example*<br>`AT+NWOTAGETADDR=mcu_fw`<br>`+NWOTAGETADDR:1f2000`<br>`OK`<br><br>`AT+NWOTAGETADDR=cert_key`<br>`+NWOTAGETADDR:1f2000`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+NWOTAREADFLASH | <sflash_addr>,<size> | Read as much as size from *sflash_addr*. |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+NWOTAREADFLASH=1f2000,128`<br>`MCU_FW ?¨ZDA16FMCUÿ23456789012345612345678901234561234567 8901234561234567890123456123456789012345612345678901234561234567890123 456`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+NWOTAERASEFLASH | <sflash_addr>,<size> | Delete as much as size from sflash_addr. |
| | *Example*<br>`AT+NWOTAERASEFLASH=1f2000,1000`<br>`+NWOTAERASEFLASH:COMPLETE`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● It will be erased in 4 kB increments | |
| AT+NWOTACOPYFLASH | <dest_sflash_addr>,<src_sflash_addr>,<size> | Copy as much as size from src_sflash_addr to dest_sflash_addr. |
| | *Example*<br>`AT+NWOTACOPYFLASH=1f2000,1f3000,1000`<br>`+NWOTACOPYFLASH:COMPLETE`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● It will be copied in 4 kB increments | |

**Table 22: OTA Response List**

| Response | Parameters | Description |
|---|---|---|
| +NWOTADWSTART | <status> | Returns the status of FW download.<br><status>: 0x00 is success. See Table 23 for other status values.<br>For example: +NWOTADWSTART:0x00 |
| +NWOTARENEW | <status> | Returns the status for FW RENEW.<br><status>: 0x00 is success. See OTA Response Code List for others.<br>For example: +NWOTARENEW:0x00 |
| +NWOTADWPROG | <progress> | Returns the percentage value (%) of the FW download progress.<br><progress>: Print download progress in percent.<br>For example: +NWOTADWPROG:100 |
| +NWOTADWSTOP | <status> | Returns the status of FW download stop.<br><status>: 0x00 is success. See Table 23 for other status values.<br>For example: +NWOTADWSTOP:0x00 |


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| Response | Parameters | Description |
|---|---|---|
| +NWOTATRANSFW | COMPLETE or FAIL | Returns result of MCU FW transmission. (Only for MCU FW)<br>For example: +NWOTATRANSFW:COMPLETE |
| +NWOTAFWNAME | <name> | String entered by a user. (Default is MCU_FW)<br>Returns “(NULL)” if there is no MCU FW.<br>(Only for MCU FW) |
| +NWOTAFWSIZE | <size> | Downloaded MCU FW size.<br>It returns 0 if there is no MCU FW.<br>(Only for MCU FW) |
| +NWOTAFWCRC | <crc> | Downloaded MCU FW CRC.<br>It returns 0 if there is no MCU FW.<br>(Only for MCU FW) |
| +NWOTAREADFW | COMPLETE or FAIL | Success: COMPLETE<br>Failure: FAIL<br>(Only for MCU FW) |
| +NWOTAERASEFW | COMPLETE or FAIL | Success: COMPLETE<br>Failure: FAIL<br>(Only for MCU FW) |
| +NWOTASETADDR | <status> | <status>: 0x00 is success. See Table 23 for other status values. |
| +NWOTAGETADDR | <sflash_addr> | Returns the value of sflash_addr. |
| (AT+NWOTAREADFLASH) | (Binary) | Returns binary data as much as entered SFLASH address and size. |
| +NWOTAERASEFLASH | COMPLETE or FAIL | Success: COMPLETE<br>Failure: FAIL |
| +NWOTACOPYFLASH | COMPLETE or FAIL | Success: COMPLETE<br>Failure: FAIL |

**Table 23: OTA Response Code List**

| Return Value | Description |
|---|---|
| 0x00 | Return success. |
| 0x01 | Return failed. |
| 0x02 | SFLASH address is wrong. |
| 0x03 | FW type is unknown. |
| 0x04 | Server URL is unknown. |
| 0x05 | FW size is too big. |
| 0x06 | CRC is not correct. |
| 0x07 | FW version is unknown. |
| 0x08 | FW version is incompatible. |
| 0x09 | FW not found on the server. |
| 0x0A | Failed to connect to the server. |
| 0x0B | All new FWs have not been downloaded.<br>Be sure to download both SLIB and RTOS in the case of SDK V2.x.x.x. |


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| | |
|---|---|
| 0x0C | Failed to allocate memory. |

### 9.5.1 OTA Download Example

SLIB download: (SDK V2.x.x.x only)

```
AT+NWOTADWSTART=slib,https://server/DA16200_SLIB-GEN01-01-1111-000000.img
```

RTOS download:

```
AT+NWOTADWSTART=rtos,https://server/DA16200_RTOS-GEN01-01-1111-000000.img
```

MCU FW download:

```
AT+NWOTADWSTART=other_fw,https://server/mcu_firmware.img
AT+NWOTADWSTART=other_fw,https://server/mcu_firmware.img,ver01
```

Cert Key download:

```
AT+NWOTADWSTART=cert_key,https://server/ca.pem
```

### 9.5.2 OTA Download Progress Example

SLIB download progress: (SDK V2.x.x.x only)

```
AT+NWOTADWPROG=slib
```

RTOS download progress:

```
AT+NWOTADWPROG=rtos
```

MCU FW download progress:

```
AT+NWOTADWPROG=mcu_fw
```

Cert Key download progress:

```
AT+NWOTADWPROG=cert_key
```

### 9.5.3 OTA Renew Example

Renew Firmware (reboot with updated FW):

```
AT+NWOTARENEW
```

### 9.5.4 MCU FW Transport Example

MCU FW transmission:

```
AT+NWOTATRANSFW
```

Get MCU FW name:

```
AT+NWOTAFWNAME
```

Get MCU FW size:

```
AT+NWOTAFWSIZE
```

Get MCU FW CRC:

```
AT+NWOTAFWCRC
```

Read MCU FW as much as specified size:

```
AT+NWOTAREADFW=1f2000,128
```

Delete MCU FW stored in the DA16200 (DA16600) SFLASH:

```
AT+NWOTAERASEFW
```


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### 9.5.5 SFLASH User Area Address Setting Example

SET ADDR:

```
AT+NWOTASETADDR=0x1f2000
```

GET ADDR:

```
AT+NWOTAGETADDR=mcu_fw
AT+NWOTAGETADDR=cert_key
```

### 9.5.6 SFLASH READ/COPY/ERASE Example

SFLASH Read:

```
AT+NWOTAREADFLASH=0x1f2000,128
```

SFLASH Copy:

```
AT+NWOTACOPYFLASH=0x1f2000,0x1f3000,128
```

SFLASH Erase:

```
AT+NWOTAERASEFLASH=0x1f2000,128
```

## 9.6 Zeroconf Commands

**Table 24: Zeroconf Command List**

<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td rowspan="4">AT+NWMDNSSTART</td><td><Mode></td><td>Start the mDNS module. The mDNS module is communicated through multicast. DA16200 (DA16600) could frequently be changed from/to DPM sleep and wake-up states. It may make power consumption more spend.<br><Mode>: The mode in which the WLAN interface is running, 0 (Station), 1 (Soft-AP).</td></tr>
<tr><td colspan="2">Prerequisite<br>DA16200 (DA16600) in STA mode should connect to an AP.<br>The host name of mDNS module should be set up.<br>Example<br>AT+NWMDNSSTART=1<br>OK<br>Note<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_ZERO_CONFIG__ is enabled in the SDK, this command will be enabled</td></tr>
<tr><td>?</td><td>Get the string representing the status of mDNS module, “started” or “not started”.</td></tr>
<tr><td colspan="2">Example<br>AT+NWMDNSSTART=?<br>+NWMDNSSTART:started</td></tr>
<tr><td>AT+NWMDNSHNREG</td><td><Host name></td><td>Register the host name in the mDNS module. mDNS supports one configured host name only, to change or set a new mDNS host name. mDNS service must be stopped and started again.<br><Host name>: The name of the host to be registered.</td></tr>
</table>


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Example<br>AT+NWMDNSSHNREG=da16x<br>OK<br>Note<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_ZERO_CONFIG__ is enabled in the SDK, this command will be enabled</td></tr>
<tr><td rowspan="2">AT+NWMDNSSRVREG</td><td><Instance name>,<Protocol>,<br><Port>,<Text record></td><td><Instance name>: The instance name of service to be registered.<br><Protocol>: The protocol and the type of the service to be registered.<br><Port>: The port number of the service to be registered.<br><Text record>: The text record of the service that must be registered and mentioned in “Key=Value” format. Multiple pairs of text records should be separated using a “,”.</td></tr>
<tr><td colspan="2">Prerequisite<br>DA16200 (DA16600) in STA mode should connect to an AP.<br>The mDNS module should be running.<br>Example<br>AT+NWMDNSSRVREG=_WEBAPP,_http,_tcp,80,LIGHT=OFF,FAN=ON<br>OK<br>Note<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_ZERO_CONFIG__ is enabled in the SDK, this command will be enabled</td></tr>
<tr><td rowspan="2">AT+NWMDNSSUPDATETXT</td><td><Text record></td><td>Update the text record of a service in the mDNS module.<br><Text record>: The text record of the service to be updated.</td></tr>
<tr><td colspan="2">Example<br>AT+NWMDNSUPDATETXT=LIGHT=OFF,FAN=ON<br>OK<br>Note<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_ZERO_CONFIG__ is enabled in the SDK, this command will be enabled</td></tr>
<tr><td>AT+NWMDNSSRVDEREG</td><td>(None)</td><td>Unregister a service in the mDNS module.<br>Response: OK or ERROR<br>For example: AT+NWMDNSSRVDEREG</td></tr>
</table>


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| Command | Parameters | Description |
|---|---|---|
| | *Prerequisite*<br>DA16200 (DA16600) in STA mode should connect to an AP.<br>The mDNS module should be running.<br>The service in the mDNS module should be registered.<br>*Example*<br>`AT+NWMDNSSRVDEREG`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_ZERO_CONFIG__ is enabled in the SDK, this command will be enabled | |
| AT+NWMDNSSTOP | (None) | Stop the mDNS module. |
| | *Prerequisite*<br>DA16200 (DA16600) in STA mode should connect to an AP.<br>The mDNS module should be running.<br>*Example*<br>`AT+NWMDNSSTOP`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_ZERO_CONFIG__ is enabled in the SDK, this command will be enabled | |
| | ? | Get the string representing the status of the mDNS module, “stopped” or “running”. |
| | *Example*<br>`AT+NWMDNSSTOP`<br>`+NWMDNSSTOP:stopped` | |

### 9.6.1 Zeroconf Example

Configure the parameters and start the mDNS module (After Wi-Fi Connection):

```
AT+NWMDNSHNREG=da16x
AT+NWMDNSSTART=0
```

If the mDNS module is started successfully, the following response is shown:

```
OK
```

Register a service in the mDNS module:

```
AT+NWMDNSSRVREG=_WEBAPP,_http._tcp,80,LIGHT=OFF,FAN=ON
```

If the service is registered successfully, the following response is shown:

```
OK
```

Registered service and host name can be discovered by other mDNS services. In this example, Bonjour service (https://support.apple.com/kb/DL999?viewlocale=en_US&locale=zh_TW) on Windows is used to discover them.

To discover DA16200 (DA16600)’s mDNS, the “-G” option can be used like the following:


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```
C:> dns-sd -G v4 da16x.local

Timestamp     A/R Flags if Hostname                Address
TTL

18:04:04.474  Add     2 24 da16x.local.            192.168.0.4
120
```

To discover the service, the “-L” option can be used like the following:

```
C:>dns-sd -L _WEBAPP _http._tcp local

Lookup _WEBAPP._http._tcp.local

18:04:29.453  _WEBAPP._http._tcp.local. can be reached at da16x.local.:80
(interface 24)

 LIGHT=OFF FAN=ON
```

## 9.7 Provision Commands

The provision commands are used for starting to provision procedure and getting provisioning status. To use these commands, __PROVISION_ATCMD__ feature should be enabled in SDK.

**Table 25: Provision Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+PROVSTART | (none) | Removed all profile data in NVRAM and configure a new station or Soft-AP profile is set.<br>Response: ERROR. |
| | *Example*<br>`AT+PROVSTART`<br>`+INIT:DONE,1`<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __PROVISION_ATCMD__ is enabled in the SDK, this command will be enabled<br>● In case of DA16200 (DA16600), __SUPPORT_FACTORY_RST_APMODE__ should be enabled. For DA16600, __SUPPORT_FACTORY_RST_STAMODE__ should be used.<br>● After restarting, the system will be ready for the provision procedure | |
| AT+PROVSTAT | (none) | Get status of provisioning.<br>Response: OK or ERROR. |
| | *Example*<br>`AT+PROVSTAT`<br>`+ATPROV=STATUS 1`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __PROVISION_ATCMD__ is enabled in the SDK, this command will be enabled<br>● The list of provision status can be found in the thread_atcmd.h. Check the `atcmd_provision_stat` enumeration. | |


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**Table 26: atcmd_provision_stat Enumeration**

| Value | Name | Description |
|---|---|---|
| 0 | ATCMD_PROVISION_IDLE | Not run or finish |
| 1 | ATCMD_PROVISION_START | Start |
| 101 | ATCMD_PROVISION_SELECTED_AP_SUCCESS | Receive AP information |
| 102 | ATCMD_PROVISION_SELECTED_AP_FAIL | |
| 103 | ATCMD_PROVISION_WORNG_PW | AP connection fail by the wrong PW |
| 104 | ATCMD_PROVISION_NETWORK_INFO | Get Network info. from Mobile App. |
| 105 | ATCMD_PROVISION_AP_FAIL | AP connection fail |
| 106 | ATCMD_PROVISION_DNS_FAIL_SERVER_FAIL | Network connection check |
| 107 | ATCMD_PROVISION_DNS_FAIL_SERVER_OK | |
| 108 | ATCMD_PROVISION_NO_URL_PING_FAIL | |
| 109 | ATCMD_PROVISION_NO_URL_PING_OK | |
| 110 | ATCMD_PROVISION_DNS_OK_PING_FAIL_N_SERVER_OK | |
| 113 | ATCMD_PROVISION_DNS_OK_PING_N_SERVER_FAIL | |
| 111 | ATCMD_PROVISION_DNS_OK_PING_OK | |
| 112 | ATCMD_PROVISION_REBOOT_ACK | Reboot after provisioning |

## 9.8 Bluetooth® LE Commands

The Bluetooth® LE commands are available when the DA16600 is running the Bluetooth® LE enabled version of the firmware. Select and build the DA16600 project in SDK to use these commands.

**Table 27: Bluetooth® LE Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+BLENAME | <device name> | Change the device name of the DA16600 / Bluetooth® LE device.<br>Response: OK or ERROR. |
| | *Example*<br>`AT+BLENAME=DA16600-BLE`<br>`OK`<br>*Note*<br>• Not supported in the ThreadX SDK<br>• Enabled by default in the SDK v3.2.3.0 or later | |
| AT+ADVSTOP | (none) | Stop advertising of DA16600 / Bluetooth® LE device.<br>Response: OK or ERROR. |
| | *Example*<br>`AT+ADVSTOP`<br>`OK`<br>*Note*<br>• Not supported in the ThreadX SDK<br>• Enabled by default in the SDK v3.2.3.0 or later | |


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| Command | Parameters | Description |
|---|---|---|
| AT+ADVSTART | (none) | Start advertising of DA16600 / Bluetooth® LE device.<br>Response: OK or ERROR. |
| | *Example*<br>`AT+ADVSTART`<br>`OK`<br>*Note*<br>● Not supported in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later | |


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# 10 Transfer Function Commands

## 10.1 Socket Commands

**Table 28: Socket Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+TRTS | <local_port> | Open a TCP server socket.<br><local_port>: Local port number of the socket.<br>Response: OK (with '+TRCTS:*** ', see Table 25) or ERROR |
| | *Prerequisite*<br>Target system should be connected to an AP.<br>*Example*<br>`AT+TRTS=10194`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+TRTC | <server_ip>,<br><server_port>,<br><local_port> | Open a TCP client socket and connect to a TCP server.<br><server_ip>: IP address of TCP server to be accessed.<br><server_port>: Port number of TCP server.<br><local_port>: Local port number of the socket (optional, 0: auto).<br>Response: OK or ERROR |
| | *Prerequisite*<br>Target system should be connected to an AP and a TCP server should be<br>running (for example, 192.168,0.18 with 1025 port)<br>*Example*<br>`AT+TRTC=192.168.0.18,1025,1024`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+TRUSE | <local_port> | Open a UDP socket.<br><local_port>: Local port number of the socket.<br>Response: OK or ERROR |
| | *Example*<br>`AT+TRTALL (optional, run this first if 'ERROR' is responded)`<br>`AT+TRUSE=10195`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+TRUR | <remote_ip>,<br><remote_port> | Set remote IP and port of the UDP socket.<br><remote_ip>: Remote IP address.<br><remote_port>: Remote port number.<br>Response: OK or ERROR |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+TRUR=192.168.0.18,1027`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+TRPRT | <cid> | Get session information by CID.<br><cid>: 0 (TCP server), 1 (TCP client), 2 (UDP)<br>Response: <cid>,[TCP\|UDP],<remote_ip>,<remote_port>,<local_port> |
| | *Prerequisite*<br>A UDP socket must be opened first in the target system (AT+TRUSE=10195)<br>*Example*<br>`AT+TRPRT=2`<br>`+TRPRT:2,UDP,192.168.0.18,10194,10195`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+TRPALL | (none) | Get all session information.<br>Response: <cid>,[TCP\|UDP],<remote_ip>,<remote_port>,<local_port><LF>… |
| | *Prerequisite*<br>The target system should be connected to any UDP or TCP server to get a proper response for this command. The example below is the result after the UDP socket is opened in the target system.<br>*Example*<br>`AT+TRPALL`<br>`+TRPALL:2,UDP,192.168.0.18,10194,10195`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+TRTRM | <cid><br>[,<remote_ip><br>,<remote_port>] | Close (terminate) a session by CID. If CID is 0(TCP server) and remote_ip and remote_port are input, the session with the specific remote will be closed.<br><cid>: 0 (TCP server), 1 (TCP client), 2 (UDP)<br><remote_ip>: Remote IP address connected to TCP server. It's only allowed when CID is 0(TCP server)<br><remote_port>: Remote port number connected to TCP server. It's only allowed when CID is 0(TCP server)<br>Response: OK or ERROR |


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| Command | Parameters | Description |
|---|---|---|
| | *Prerequisite*<br>The target system should have any connections to UDP or TCP server.<br>*Example*<br>`AT+TRTRM=2`<br>`OK`<br><br>`AT+TRTRM=0,192.168.0.18,10194`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● The remote_ip and remote_port parameters are only supported in SDK v3.2.3.0 or later. | |
| AT+TRTALL | (none) | Close (terminate) all sessions.<br>Response: OK or ERROR |
| | *Example*<br>`AT+TRTALL`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+TRSAVE | (none) | Save status of all sessions to NVRAM.<br>Response: OK or ERROR |
| | *Example*<br>`AT+TRSAVE`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK | |
| AT+TCPDATAMODE | <mode> | Set the mode of the received TCP data.<br><mode><br>0: text mode (default)<br>1: hex string mode<br>In text mode, data is returned as an ascii string: “0123ABCD”.<br>In hex mode, the data is returned as a hex encoded text string “3031323341424344”.<br>Response: OK or ERROR |
| | *Example*<br>`AT+TCPDATAMODE=1`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK v3.2.2.1 or later | |

**Table 29: Socket Connection Response List**

| Response | Parameters | Description |
|---|---|---|
| +TRCTS | <cid>,<br><remote_ip>,<br><remote_port> | A remote TCP client is connected to the TCP server that was opened by AT+TRTS.<br><cid>: 0 (TCP Server).<br><remote_ip>: TCP client IP address.<br><remote_port>: TCP client port number. |


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| Response | Parameters | Description |
|---|---|---|
| | *Prerequisite*<br>When you send the AT command (AT+TRTS=40000), you can receive this result if there is no error.<br>*Example*<br>`+TRCTS:0,192.168.0.18,41014` | |
| +TRXTS | <cid>,<br><remote_ip>,<br><remote_port> | A remote TCP client is disconnected from the TCP server that was opened by AT+TRTS.<br><cid>: 0 (TCP Server).<br><remote_ip>: TCP client IP address.<br><remote_port>: TCP client port number. |
| | *Example*<br>; When a remote peer is disconnected<br>`+TRXTS:0,192.168.0.18,41014` | |
| +TRXTC | <cid>,<br><remote_ip>,<br><remote_port> | The TCP client socket that was opened by AT+TRTC is disconnected.<br><cid>: 1 (TCP Client).<br><remote_ip>: TCP server IP address.<br><remote_port>: TCP server port number. |
| | *Example*<br>; When the TCP client socket is disconnected<br>`+TRXTC:1,192.168.0.18,1025` | |


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### 10.1.1 Data Transfer Commands

**Table 30: Data Transmission Command**

| Escape Sequence | Parameters | Description |
|---|---|---|
| <ESC>S | <cid><length>,<br><remote_ip >,<br><remote_port >,[<mode>,]<data> | Transmit data through a socket with the CID specified.<br><ESC>S: To enter data input mode, type in <ESC> (0x1B) and S keys together.<br><cid>: 0 (TCP server), 1 (TCP client), 2 (UDP)<br><length>: Data length. Data length can be 0 in only text mode. If this is 0, data is read until "\r" or "\n" is met. In raw mode, data is read until the length.<br><remote_ip>: Remote IP address.<br><remote_port>: Remote port number.<br>● For TCP Server, <remote_ip> and <remote_port> of a TCP Client should be given. Maximum four TCP Clients can be connected to the TCP Server<br>● For TCP Client, 0, 0 is given (as the destination is the server)<br>● For UDP: if you give 0,0, the data is sent to the destination that AT+TRUR has specified. if non-0 <remote_ip> and <remote_port> are given, UDP temporarily sends to the destination <remote_ip> and <remote_port> specifies<br><mode>: Mode to transmit data in raw or text mode. It's optional. If there is no option, data will be transmitted in text mode. This option is allowed only for UART communication.<br>● r: The raw mode is active. In raw mode, Data is read until data length. The data length is specified in <length> parameter<br>● t: The text mode is active. In text mode, the data can be affected if it has unprintable control codes like backspace(0x08)<br>Response: OK or ERROR |


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| Escape Sequence | Parameters | Description |
|---|---|---|
| | *Prerequisite*<br>The connections should be set up first to run this command, then press the "ESC" key and type the rest of the string after <ESC>.<br>*Example1* – To send data to TCP client<br>`<ESC>S010,192.168.0.18,43110,abcde12345`<br>`OK`<br>*Example2* – To send data to TCP server<br>`<ESC>S110,192.168.0.18,1025,abcde12345`<br>`OK`<br>*Example3* – To send data to TCP server with '0, 0' as the destination/server<br>`<ESC>S110,0,0,abcde12345`<br>`OK`<br>*Example4* – To send data to UDP receiver<br>`<ESC>S110,192.168.0.18,1024,abcde12345`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK<br>● The maximum length of data depends on TX_PAYLOAD_MAX_SIZE definition. It is defined in atcmd.h for SDK v3.x, and thread_atcmd.h for SDK v2.x. TX_PAYLOAD_MAX_SIZE includes all parameters of AT-Command. Therefore, the maximum length of 'length' parameter depends on length of other parameters.<br>● TX_PAYLOAD_MAX_SIZE is defined 4,096 bytes in the SDK v3.2.3.0 or later, and 2,048 bytes in the SDK v2.4.x.x.<br>● In the case of ATCMD over SPI or SDIO:<br>The result for this command is sent to the host as the "response" field of "struct _st_host_response". The "response" field is a 1 byte decimal value. A value of 0x20 is a result of "OK". All other values are an "ERROR". | |

**Table 31: Data Reception Responses**

| Response | Parameters | Description |
|---|---|---|
| +TRDTS | <cid>,<br><src_ip>,<src_port>,<br><length>,<data> | Receive data through TCP server socket.<br><cid>: 0 (TCP Server).<br><src_ip>: Source IP address.<br><src_port>: Source port number.<br><length>: Data length.<br><data>: Received data. |
| | *Prerequisite*<br>A remote TCP client should be connected to the TCP server of your target system.<br>*Example*<br>; When data is sent from a TCP client<br>`+TRDTC:1,192.168.0.18,1025,4,test` | |
| +TRDTC | <cid>,<br><src_ip>,<src_port>,<br><length>,<data> | Receive data through TCP client socket.<br><cid>: 1 (TCP Client).<br><src_ip>: Source IP address.<br><src_port>: Source port number.<br><length>: Data length.<br><data>: Received data |


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| Response | Parameters | Description |
|---|---|---|
| | *Prerequisite*<br>A TCP server should be set up in another system and the TCP Client of your target system should be connected (AT+TRTC=192.168.0.18,1025,1024), then the TCP server sends the data (test).<br>*Example*<br>; When TCP client receives data,<br>`+TRDTC:1,192.168.0.18,1025,4,test` | |
| +TRDUS | <cid>,<br><src_ip>,<src_port>,<br><length>,<data> | Receive data through UDP socket.<br><cid>: 2 (UDP Session).<br><src_ip>: Source IP address.<br><src_port>: Source port number.<br><length>: Data length.<br><data>: Received data. |
| | *Prerequisite*<br>A UDP socket (AT+TRUSE=10194) must be opened first in the target system. Then UDP socket in another system should send the datagram (test).<br>*Example*<br>; When UDP session receives data,<br>`+TRDUS:2,192.168.0.18,10194,4,test` | |

### 10.1.2 Data Transfer with DPM

#### 10.1.2.1 TCP Server

After a connection to an AP is made in the normal BOOT state, open a TCP server socket and save the config to NVRAM.

```
AT+TRTS=32000
AT+TRSAVE
```

The TCP server socket that has been opened should be closed before switching to DPM mode.

```
AT+TRTRM=0
```

Change the DA16200 (DA16600) state to DPM mode (AT+DPM=1). When the DA16200 (DA16600) starts the session on DPM mode successfully, the following is shown:

```
+INIT:DONE,0
+WFJAP:1,'WI-FI_AP',192.168.5.19
+TRPALL:0,TCP,0.0.0.0,0,32000
```

When a TCP client connects to DA16200 (DA16600), the following is shown:

```
+INIT:WAKEUP,UC
```

To receive +TRCTS message, send AT+MCUWUDONE immediately after “+INIT:WAKEUP,UC”

```
+TRCTS:0,192.168.0.1,42000
```

When the DA16200 (DA16600) receives a message from a client, the following is shown:

```
+INIT:WAKEUP,UC
+TRDTS:0,192.168.0.1,42000,10,1234567890
```

To send a TCP message, send AT+MCUWUDONE immediately after “external wake-up” is triggered (+INIT:WAKEUP,EXT). To prevent that DA16200 (DA16600) enters DPM Sleep mode, MCU should


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send AT+CLRDPMSLPEXT before a message is sent. The DA16200 (DA16600) can send data to a TCP client with the command "<ESC>S". Finally, to enter DPM sleep mode, send "AT+SETDPMSLTEXT".

```
+INIT:WAKEUP,EXT  // external wake-up
AT+MCUWUDONE
AT+CLRDPMSLPEXT
…
<ESC>S…
…
AT+SETDPMSLPEXT
```

When a TCP client disconnects from DA16200 (DA16600), the following is shown:

```
+INIT:WAKEUP,UC
```

To receive +RTXTS message, send AT+MCUWUDONE immediately after "+INIT:WAKEUP,UC"

```
+TRXTS:0,192.168.0.1,42000
```

#### 10.1.2.2 TCP Client

After a connection is made to an AP in the normal BOOT state, connect the TCP client of the DA16200 (DA16600) to a TCP server and save the config to NVRAM. (To save TCP client config information, the DA16200 (DA16600) should connect to the server successfully beforehand.)

```
AT+TRTC=192.168.5.1,34000
AT+TRSAVE
```

Before switching to DPM mode, disconnect the TCP Client:

```
AT+TRTRM=1
```

Change the DA16200 (DA16600) state to DPM mode (AT+DPM=1). When the DA16200 (DA16600) starts the session on DPM mode successfully, the following is shown:

```
+INIT:DONE,0
+WFJAP:1,'WI-FI_AP',192.168.5.19
+TRPALL:1,TCP,192.168.5.1,34000,30000
```

The procedure to exchange TCP data is the same as in Section 10.1.2.1. When the DA16200 (DA16600) receives a message from the server, the following is shown:

```
+INIT:WAKEUP,UC
+TRDTC:1,192.168.5.1,34000,10,1234567890
```

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#### 10.1.2.3 UDP Session

After a connection is made to an AP in the normal BOOT state, open a UDP socket and save the config to NVRAM:

```
AT+TRUSE=48000
AT+TRSAVE
```

Before switching to DPM mode, disconnect TCP Client:

```
AT+TRTRM=2
```

Change the DA16200 (DA16600) state to DPM mode. When the DA16200 (DA16600) starts the session in DPM mode successfully, the following is shown:

```
+INIT:DONE,0
+WFJAP:1,'WI-FI_AP',192.168.5.19
+TRPALL:2,UDP,0.0.0.0,0,48000
```

The procedure to exchange UDP data is the same as in Section 10.3.1. When the DA16200 (DA16600) receives a message from the server, the following is shown:

```
+INIT:WAKEUP,UC
+TRDUS:2,192.168.5.23,35000,10,1234567890
```

## 10.2 Secure Socket Commands

**Table 32: Secure Socket Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+TRSSLINIT | <Role> | Initialize the SSL module. DA16200 (DA16600) allows to create two SSL modules.<br><Role>: The role of SSL, 1 – Client. |
| | *Example*<br>`AT+TRSSLINIT=1`<br>`+TRSSLINIT:0`<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_ATCMD_TLS__ is enabled in the SDK, this command will be enabled | |


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td rowspan="2">AT+TRSSLCFG</td><td><CID>,<Configuration ID>,<Configuration value></td><td>Configure SSL connection.<br><CID>: The CID obtained after issuing the AT+TRSSLINIT command.<br><Configuration ID>: The configuration ID available in the below list of configurations:<br>0 - Invalid configuration parameter<br>2 - To set SSL CA Certificate<br>3 - To set SSL Certificate<br>6 - To set the SNI<br>9 - To enable/disable server validation<br>10 - To set the Incoming buffer Length<br>11 - To set the Outgoing buffer Length<br><Configuration value>: Value to the configuration provided in configuration ID.<br>CONF_ID:CONF_VAL<br>0 - Invalid<br>2 - SSL CA Certificate Name<br>3 - SSL Certificate Name<br>6 - To Set the SNI (supported only for TLS client)<br>9 - To enable/disable server validation<br>0: Disables server validation (Default)<br>1: Enables server validation<br>10 - To set the Incoming buffer length<br>11 - To set the outgoing buffer length</td></tr>
<tr><td colspan="2">Prerequisite<br>The CID obtained after issuing the AT+TRSSLINIT command.<br>SSL CA certificate and the certificate should be set up to configure them.<br>Example<br><pre>0,2,CA_CERT
OK
0,3,CERT
OK
0,6,da16x
OK
0,9,0
OK
0,10,6144
OK
0,11,6144
OK</pre>Note<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_ATCMD_TLS__ is enabled in the SDK, this command will be enabled</td></tr>
</table>


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td rowspan="2">AT+TRSSLCO</td><td><CID>,<Server IP Address>,<Server Port number></td><td>Connect to an SSL server.<br><CID>: The CID obtained after issuing the AT+TRSSLINIT command.<br><Server IP Address>: The IP Address of the server to connect. Only supported IPv4 address.<br><Server Port>: The port number of the SSL server to connect.</td></tr>
<tr><td colspan="2">Prerequisite<br>The CID obtained after issuing the AT+TRSSLINIT command.<br>Example<br><pre>AT+TRSSLCO=0,192.168.0.11,30000
OK</pre>Note<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_ATCMD_TLS__ is enabled in the SDK, this command will be enabled</td></tr>
<tr><td>AT+TRSSLWR</td><td><CID>,<Server IP Address>,<Server Port number>,[<mode>,]<Data length>,<Data></td><td>Send the data to the SSL server is already established.<br><CID>: The CID obtained after issuing the AT+TRSSLINIT command.<br><Server IP Address>: The IP Address of the SSL server is already established. If no input, the IP address would internally be used to the SSL server IP address.<br><Server Port number>: The port number of the SSL server is already established. If no input, the Port number is internally used to the SSL server port number.<br><mode>: Mode to transmit data in raw or text mode. It's optional. If there is no option, data will be transmitted in text mode.<br>● r: The raw mode is active. In raw mode, Data is read until data length. The data length is specified in <length> parameter<br>● t: The text mode is active. In text mode, the data can be affected if it has unprintable control codes like backspace(0x08)<br><Data length>: The length of data to send.<br><Data>: The data to send. The input can be closed by <Ctrl>+C or reaching data length.</td></tr>
</table>

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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Prerequisite<br>The CID obtained after issuing the AT+TRSSLINIT command.<br>Example<br><pre>AT+TRSSLWR=0,10,0123456789<Ctrl> + C
OK</pre>Note<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_ATCMD_TLS__ is enabled in the SDK, this command will be enabled<br>● The maximum length of data depends on TX_PAYLOAD_MAX_SIZE definition. It is defined in atcmd.h for SDK v3.x, and thread_atcmd.h for SDK v2.x. TX_PAYLOAD_MAX_SIZE includes all parameters of AT-Command. Therefore, the maximum length of 'length' parameter depends on length of other parameters.</td></tr>
<tr><td rowspan="2">AT+TRSSLCL</td><td><CID></td><td>Close the SSL connection.<br><CID>: The CID obtained after issuing AT+TRSSLINIT command.</td></tr>
<tr><td colspan="2">Prerequisite<br>The CID obtained after issuing the AT+TRSSLINIT command.<br>Example<br><pre>AT+TRSSLCL=0
OK</pre>Note<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_ATCMD_TLS__ is enabled in the SDK, this command will be enabled</td></tr>
<tr><td rowspan="2">AT+TRSSLCERTLIST</td><td><Certificate Type></td><td>Show a list of certificates or a list of CA data available in sflash memory.<br><Certificate Type>: The type of the certificate. 0 – CA Certificates, 1 – Client/Server Certificates.</td></tr>
<tr><td colspan="2">Example<br><pre>AT+TRSSLCERTLIST=0
+TRSSLCERTLIST=0,CA_CERT</pre>Note<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_ATCMD_TLS__ is enabled in the SDK, this command will be enabled</td></tr>
</table>


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td rowspan="2">AT+TRSSLCERTSTORE</td><td><Certificate Type>, <Sequence>, <Format>, <Name>, <Data></td><td>Store a certificate and CA list data in sflash memory.<br><Certificate Type>: The type of the certificate. 0 – CA Certificates, 1 – Client/Server Certificates.<br><Sequence>: If the certificate type is 0 (CA), the number of certificates in the sequence is 1-5. If the certificate type is 1 (Client/Server certificate), then a number of certificates in a sequence is 1-SSL cert, 2-SSL key.<br><Format>: The format of the CA/Certificate/Key where: 0: DER, 1: PEM.<br><Name>: The name of the certificate. While loading certificate and key file separately, the name should be same in both commands.<br><Data>: The certificate data to be store.</td></tr>
<tr><td colspan="2">Example<br>AT+TRSSLCERTSTORE=0,0,0,CA_CERT, -----BEGIN CERTIFICATE-----<br>…<br>-----END CERTIFICATE-----<Ctrl>+C<br>OK<br>Note<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_ATCMD_TLS__ is enabled in the SDK, this command will be enabled</td></tr>
<tr><td rowspan="2">AT+TRSSLCERTDELETE</td><td><Certificate Type>,<Name></td><td>Delete a certificate or CA list data in sflash memory.<br><Certificate Type>: The type of the certificate. 0 – CA Certificates, 1 – Client/Server Certificates.<br><Name>: The name of the certificate.</td></tr>
<tr><td colspan="2">Example<br>AT+TRSSLCERTDELETE=0,CA_CERT<br>OK<br>Note<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_ATCMD_TLS__ is enabled in the SDK, this command will be enabled</td></tr>
<tr><td rowspan="2">AT+TRSSLSAVE</td><td>(none)</td><td>Store the current SSL module’s configuration in NVRAM.</td></tr>
<tr><td colspan="2">Example<br>AT+TRSSLSAVE<br>OK<br>Note<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_ATCMD_TLS__ is enabled in the SDK, this command will be enabled</td></tr>
<tr><td>AT+TRSSLDELETE</td><td>(none)</td><td>Delete the stored SSL module’s configuration in NVRAM.</td></tr>
</table>

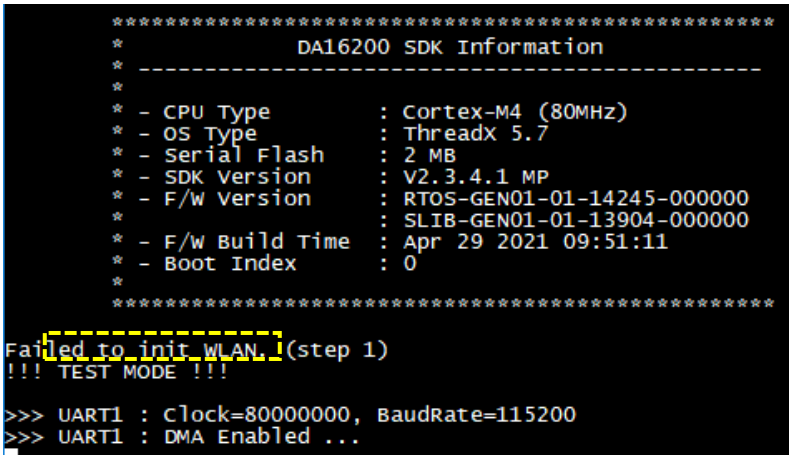

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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+TRSSLDELETE`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_ATCMD_TLS__ is enabled in the SDK, this command will be enabled | |


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# 11 RF Test Function Commands

**Table 33: RF Test Command List**

<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td rowspan="2">AT+TMRFNOINIT</td><td><flag></td><td>Set boot mode.<br><flag>: 0 (normal boot), 1 (RF test mode boot)<br>Response: OK or ERROR</td></tr>
<tr><td colspan="2">Example<br>AT+TMRFNOINIT=1<br>OK<br><br>AT+RESTART<br>OK<br>Note<br>● Disabled by default in the ThreadX SDK. To use, manufacture image is needed or contact Renesas Electronics<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● To test RF performance, set the boot mode as RF test mode (AT+TMRFNOINIT=1) and restart the DA16200 (DA16600) (AT+RESTART)<br>● After DA16200 (DA16600) is restarted, “!!! TEST MODE !!!” log is displayed<br>  </td></tr>
<tr><td rowspan="2">AT+TMLMACINIT</td><td>(none)</td><td>Initialize LMAC (for test mode).<br>Response: OK or ERROR</td></tr>
<tr><td colspan="2">Prerequisite<br>Boot as RF test mode (AT+TMRFNOINIT=1)<br>Example<br>AT+TMLMACINIT<br>OK<br>Note<br>● Disabled by default in the ThreadX SDK. To use, manufacture image is needed or contact Renesas Electronics<br>● Enabled by default in the SDK v3.2.3.0 or later</td></tr>
<tr><td>AT+RFTESTSTART</td><td>(none)</td><td>Start RF test mode.</td></tr>
</table>

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| Command | Parameters | Description |
|---|---|---|
| | *Prerequisite*<br>Boot as RF test mode (AT+TMRFNOINIT=1)<br>Restart the DA16200 (DA16600) (AT+RESTART)<br>*Example*<br>`AT+RFTESTSTART`<br>`OK`<br>*Note*<br>• Disabled by default in the ThreadX SDK. To use, manufacture image is needed or contact Renesas Electronics<br>• Enabled by default in the SDK v3.2.3.0 or later | |
| AT+RFTX | <Ch>,<br><BW>,<br><numFrames>,<br><frameLen>,<br><txRate>,<br><txPower>,<br><destAddr>,<br><bssid>,<br><htEnable>,<br><GI>,<br><greenField>,<br><preambleType>,<br><qosEnable>,<br><ackPolicy>,<br><scrambler>,<br><aifsnVal>,<br><ant> | Start RF TX test.<br><Ch>: Carrier frequency (2412 ~ 2484 MHz).<br><BW>: [0]: Fixed. Carrier bandwidth. 20 MHz fixed.<br><numFrames>: Number of frames to transmit.<br><frameLen>: Length of frame (bytes).<br><txRate>: Data rate.<br>b1: 11b DSSS 1 Mbps<br>b2: 11b DSSS 2 Mbps<br>b5_5: 11b DSSS 5.5 Mbps<br>b11: 11b DSSS 11 Mbps<br>g6: 11g 6 Mbps<br>g9: 11g 9 Mbps<br>g12: 11g 12 Mbps<br>g18: 11g 18 Mbps<br>g24: 11g 24 Mbps<br>g36: 11g 36 Mbps<br>g48: 11g 48 Mbps<br>g54: 11g 54 Mbps<br>n6_5: 11n 6.5 Mbps (7.2 Mbps @Short GI)<br>n13: 11n 13 Mbps (14.4 Mbps @Short GI)<br>n19_5: 11n 19.5 Mbps (21.7 Mbps @Short GI)<br>n26: 11n 26 Mbps (28.9 Mbps @Short GI)<br>n39: 11n 39 Mbps (43.3 Mbps @Short GI)<br>n52: 11n 52 Mbps (57.8 Mbps @Short GI)<br>n58_5: 11n 58.5 Mbps (65 Mbps @Short GI)<br>n65: 11n 65 Mbps (72.2 Mbps @Short GI)<br><txPower>: TX power (0 ~ 15), 0.8 dB step.<br><destAddr>: MAC address to send packet.<br><bssid>: BSSID.<br><htEnable>: N/A<br><GI>: [short\|long]. Guad interval. 11n mode only.<br><greenField>: [on\|off]. Set greenfield mode on/off.<br><preambleType>: [short\|long]. Preamble type @DSSS mode.<br><qosEnable>: [on\|off]. MAC header QoS control.<br><ackPolicy>: [NO\|NORM\|BA\|CBA]<br><scrambler>: N/A |


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| Command | Parameters | Description |
|---|---|---|
| | | <aifsnVal>: [0 ~ 15]. Indicate the AIFS in units of slots after SIFS that HW should wait for before starting backoff, for access category.<br><ant>: [0]. Fixed.<br>Response: OK or ERROR |
| | *Prerequisite*<br>Start RF test mode (AT+RFTESTSTART)<br>*Example*<br>; Tx test with 11N MCS7, 2412MHz and power grade as '0' (max power)<br>`AT+RFTX 2412,0,0,1000,n65,0`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK. To use, manufacture image is needed or contact Renesas Electronics<br>● Enabled by default in the SDK v3.2.3.0 or later | |
| AT+RFTXSTOP | (none) | Stop RF TX test. |
| | *Prerequisite*<br>Start RF TX test (AT+RFTX)<br>*Example*<br>`AT+RFTX 2412,0,0,1000,n65,0`<br>`OK`<br><br>`AT+RFTXSTOP`<br>`OK`<br><br>`AT+RFTX 2442,0,0,1000,n65,0`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK. To use, manufacture image is needed or contact Renesas Electronics<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● AT+RFTXSTOP is required before testing other items | |
| AT+RFCWTEST | <Ch>,<br><BW>,<br><txPower>,<br><ant>,<br><CWCycle> | Start CW test.<br><Ch>: Carrier frequency (2412 ~ 2484 MHz).<br><BW>: [0]: Fixed. Carrier bandwidth. 20 MHz fixed.<br><txPower>: TX power (0 ~ 15), 0.8 dB step.<br><ant>: [0]. Fixed.<br><CWCycle>: 1 MHz fixed.<br>Response: OK or ERROR |
| | *Prerequisite*<br>Start RF test mode (AT+RFTESTSTART)<br>*Example*<br>`AT+RFCWTEST 2442,0,2`<br>`OK`<br>`CW Tx test with 2442 MHz and power grade as 2`<br>*Note*<br>● Disabled by default in the ThreadX SDK. To use, manufacture image is needed or contact Renesas Electronics<br>● Enabled by default in the SDK v3.2.3.0 or later | |


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| Command | Parameters | Description |
|---|---|---|
| AT+RFCWSTOP | (none) | Stop CW test.<br>Response: OK or ERROR |
| | *Prerequisite*<br>Start RF CW test (AT+RFCWTEST)<br>*Example*<br>`AT+RFCWTEST 2442,0,2`<br>`OK`<br><br>`AT+RFCWSTOP`<br>`OK`<br><br>`AT+RFCWTEST 2472,0,2`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK. To use, manufacture image is needed or contact Renesas Electronics<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● AT+RFCWSTOP is required before testing other items | |
| AT+RFCONTSTART | <txRate>,<br><txPower>,<br><Ch> | Start RF continuous TX test.<br><txRate>: Data rate. Refer to AT+RFTX command.<br><txPower>: TX power (0 ~ 15), 0.8 dB step.<br><Ch>: Carrier frequency (2412 ~ 2484 MHz).<br>Response: OK or ERROR |
| | *Prerequisite*<br>Start RF test mode (AT+RFTESTSTART)<br>*Example*<br>; Continuous Tx test with 11G 54 MHz, 2472 MHz and power grade as 2<br>`AT+RFCONTSTART g54,2,2472`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK. To use, manufacture image is needed or contact Renesas Electronics<br>● Enabled by default in the SDK v3.2.3.0 or later | |
| AT+RFCONTSTOP | (none) | Stop RF continuous TX test.<br>Response: OK or ERROR |


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| Command | Parameters | Description |
|---|---|---|
| | *Prerequisite*<br>Start RF continuous TX test (AT+RFCONTSTART)<br>*Example*<br>`AT+RFCONTSTART g54,2,2412`<br>`OK`<br><br>`AT+RFCONTSTOP`<br>`OK`<br><br>`AT+RFCONTSTART g54,2,2472`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK. To use, manufacture image is needed or contact Renesas Electronics<br>● Enabled by default in the SDK v3.2.3.0 or later | |
| AT+RFCHANNEL | <Ch> | Change RF channel for PER test.<br><Ch>: Carrier frequency (2412 ~ 2484 MHz).<br>Response: OK or ERROR |
| | *Prerequisite*<br>Start RF test mode (AT+RFTESTSTART)<br>*Example*<br>`AT+RFCHANNEL 2412`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK. To use, manufacture image is needed or contact Renesas Electronics<br>● Enabled by default in the SDK v3.2.3.0 or later | |
| AT+RFPERRESET | (none) | Reset PER count.<br>Response: OK or ERROR |
| | *Prerequisite*<br>Start RF test mode (AT+RFTESTSTART)<br>*Example*<br>`AT+RFPERRESET`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK. To use, manufacture image is needed or contact Renesas Electronics<br>● Enabled by default in the SDK v3.2.3.0 or later | |
| AT+RFPER | (none) | Display PER state.<br>Indicate a number of Valid packets, FCS Errors packets, PHY Errors packets, Overflow Errors.<br>Response: OK or ERROR |


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| Command | Parameters | Description |
|---|---|---|
| | *Prerequisite*<br>Start RF test mode (AT+RFTESTSTART)<br>*Example*<br>`AT+RFPER`<br>`20 0 0 0`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK. To use, manufacture image is needed or contact Renesas Electronics<br>● Enabled by default in the SDK v3.2.3.0 or later | |
| AT+RFTESTSTOP | (none) | Stop RF test mode. |
| | *Example*<br>`AT+RFTESTSTOP`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK. To use, manufacture image is needed or contact Renesas Electronics<br>● Enabled by default in the SDK v3.2.3.0 or later | |

**Table 34: RF Test Examples**

| Test Step | Command | Description |
|---|---|---|
| ECHO on | ATE | ECHO ON |
| Set boot mode | AT+TMRFNOINIT=1 | Set boot mode as RF test mode |
| Restart the DA16200 (DA16600) | AT+RESTART | Reboot as RF test mode |
| ECHO on | ATE | ECHO ON |
| Start RF test mode | AT+RFTESTSTART | Start RF test mode. |
| Tx Test @11B 1Mbps | AT+RFTX 2412,0,0,200,b1,0 | 11B 1Mbps/Channel 1 |
| | AT+RFTXSTOP | Stop Tx |
| | AT+RFTX 2442,0,0,200,b1,0 | 11B 1Mbps/Channel 7 |
| | AT+RFTXSTOP | Stop Tx |
| | AT+RFTX 2472,0,0,200,b1,0 | 11B 1Mbps/Channel 13 |
| | AT+RFTXSTOP | Stop Tx |
| Tx Test @11G 54Mbps | AT+RFTX 2412,0,0,1000,g54,0 | 11G 54Mbps/Channel 1 |
| | AT+RFTXSTOP | Stop Tx |
| | AT+RFTX 2442,0,0,1000,g54,0 | 11G 54Mbps/Channel 7 |
| | AT+RFTXSTOP | Stop Tx |
| | AT+RFTX 2472,0,0,1000,g54,0 | 11G 54Mbps/Channel 13 |
| | AT+RFTXSTOP | Stop Tx |
| Tx Test @11N MCS7 | AT+RFTX 2412,0,0,1000,n65,0 | 11N MCS7/Channel 1 |
| | AT+RFTXSTOP | Stop Tx |
| | AT+RFTX 2442,0,0,1000,n65,0 | 11N MCS7/Channel 7 |


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| Test Step | Command | Description |
|---|---|---|
| | AT+RFTXSTOP | Stop Tx |
| | AT+RFTX 2472,0,0,1000,n65,0 | 11N MCS7/Channel 13 |
| | AT+RFTXSTOP | Stop Tx |
| Rx Test | AT+RFCHANNEL 2412 | Change RF channel to 1 |
| | AT+RFPERRESET | Reset PER count |
| | AT+RFPER | Display PER state |
| | AT+RFCHANNEL 2442 | Change RF channel to 7 |
| | AT+RFPERRESET | Reset PER count |
| | AT+RFPER | Display PER state |
| | AT+RFCHANNEL 2472 | Change RF channel to 13 |
| | AT+RFPERRESET | Reset PER count |
| | AT+RFPER | Display PER state |
| Stop RF test mode | AT+RFTESTSTOP | Stop RF test mode |

**NOTE**

Renesas Electronics provides the AT-GUI tool to test RF performance easily. The tool and manual are available on the Renesas website. (https://www.dialog-semiconductor.com/products/wi-fi/da16200#tab-field_tab_content_resources)

- UM-004 DA16200 AT GUI Tool User Manual
- DA16200 AT GUI Tool

# 12 System and Peripheral Function Commands

**NOTE**

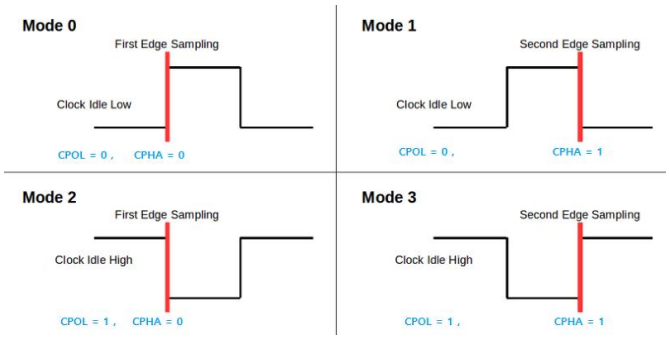
Peripheral function commands are disabled in the ThreadX SDK. To enable, define the feature __SUPPORT_PERI_CMD__ in sys_common_features.h file. These commands are enabled by default in manufacture image.

## 12.1 SPI Commands

**Table 35: SPI Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+SPICONF | <clockpol>,<br><clockpha> | Configure SPI.<br><clockpol>: Clock polarity [0\|1].<br><clockpha>: Clock phase [0\|1]. |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+SPICONF=1,1`<br>`OK`<br><br>`AT+SPICONF=0,1`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_PERI_CMD__ is enabled in the SDK, this command will be enabled<br>● *The <clockpol> sets the polarity of the clock signal during the idle state. The idle state is defined as the period when CS is high and transitioning to low at the start of the transmission and when CS is low and transitioning to high at the end of the transmission. The <clockpha> selects the clock phase. Depending on the <clockpha>, the rising or falling clock edge is used to sample the data. The default value of DA16200 (DA16600) are clockpol,0 and clockpha 0*<br>  <br> | |

## 12.2 OTP Commands

**Table 36: OTP Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+UOTPRDASC | <addr>,<cnt> | Read OTP data.<br><addr>: OTP address to read four-byte aligned.<br><cnt>: Bytes to read.<br>Response: OK or Error<br>A string of four-bit HEXA value represented by the ASCII code. |


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td></td><td colspan="2">Example<br>; Reading four bytes at offset h180 ➔ h180 * 4 = h600<br>; If data "12345678" is written to 0x600, can read the values<br>AT+UOTPRDASC=600,4<br>12345678<br>OK<br>Note<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_PERI_CMD__ is enabled in the SDK, this command will be enabled<br>● Physical OTP offset range of DA16200 (DA16600) is h0~h1FF; at each offset, four bytes are stored or read<br>● For accessing OTP using this command, a four-byte aligned address should be given. For example: h0, h4, h8 ...</td></tr>
<tr><td rowspan="2">AT+UOTPWRASC</td><td><addr>,<cnt>,<value></td><td>Write OTP data.<br><addr>: OTP address to write four-byte aligned.<br><cnt>: Bytes to write.<br><value>: A string of four-bit HEXA value represented by the ASCII code<br>Response: OK or Error<br>Important<br>For MAC address read or write, AT+WFOTP (write) and AT+WFMAC (read) must be used. Do not use AT+UOTPRDASC or AT+UOTPWRASC for this purpose.<br>OTP offset from 0x00 ~ 0x2b should not be written as this section is for “secure” boot.</td></tr>
<tr><td colspan="2">Example<br>; Writing h12345678 to OTP Address 0x600:<br>; To write “12345678” data into the 0x600,<br>AT+UOTPWRASC=600,4,12345678<br>OK<br><br>; To read written Data via UOTPRDASC<br>AT+UOTPRDASC=600,4<br>12345678<br>OK<br>Note<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_PERI_CMD__ is enabled in the SDK, this command will be enabled<br>● Physical OTP offset range of DA16200 (DA16600) is h0~h1FF; at each offset, four bytes are stored or read<br>● For accessing OTP using this command, a four-byte aligned address should be given. For example: h0, h4, h8 ...</td></tr>
</table>

DA16200 (DA16600) provides four slots to store MAC addresses and eight bytes are allocated for each slot.


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**Table 37: OTP Memory Address for Writing MAC Address**

| Slot | OTP Address | Description | Size (Byte) |
|---|---|---|---|
| MAC Address #0 | 0x100 | MAC Address Low | 4 |
| | 0x101 | MAC Address High | 4 |
| MAC Address #1 | 0x102 | MAC Address Low | 4 |
| | 0x103 | MAC Address High | 4 |
| MAC Address #2 | 0x104 | MAC Address Low | 4 |
| | 0x105 | MAC Address High | 4 |
| MAC Address #3 | 0x106 | MAC Address Low | 4 |
| | 0x107 | MAC Address High | 4 |

DA16200 (DA16600) provides two slots to store XTAL offset in the OTP memory. Slot #0 is the primary slot while Slot#1 is for back-up, which is used when overriding Slot #0.

**Table 38: Size of Memory by XTAL Offset**

| Slot | OTP Address | Description | Size (Bytes) |
|---|---|---|---|
| XTAL Offset #0 | 0x10A | XTAL Offset #0 value | 2 |
| XTAL Offset #1 | 0x10B | XTAL Offset #1 value | 2 |


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## 12.3 XTAL Commands

These commands are used for XTAL calibration and the usage is described in DA16200 (DA16600) Module Production Guide.

**Table 39: XTAL Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+XTALWR | <value> | Write XTAL Offset to DA16200 (DA16600) system register.<br><value>: Seven-bits to write [h'1 ~ h'7f].<br>Response: OK or Error |
| | *Example*<br>`AT+XTALWR=7f`<br>`OK`<br><br>`AT+XTALWR=80`<br>`ERROR`<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_PERI_CMD__ is enabled in the SDK, this command will be enabled | |
| AT+XTALRD | (none) | Read XTAL Offset from DA16200 (DA16600) System.<br>Response:<br><CR><LF><A string of seven-bit HEXA value represented by the ASCII Code><CR><LF>OK<CR><LF><br>or Error |
| | *Example*<br>`AT+XTALRD`<br>`0x7f`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_PERI_CMD__ is enabled in the SDK, this command will be enabled | |

## 12.4 Flash Dump Commands

**Table 40: Flash Dump Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+FLASHDUMP | <address>,<br><length> | Dump serial flash data.<br><address>: Start address [h'0 ~ h'3fffff].<br><length>: Data length [d'].<br>Response:<br><CR><LF><br><dump data><br><CR><LF>OK<CR><LF><br>or Error |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>; The following example reads 32 kB from 0x00, (1024*32 = 32768)<br>`AT+FLASHDUMP=0,32768`<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_PERI_CMD__ is enabled in the SDK, this command will be enabled | |

## 12.5 GPIO Commands

**Table 41: GPIO Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+GPIOSTART | <port>,<br><pin >,<br><direction> | Configures the GPIO pin mux and the direction of a GPIO.<br><port>: GPIO port number.<br>● 0: GPIOA<br>● 2: GPIOC<br><pin>: GPIO pin number. This is a hexadecimal value and indicates a GPIO bitmap.<br>● GPIOA: GPIOA0 ~ GPIOA11<br>● GPIOC: GPIOC6 ~ GPIOC8<br><direction>: GPIO pin direction.<br>● 0: Sets the pin as an input<br>● 1: Sets the pin as an output<br>Response: OK or Error |
| | *Example*<br>; To configure GPIOA [3:0] output:<br>; GPIOs 0,1,2,3 is set to binary 1(0000 0000 0000 1111).<br>`AT+GPIOSTART=0,f,1`<br>`OK`<br><br>; To configure GPIOC [8:6] input:<br>; GPIOs 6, 7, 8 is set to binary 1(0000 0001 1100 0000).<br>`AT+GPIOSTART=2,1c0,0`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_PERI_CMD__ is enabled in the SDK, this command will be enabled | |


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<table>
<tr><th>Command</th><th>Parameters</th><th>Description</th></tr>
<tr><td rowspan="2">AT+GPIORD</td><td><port>,<br><pin></td><td>Reads the GPIO input level.<br><port>: GPIO port number.<br>● 0: GPIOA<br>● 2: GPIOC<br><pin>: GPIO pin number. This is a hexadecimal value and indicates a GPIO bitmap.<br>Response:<br><Read value>: [h'0 ~ h'1fff]<br>OK or Error</td></tr>
<tr><td colspan="2">Example<br>; Configure GPIOC[8:6] as output and set to high.<br>AT+GPIOSTART=2,1c0,1<br>OK<br><br>AT+GPIOWR=2,1c0,1<br>OK<br><br>; Read back the status of the pins:<br>AT+GPIORD=2,1c0<br>0x01c0<br>OK<br>Note<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_PERI_CMD__ is enabled in the SDK, this command will be enabled<br>● The read value indicates GPIO bitmap. If a value is 0x1c0, it means GPIO #6, #7, #8 high</td></tr>
<tr><td>AT+GPIOWR</td><td><port>,<br><pin>,<br><level></td><td>Configures the output level of GPIO pins.<br><port>: GPIO port number.<br>● 0: GPIOA<br>● 2: GPIOC<br><pin>: GPIO pin number. This is a hexadecimal value and indicates a GPIO bitmap.<br><level>: GPIO output level.<br>● 0: Low<br>● 1: High<br>Response: OK or Error.</td></tr>
</table>


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| Command | Parameters | Description |
|---|---|---|
| | *Prerequisite*<br>The GPIO pin that wants to change output status should be set a direction as output.<br>*Example*<br>; Configure GPIOC[8:6] as output and set to high.<br>`AT+GPIOSTART=2,1c0,1`<br>`OK`<br><br>`AT+GPIOWR=2,1c0,1`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_PERI_CMD__ is enabled in the SDK, this command will be enabled | |
| AT+SAVE_PININFO | (none) | Save pin mux information.<br>Response: OK or Error. |
| | *Example*<br>`AT+SAVE_PININFO`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_PERI_CMD__ is enabled in the SDK, this command will be enabled<br>● It is to save a current PIN mux configured | |
| AT+RESTORE_PININFO | (none) | Restore pin mux information.<br>Response: OK or Error. |
| | *Example*<br>`AT+RESTORE_PININFO`<br>`OK`<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_PERI_CMD__ is enabled in the SDK, this command will be enabled<br>● It is to restore the PIN multiplexing status saved through the AT+SAVE_PININFO command | |

## 12.6 LED Commands

**Table 42: LED Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+LEDINIT | <none> | Configure GPIOC_6(LED1) / GPIOC_7(LED2) / GPIOC_8(LED3) pins to GPIO output<br>Response: OK or Error |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+LEDINIT`<br>`+OK`<br>*Note*<br>● Enabled by default in the SDK v3.2.2.1 or later | |
| AT+LEDCTRL | <port >,<br><status> | Set LED1/2/3(GPIOC_6/7/8) pin to output High or Low<br><port>: GPIO port number.<br>● 1: GPIOC_6<br>● 2: GPIOC_7<br>● 3: GPIOC_8<br><status>: LED status<br>● off: LED off<br>● on: LED on<br>Response: OK or Error. |
| | *Example*<br>`AT+LEDCTRL=1,off`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK v3.2.2.1 or later | |

## 12.7 PWM Commands

**Table 43: PWM Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+PWMINIT | <none> | Configure GPIOA_10 pin to PWM output<br>Response: OK or Error |
| | *Example*<br>`AT+PWMINIT`<br>`+OK`<br>*Note*<br>● Enabled by default in the SDK v3.2.2.1 or later<br>● If __ATCMD_IF_UART1__ is enabled in the SDK, this command will be enabled | |
| AT+PWMSTART | <channel >,<br><period>,<br><duty><br><mode cycle> | Start PWM output(GPIOA_10) with given period and duty<br><channel>: PWM channel, fixed as 0<br><period>: period of one clock (micro second).<br><duty>: duty as percentage.<br><mode cycle>: fixed as 0.<br>Response: OK or Error. |
| | *Example*<br>`AT+PWMSTART=0,40,50,0`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK v3.2.2.1 or later<br>● If __ATCMD_IF_UART1__ is enabled in the SDK, this command will be enabled | |


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| Command | Parameters | Description |
|---|---|---|
| AT+PWMSTOP | <none> | Stop PWM output<br>Response: OK or Error |
| | *Example*<br>AT+PWMSTOP<br>OK<br>*Note*<br>● Enabled by default in the SDK v3.2.2.1 or later<br>● If __ATCMD_IF_UART1__ is enabled in the SDK, this command will be enabled | |

## 12.8 ADC Commands

**Table 44: ADC Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+ADCINIT | <none> | Configure GPIOA_0, GPIOA_1, GPIOA_2, GPIOA_3 to analog input pins for ADC<br>GPIOA_0 : ADC channel 0<br>GPIOA_1 : ADC channel 1<br>GPIOA_2 : ADC channel 2<br>GPIOA_3 : ADC channel 3<br>Response: OK or Error |
| | *Example*<br>AT+ADCINIT<br>OK<br>*Note*<br>● Enabled by default in the SDK v3.2.2.1 or later<br>● If __ATCMD_IF_UART1__ is enabled in the SDK, this command will be enabled | |
| AT+ADCCHEN | <channel >,<br>< resolution > | Enable given ADC channel<br><channel>: ADC channel number [0\|1\|2\|3]<br><resolution>: ADC resolution. fixed as 12-bit.<br>Response: OK or Error. |
| | *Example*<br>AT+ADCCHEN=0,12<br>OK<br>*Note*<br>● Enabled by default in the SDK v3.2.2.1 or later<br>● If __ATCMD_IF_UART1__ is enabled in the SDK, this command will be enabled | |
| AT+ADCSTART | <divider> | Start ADC function.<br><divider>: divider ADC sampling rate<br>For example, when divider is 1, 1MHz / (divider(1)+1) = 500khz<br>Response: OK or Error. |


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| Command | Parameters | Description |
|---|---|---|
| | *Example*<br>`AT+ADCSTART=1`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK v3.2.2.1 or later<br>● If __ATCMD_IF_UART1__ is enabled in the SDK, this command will be enabled | |
| AT+ADCREAD | <channel>,<br><sample count> | Read ADC value.<br><channel>: ADC channel number [0\|1\|2\|3].<br><sample count>: count of sample to read.<br>Response:<br><Read value>: [sample count]<br><br>Response: OK or Error. |
| | *Example*<br>`AT+ADCREAD=0,16`<br>`[ 279 275 269 271 270 268 268 274 274 277 276 269 271 276 264 274 ]`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK v3.2.2.1 or later<br>● If __ATCMD_IF_UART1__ is enabled in the SDK, this command will be enabled | |
| AT+ADCSTOP | (none) | Stop ADC function.<br>Response: OK or Error. |
| | *Example*<br>`AT+ADCSTOP`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK v3.2.2.1 or later<br>● If __ATCMD_IF_UART1__ is enabled in the SDK, this command will be enabled | |

## 12.9 I2C Commands

**Table 45: I2C Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+I2CINIT | <none> | Configure GPIOA_8 (I2C_SDA), GPIOA_9 (I2C_SCL) pins to I2C pins<br>Response: OK or Error |
| | *Example*<br>`AT+I2CINIT`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK v3.2.2.1 or later | |


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| Command | Parameters | Description |
|---|---|---|
| AT+I2CREAD | <slave address>,<br><register>,<br><length> | Read values from registers of I2C device.<br><slave address>: 8bit slave address of I2C device (hex)<br><register>,: register value to read (hex)<br><length>: data length to read (decimal)<br>Response:<br><Read values> (hex)<br><br>Response: OK or Error. |
| | *Example*<br>`AT+I2CREAD=d0,10,1`<br>`66`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK v3.2.2.1 or later | |
| AT+ I2CWRITE | <slave address>,<br><register>,<br><length>,<br><values> | Write values to I2C register of I2C device.<br><slave address>: 8bit slave address of I2C device (hex).<br><register>: register value to write (hex)<br><length>: data length to write (decimal)<br><values>: data to write (hex)<br>Response: OK or Error. |
| | *Example*<br>`AT+I2CWRITE=d0,10,3,670292`<br>`OK`<br>*Note*<br>● Enabled by default in the SDK v3.2.2.1 or later | |

## 12.10 Sleep Commands

**Table 46: Sleep Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+SLEEPMS | <period> | Make DA16200 (DA16600) go to Sleep mode 3 and wake up after <period> milliseconds.<br><period>: Wake-up time in milliseconds.<br>Response: OK or Error |
| | *Example*<br>`AT+SLEEPMS=5000`<br>`+INIT:DONE,0`<br>*Note*<br>● Disabled by default in the ThreadX SDK<br>● Enabled by default in the SDK v3.2.3.0 or later<br>● If __SUPPORT_PERI_CMD__ is enabled in the SDK, this command will be enabled | |


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## 12.11 CALWL Commands

**Table 47: CALWL Command List**

| Command | Parameters | Description |
|---|---|---|
| AT+CALWR | <gmode_tx_rf_proc>,<br><txpga_gmode_cal> | Change RF TX GAIN Calibration register for test.<br><gmode_tx_rf_proc><br>● 7 bits hexadecimal without “0x” prefix.<br>● Offset = bit[5:0] x 0.8 dB.<br>● MSB[6] bits 0 then TX gain is decreased.<br>● MSB[6] bits 1 then TX gain is increased.<br><br><txpga_gmode_cal><br>● 0 : 0dB offset<br>● 1 : -0.2dB offset<br>● 2 : -0.4dB offset<br>● 3 : -0.6dB offset |
| | *Example*<br>1. TX measured +14 dBm and change to +13 dBm<br><br>`AT+CALWR=1,1`<br>`OK`<br>`gmade_tx_rf_proc = 1, -0.8 dB`<br>`txpga_gmode_cal = 1, -0.2 dB`<br><br>`Changed TX Gain: 14dBm - 0.8 dB - 0.2 dB = 13.0 dBm`<br><br>2. TX measured +13 dBm and change to +13.6 dBm<br><br>`AT+CALWR=41,3`<br>`OK`<br>`gmade_tx_rf_proc = 41, +0.8 dB`<br>`txpga_gmode_cal = 1, -0.2 dB`<br><br>`Changed TX Gain: 13dBm + 0.8 dB - 0.2 dB = 13.6 dBm`<br><br>*Note*<br>● This setting is not saved to system and changed when reboot system.<br>● If __SUPPORT_PERI_CMD__ is enabled in the SDK, this command will be enabled | |


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# 13 Examples

## 13.1 Data Transfer Test

This section describes how to test the transfer function commands with a data terminal emulator. Some of the terminal applications to use for this purpose are:

- IO Ninja: http://ioninja.com/
  - HEXA data and file transmitting function
- Socket Test: http://sockettest.sourceforge.net/
  - Text data transmittable only
- Script Communicator: http://sourceforge.net/projects/scriptcommunicator
  - Socket communication, UART Rx/Tx data color-distinguished output function and HEXA data transmission

The following sections describe test procedures for socket communication between the DA16200 (DA16600) and a PC with IO Ninja. You can run DA16200 (DA16600) AT commands on a serial terminal application on your PC. The terminal must be connected to the UART1 interface of the DA16200 (DA16600).


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### 13.1.1 TCP Server Socket Test

1. DA16200 (DA16600) AT command:
   a. `AT+TRTS=1234` ← Open a TCP server socket of which the port is 1234.
2. PC:
   a. Select **TCP Connection Socket** (1, Figure 8).
   b. Enter the IP address and the port number of DA16200 (DA16600) (2, Figure 8).
   c. Click  to connect the socket (3, Figure 8).
3. DA16200 (DA16600) AT command:
   a. `+TRCTS:0,192.168.0.5,3713` ← A TCP client socket connected, and IP address is 192.168.0.5 and port is 3713.
4. PC:
   a. Send data (4, Figure 8).
5. DA16200 (DA16600) AT command:
   a. `+TRDTS:0,192.168.0.5,3713,10,DIA_AT_TCP` ← Received ten bytes of data “DIA_AT_TCP”.

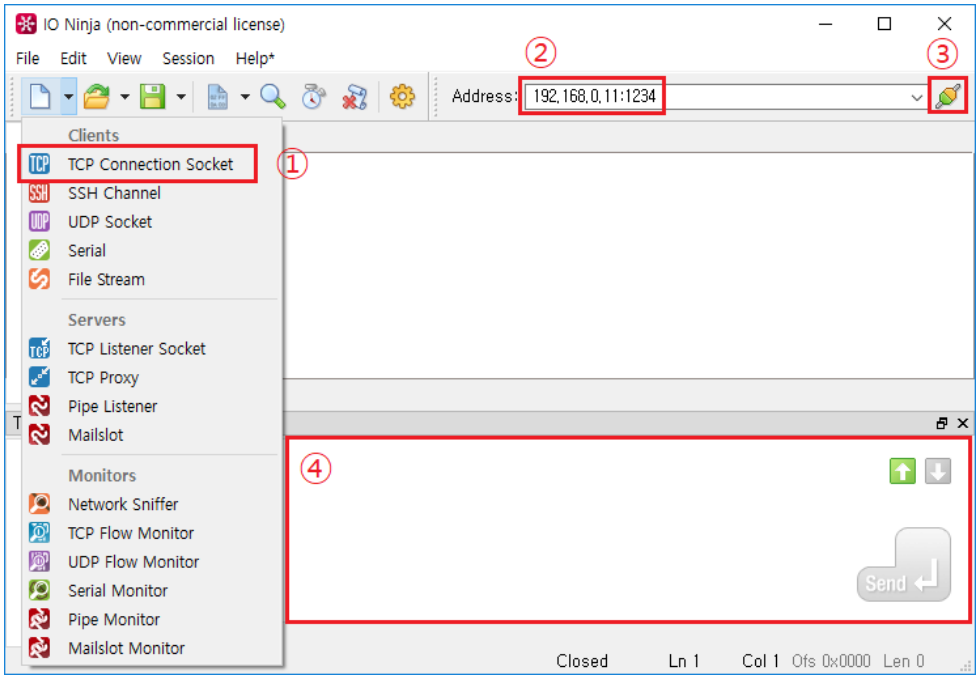



**Figure 10: IO Ninja – TCP Client Socket Setting**


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### 13.1.2 TCP Client Socket Test

1. PC:
   a. Select **TCP Listener Socket** (1, Figure 9).
   b. Enter the port number to be used (2, Figure 9).
   c. Click  to start to “Listen” (3, Figure 9).
2. DA16200 (DA16600) AT command:
   a. `AT+TRTC=192.168.0.5,1234,2300` ← Open a TCP client socket and set the server IP (192.168.0.5), port (1234), and the local port (2300).
   b. `<ESC>S18,0,0,12345678` ← Send eight bytes of data “12345678”.
3. PC:
   a. Received data.
   b. Send data (4, Figure 9).
4. DA16200 (DA16600) AT command:
   a. `+TRDTC:1,192.168.0.5,1234,10,DIA_AT_TCP` ← Received ten bytes of data “DIA_AT_TCP”.

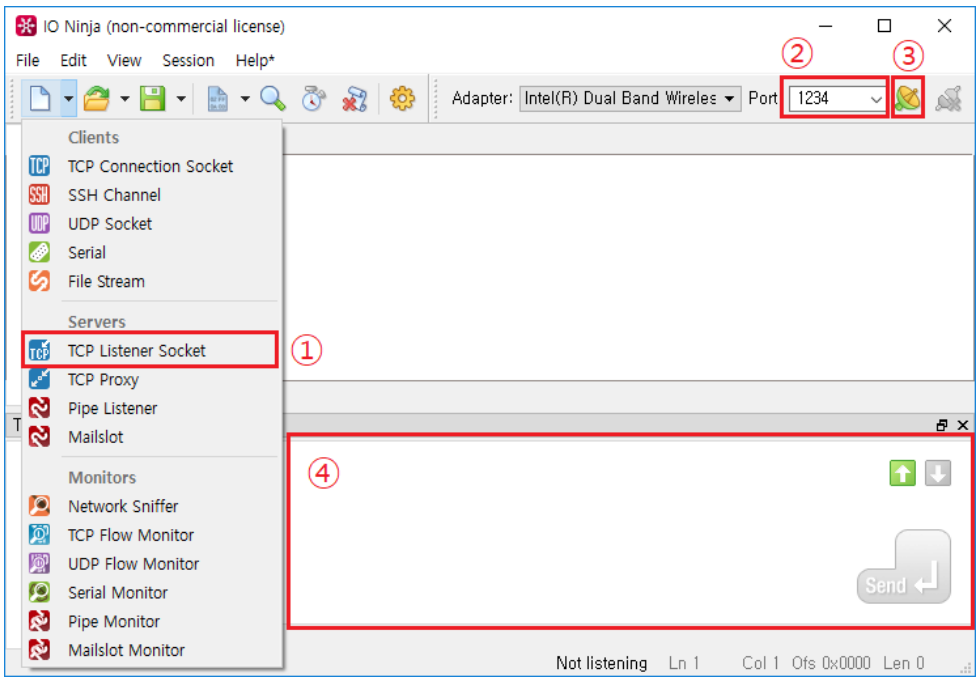



**Figure 11: IO Ninja – TCP Server Socket Setting**


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### 13.1.3 UDP Socket Test

1. PC:
   a. Select **UDP Socket** (1, Figure 10).
   b. Enter the port number to be used and click  to open the socket (2, Figure 10).
   c. Enter the IP address and port of the counterpart's UDP socket, click  and get ready for data transmission (3, Figure 10).
   d. Enter data and click **Send** to transmit (4, Figure 10).
2. DA16200 (DA16600) AT command:
   a. `AT+TRUSE=4567` ← Open a UDP socket and set the local port (4567).
   b. `AT+TRUR=192.168.0.5,1234` ← Set the remote IP (192.168.0.5) and port (1234).
   c. `<ESC>S210,0,0,1234567890` ← Send ten bytes of data "1234567890".
   d. `+TRDTC:0,10,DIA_AT_UDP` ← Received ten bytes of data "DIA_AT_UDP".

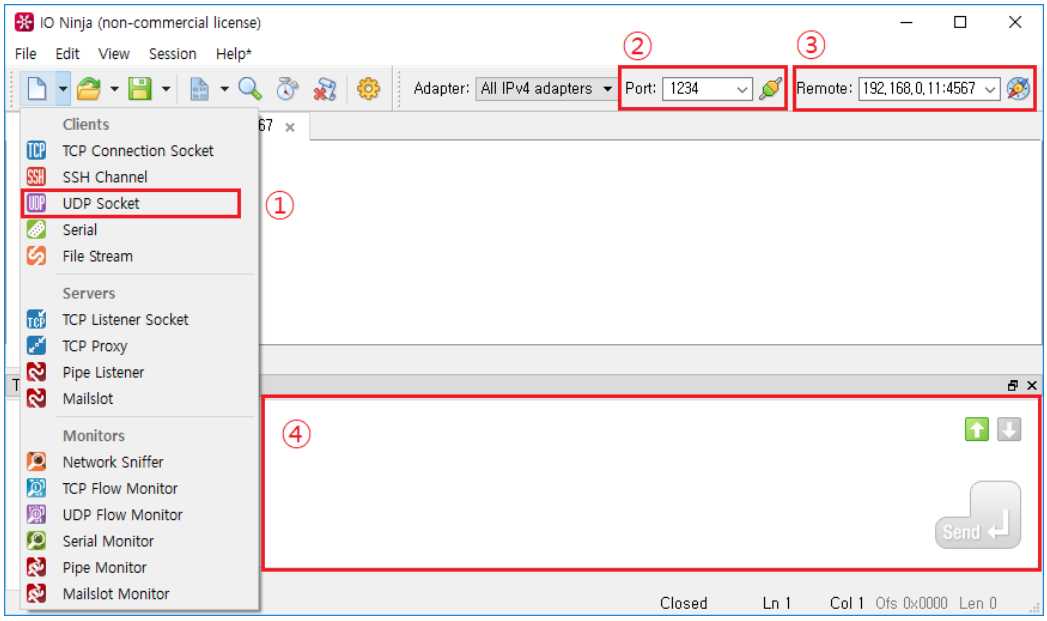



**Figure 12: IO Ninja – UDP Socket Setting**


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# Appendix A License Information

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Mosquitto1.4.14 License

Eclipse Distribution License 1.0

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UMAC GPL License

Linux kernel 3.9.0 rc3 version (backport 4.2.6-1)

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# Appendix B HTTP API Return Values

## B.1 Return value as defined by NetX Duo HTTP

| Define | Value | Define | Value |
|---|---|---|---|
| NX_SUCCESS | 0x00 | NX_RESERVED_CODE1 | 0x25 |
| NX_NO_PACKET | 0x01 | NX_SOCKET_UNBOUND | 0x26 |
| NX_UNDERFLOW | 0x02 | NX_NOT_CREATED | 0x27 |
| NX_OVERFLOW | 0x03 | NX_SOCKETS_BOUND | 0x28 |
| NX_NO_MAPPING | 0x04 | NX_NO_RESPONSE | 0x29 |
| NX_DELETED | 0x05 | NX_POOL_DELETED | 0x30 |
| NX_POOL_ERROR | 0x06 | NX_ALREADY_RELEASED | 0x31 |
| NX_PTR_ERROR | 0x07 | NX_RESERVED_CODE2 | 0x32 |
| NX_WAIT_ERROR | 0x08 | NX_MAX_LISTEN | 0x33 |
| NX_SIZE_ERROR | 0x09 | NX_DUPLICATE_LISTEN | 0x34 |
| NX_OPTION_ERROR | 0x0A | NX_NOT_CLOSED | 0x35 |
| NX_DELETE_ERROR | 0x10 | NX_NOT_LISTEN_STATE | 0x36 |
| NX_CALLER_ERROR | 0x11 | NX_IN_PROGRESS | 0x37 |
| NX_INVALID_PACKET | 0x12 | NX_NOT_CONNECTED | 0x38 |
| NX_INVALID_SOCKET | 0x13 | NX_WINDOW_OVERFLOW | 0x39 |
| NX_NOT_ENABLED | 0x14 | NX_ALREADY_SUSPENDED | 0x40 |
| NX_ALREADY_ENABLED | 0x15 | NX_DISCONNECT_FAILED | 0x41 |
| NX_ENTRY_NOT_FOUND | 0x16 | NX_STILL_BOUND | 0x42 |
| NX_NO_MORE_ENTRIES | 0x17 | NX_NOT_SUCCESSFUL | 0x43 |
| NX_ARP_TIMER_ERROR | 0x18 | NX_UNHANDLED_COMMAND | 0x44 |
| NX_RESERVED_CODE0 | 0x19 | NX_NO_FREE_PORTS | 0x45 |
| NX_WAIT_ABORTED | 0x1A | NX_INVALID_PORT | 0x46 |
| NX_IP_INTERNAL_ERROR | 0x20 | NX_INVALID_RELISTEN | 0x47 |
| NX_IP_ADDRESS_ERROR | 0x21 | NX_CONNECTION_PENDING | 0x48 |
| NX_ALREADY_BOUND | 0x22 | NX_TX_QUEUE_DEPTH | 0x49 |
| NX_PORT_UNAVAILABLE | 0x23 | NX_NOT_IMPLEMENTED | 0x4A |
| NX_NOT_BOUND | 0x24 | NX_NOT_SUPPORTED | 0x4B |
| NX_INVALID_INTERFACE | 0x4C | NX_DUPLICATED_ENTRY | 0x52 |
| NX_INVALID_PARAMETERS | 0x4D | NX_PACKET_OFFSET_ERROR | 0x53 |
| NX_NOT_FOUND | 0x4E | NX_OPTION_HEADER_ERROR | 0x54 |
| NX_CANNOT_START | 0x4F | NX_CONTINUE | 0x55 |
| NX_NO_INTERFACE_ADDRESS | 0x50 | NX_PARAMETER_ERROR | 0xFF |
| NX_INVALID_MTU_DATA | 0x51 | | |

## B.2 Return value as defined by LWIP HTTP

| Define | Value | Define | Value |
|---|---|---|---|
| ERR_OK | 0 | ERR_ISCONN | -10 |
| ERR_MEM | -1 | ERR_CONN | -11 |
| ERR_BUF | -2 | ERR_IF | -12 |
| ERR_TIMEOUT | -3 | ERR_ABRT | -13 |
| ERR_RTE | -4 | ERR_RST | -14 |
| ERR_INPROGRESS | -5 | ERR_CLSD | -15 |
| ERR_VAL | -6 | ERR_ARG | -16 |
| ERR_WOULDBLOCK | -7 | ERR_UNKNOWN | -17 |
| ERR_USE | -8 | ERR_NOT_FOUND | -18 |


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| | | | |
|---|---|---|---|
| ERR_ALREADY | -9 | | |

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# Appendix C User UART Configuration

## C.1 How to Run AT-CMD on UART2

AT-CMD is configured to use the UART1 interface by default.

AT-CMD can be configured to use the UART2 interface.

To configure AT-CMD to use the UART2 interface, modify config_generic_sdk.h as shown in bold below:

```
      // AT-CMD service
      #define  __SUPPORT_ATCMD__
...
      #if defined ( __SUPPORT_ATCMD__ )
              #undef __ATCMD_IF_UART1__           // AT-CMD over UART1
              #define __ATCMD_IF_UART2__          // AT-CMD over UART2
              ...
              #undef __USER_UART_CONFIG__ // Support Customer's UART configuration
              #undef __ATCMD_IF_SPI__             // AT-CMD over SPI
              #undef __ATCMD_IF_SDIO__            // AT-CMD over SDIO
       #endif /* __SUPPORT_ATCMD__ */
...
```

## C.2 User UART Configuration

There's a feature called "User UART Configuration" that is enabled by __USER_UART_CONFIG__ (available in SDK V2.3.4.0 and later version).

When the SDK is built with __USER_UART_CONFIG__ defined, the UART settings for the AT-CMD interface can be configured. In this case, ATB will not be available.

For example, to run AT-CMD on UART2 with a static baud rate of 230400, the SDK should be configured as shown in **bold** below.

```
// config_generic_sdk.h
...
      // AT-CMD service
      #define  __SUPPORT_ATCMD__
...
      #if defined ( __SUPPORT_ATCMD__ )
              #undef __ATCMD_IF_UART1__    // AT-CMD over UART1
              #define __ATCMD_IF_UART2__   // AT-CMD over UART2
              ...
              #define __USER_UART_CONFIG__ // Support Customer's UART configuration
              ...
       #endif /* __SUPPORT_ATCMD__ */
...

// user_interface.c
...
#if defined ( __USER_UART_CONFIG__ )
/*
 * Customer configuration for AT-CMD UART
 */
uart_info_t ATCMD_UART_config_info =
{
      UART_BAUDRATE_230400,         /* baud */
      UART_DATABITS_8,              /* bits */
```

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```
      UART_PARITY_NONE,             /* parity */
      UART_STOPBITS_1,              /* stopbit */
      UART_FLOWCTL_OFF              /* flow control */
};
#endif // __USER_UART_CONFIG__
...
```

With changes above, when DA16200 (DA16600) boots, AT-CMD is initialized in baud rate 230400 by default and cannot change at run time.

## C.3 Use Case

// __USER_UART_CONFIG__ **disabled**

- Baud rate (and other parameters) configurable by NVRAM
- ATB available, UART Setting can change at run-time without SDK rebuild
- Example Use case
  - MCU: Run on UART in baud rate 115200
  - MCU: Run ATF
  - DA16200 (DA16600): AT-CMD is initialized in 115200
  - MCU: ATB=230400
  - MCU: Now it should change its UART baud rate to 230400 to communicate with DA16200 (DA16600)

// __USER_UART_CONFIG__ **enabled**

- AT-CMD UART's baud rate (and other parameters) is configurable statically
- ATB NOT available
- Example Use Case
  - DA16200 (DA16600): DA16200 (DA16600) boots and AT-CMD is initialized in 230400 by default now.
  - MCU: Start on UART in baud rate 230400
  - MCU: AT-CMD operation …


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# Appendix D Example Sequence to Initiate AT-CMD through SPI or SDIO

## D.1 How to Run AT-CMD on SPI

AT-CMD is configured to use the UART1 interface by default.

AT-CMD can be configured to use the SPI interface.

To configure AT-CMD to use the SPI interface, modify config_generic_sdk.h as shown in bold below:

```
...
      // AT-CMD service
      #define  __SUPPORT_ATCMD__
...
      #if defined ( __SUPPORT_ATCMD__ )
              #undef __ATCMD_IF_UART1__           // AT-CMD over UART1
              #undef __ATCMD_IF_UART2__           // AT-CMD over UART2
              ...
              #undef __USER_UART_CONFIG__ // Support Customer's UART configuration
              #define __ATCMD_IF_SPI__            // AT-CMD over SPI
              #undef __ATCMD_IF_SDIO__            // AT-CMD over SDIO
       #endif /* __SUPPORT_ATCMD__ */
...
```

## D.2 How to Run AT-CMD on SDIO

AT-CMD can also be configured to use the SDIO interface.

To configure AT-CMD to use the SDIO interface, modify config_generic_sdk.h as shown in bold below:

```
...
      // AT-CMD service
      #define  __SUPPORT_ATCMD__
...
      #if defined ( __SUPPORT_ATCMD__ )
              #undef __ATCMD_IF_UART1__           // AT-CMD over UART1
              #undef __ATCMD_IF_UART2__           // AT-CMD over UART2
              ...
              #undef __USER_UART_CONFIG__ // Support Customer's UART configuration
              #undef __ATCMD_IF_SPI__             // AT-CMD over SPI
              #define __ATCMD_IF_SDIO__           // AT-CMD over SDIO
       #endif /* __SUPPORT_ATCMD__ */
...
```

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## D.3 Example Sequence for SDIO interface

An example of the sequence used to initiate a command through the SDIO interface is shown in Figure 13.

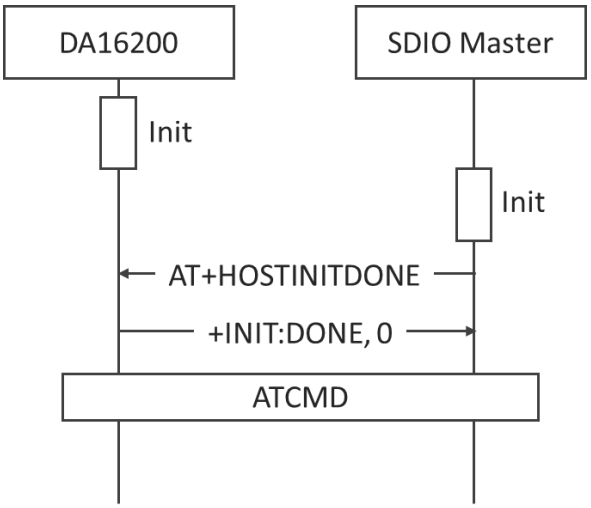



**Figure 13: Example Sequence to Initiate AT-CMD through SDIO Interface**

When using the SDIO interface, both the DA16200 and the SDIO master device must be initialized before initiating an AT-CMD. The SDIO Master device must send ‘AT+HOSTINITDONE’ immediately after initialization is complete.


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# Appendix E DA16200 (DA16600) Cipher Suites

| No. | Cipher Suite Supported by DA16200 (DA16600) | Hex Code |
|---|---|---|
| 1 | TLS_RSA_WITH_AES_128_CBC_SHA | 2F |
| 2 | TLS_RSA_WITH_AES_256_CBC_SHA | 35 |
| 3 | TLS_RSA_WITH_AES_128_CBC_SHA256 | 3C |
| 4 | TLS_RSA_WITH_AES_256_CBC_SHA256 | 3D |
| 5 | TLS_RSA_WITH_AES_128_GCM_SHA256 | 9C |
| 6 | TLS_RSA_WITH_AES_256_GCM_SHA384 | 9D |
| 7 | TLS_RSA_WITH_AES_128_CCM | C09C |
| 8 | TLS_RSA_WITH_AES_256_CCM | C09D |
| 9 | TLS_RSA_WITH_AES_128_CCM_8 | C0A0 |
| 10 | TLS_RSA_WITH_AES_256_CCM_8 | C0A1 |
| 11 | TLS_RSA_WITH_DES_CBC_SHA | 9 |
| 12 | TLS_DHE_RSA_WITH_AES_128_CBC_SHA | 33 |
| 13 | TLS_DHE_RSA_WITH_AES_256_CBC_SHA | 39 |
| 14 | TLS_DHE_RSA_WITH_AES_128_CBC_SHA256 | 67 |
| 15 | TLS_DHE_RSA_WITH_AES_256_CBC_SHA256 | 6B |
| 16 | TLS_DHE_RSA_WITH_AES_128_GCM_SHA256 | 9E |
| 17 | TLS_DHE_RSA_WITH_AES_256_GCM_SHA384 | 9F |
| 18 | TLS_DHE_RSA_WITH_AES_128_CCM | C09E |
| 19 | TLS_DHE_RSA_WITH_AES_256_CCM | C09F |
| 20 | TLS_DHE_RSA_WITH_AES_128_CCM_8 | C0A2 |
| 21 | TLS_DHE_RSA_WITH_AES_256_CCM_8 | C0A3 |
| 22 | TLS_DHE_RSA_WITH_3DES_EDE_CBC_SHA | 16 |
| 23 | TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA | C011 |
| 24 | TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA | C014 |
| 25 | TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256 | C027 |
| 26 | TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384 | C028 |
| 27 | TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 | C02F |
| 28 | TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 | C030 |
| 29 | TLS_ECDHE_RSA_WITH_3DES_EDE_CBC_SHA | C012 |
| 30 | TLS_ECDH_RSA_WITH_AES_128_CBC_SHA | C00E |
| 31 | TLS_ECDH_RSA_WITH_AES_256_CBC_SHA | C00F |
| 32 | TLS_ECDH_RSA_WITH_AES_128_CBC_SHA256 | C029 |
| 33 | TLS_ECDH_RSA_WITH_AES_256_CBC_SHA384 | C02A |
| 34 | TLS_ECDH_RSA_WITH_AES_128_GCM_SHA256 | C031 |
| 35 | TLS_ECDH_RSA_WITH_AES_256_GCM_SHA384 | C032 |
| 36 | TLS_ECDH_RSA_WITH_3DES_EDE_CBC_SHA | C00D |
| 37 | TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA | C009 |


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| No. | Cipher Suite Supported by DA16200 (DA16600) | Hex Code |
|---|---|---|
| 38 | TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA | C00A |
| 39 | TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256 | C023 |
| 40 | TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384 | C024 |
| 41 | TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256 | C02B |
| 42 | TLS_ECDHE_ECDSA_WITH_AES_256_GCM_SHA384 | C02C |
| 43 | TLS_ECDHE_ECDSA_WITH_AES_128_CCM | C0AC |
| 44 | TLS_ECDHE_ECDSA_WITH_AES_256_CCM | C0AD |
| 45 | TLS_ECDHE_ECDSA_WITH_AES_128_CCM_8 | C0AE |
| 46 | TLS_ECDHE_ECDSA_WITH_AES_256_CCM_8 | C0AF |
| 47 | TLS_ECDHE_ECDSA_WITH_3DES_EDE_CBC_SHA | C008 |
| 48 | TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA | C004 |
| 49 | TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA | C005 |
| 50 | TLS_ECDH_ECDSA_WITH_AES_128_CBC_SHA256 | C025 |
| 51 | TLS_ECDH_ECDSA_WITH_AES_256_CBC_SHA384 | C026 |
| 52 | TLS_ECDH_ECDSA_WITH_AES_128_GCM_SHA256 | C02D |
| 53 | TLS_ECDH_ECDSA_WITH_AES_256_GCM_SHA384 | C02E |
| 54 | TLS_ECDH_ECDSA_WITH_3DES_EDE_CBC_SHA | C003 |


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# Appendix F Reason Code For Wi-Fi Connection Failure or Disconnection

SDK v3.x.x.x: core\wifistack\supplicant\src\common\ieee802_11_defs.h

SDK v2.x.x.x: core\supplicant\src\common\ieee802_11_defs.h

```
/* Reason codes (IEEE Std 802.11-2016, 9.4.1.7, Table 9-45) */
#define WLAN_REASON_UNSPECIFIED                                 1
#define WLAN_REASON_PREV_AUTH_NOT_VALID                         2
#define WLAN_REASON_DEAUTH_LEAVING                              3
#define WLAN_REASON_DISASSOC_DUE_TO_INACTIVITY                  4
#define WLAN_REASON_DISASSOC_AP_BUSY                            5
#define WLAN_REASON_CLASS2_FRAME_FROM_NONAUTH_STA               6
#define WLAN_REASON_CLASS3_FRAME_FROM_NONASSOC_STA              7
#define WLAN_REASON_DISASSOC_STA_HAS_LEFT                       8
#define WLAN_REASON_STA_REQ_ASSOC_WITHOUT_AUTH                  9
/* IEEE 802.11h */
#define WLAN_REASON_PWR_CAPABILITY_NOT_VALID                    10
#define WLAN_REASON_SUPPORTED_CHANNEL_NOT_VALID                 11
#define WLAN_REASON_BSS_TRANSITION_DISASSOC                     12
/* IEEE 802.11i */
#define WLAN_REASON_INVALID_IE                                  13
#define WLAN_REASON_MICHAEL_MIC_FAILURE                         14
#define WLAN_REASON_4WAY_HANDSHAKE_TIMEOUT                      15
#define WLAN_REASON_GROUP_KEY_UPDATE_TIMEOUT                    16
#define WLAN_REASON_IE_IN_4WAY_DIFFERS                          17
#define WLAN_REASON_GROUP_CIPHER_NOT_VALID                      18
#define WLAN_REASON_PAIRWISE_CIPHER_NOT_VALID                   19
#define WLAN_REASON_AKMP_NOT_VALID                              20
#define WLAN_REASON_UNSUPPORTED_RSN_IE_VERSION                  21
#define WLAN_REASON_INVALID_RSN_IE_CAPAB                        22
#define WLAN_REASON_IEEE_802_1X_AUTH_FAILED                     23
#define WLAN_REASON_CIPHER_SUITE_REJECTED                       24
#define WLAN_REASON_TDLS_TEARDOWN_UNREACHABLE                   25
#define WLAN_REASON_TDLS_TEARDOWN_UNSPECIFIED                   26
#define WLAN_REASON_SSP_REQUESTED_DISASSOC                      27
#define WLAN_REASON_NO_SSP_ROAMING_AGREEMENT                    28
#define WLAN_REASON_BAD_CIPHER_OR_AKM                           29
#define WLAN_REASON_NOT_AUTHORIZED_THIS_LOCATION                30
#define WLAN_REASON_SERVICE_CHANGE_PRECLUDES_TS                 31
#define WLAN_REASON_UNSPECIFIED_QOS_REASON                      32
#define WLAN_REASON_NOT_ENOUGH_BANDWIDTH                        33
#define WLAN_REASON_TDLS_TEARDOWN_UNSPECIFIED                   26
/* IEEE 802.11e */
#define WLAN_REASON_DISASSOC_LOW_ACK                            34
#define WLAN_REASON_EXCEEDED_TXOP                               35
```


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```
#define WLAN_REASON_STA_LEAVING                                  36
#define WLAN_REASON_END_TS_BA_DLS                                37
#define WLAN_REASON_UNKNOWN_TS_BA                                38
#define WLAN_REASON_TIMEOUT                                      39
#define WLAN_REASON_PEERKEY_MISMATCH                             45
#define WLAN_REASON_AUTHORIZED_ACCESS_LIMIT_REACHED              46
#define WLAN_REASON_EXTERNAL_SERVICE_REQUIREMENTS                47
#define WLAN_REASON_INVALID_FT_ACTION_FRAME_COUNT                48
#define WLAN_REASON_INVALID_PMKID                                49
#define WLAN_REASON_INVALID_MDE                                  50
#define WLAN_REASON_INVALID_FTE                                  51
#define WLAN_REASON_MESH_PEERING_CANCELLED                       52
#define WLAN_REASON_MESH_MAX_PEERS                               53
#define WLAN_REASON_MESH_CONFIG_POLICY_VIOLATION                 54
#define WLAN_REASON_MESH_CLOSE_RCVD                              55
#define WLAN_REASON_MESH_MAX_RETRIES                             56
#define WLAN_REASON_MESH_CONFIRM_TIMEOUT                         57
#define WLAN_REASON_MESH_INVALID_GTK                             58
#define WLAN_REASON_MESH_INCONSISTENT_PARAMS                     59
#define WLAN_REASON_MESH_INVALID_SECURITY_CAP                    60
#define WLAN_REASON_MESH_PATH_ERROR_NO_PROXY_INFO                61
#define WLAN_REASON_MESH_PATH_ERROR_NO_FORWARDING_INFO           62
#define WLAN_REASON_MESH_PATH_ERROR_DEST_UNREACHABLE             63
#define WLAN_REASON_MAC_ADDRESS_ALREADY_EXISTS_IN_MBSS           64
#define WLAN_REASON_MESH_CHANNEL_SWITCH_REGULATORY_REQ           65
#define WLAN_REASON_MESH_CHANNEL_SWITCH_UNSPECIFIED              66
```


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# Appendix G Fast-reconnect on SLEEP mode 2

In the case where Wi-Fi STA tries to connect to an AP after a power-on by power-switch or RTC timer wakeup on SLEEP mode 2, it requires some time to make a Wi-Fi connection.

To reduce this Wi-Fi connection time between Wi-Fi AP and STA, the DA16200/DA16600 provides "Fast-connection" function.

In AT-CMD of the DA16200/DA16600, this function is enabled by default.

## G.1 Technical overview

- **Direct Probe request for Wi-Fi SCAN**

  During the Wi-Fi SETUP processing after factory status, associated channel number will be saved in NVRAM automatically after success Wi-Fi connect.

  After saving the connected channel number to NVRAM, when reconnecting to Wi-Fi is attempted, the total Wi-Fi SCAN time to connect is reduced because only the registered channel number is scanned without performing full-channel scan for Wi-Fi connection.

- **Network Address without DHCP Client procedure**

  The DA16200/DA16600 obtains an IP address by DHCP Client procedure when the first Wi-Fi connection is completed after Factory-reset. The DHCP Client operation may take a lot of time in some cases.

  To reduce DHCP Client procedure time, the DA16200/DA16600 saves the IP address, subnet mask, gateway address, and DNS address information in NVRAM after a successful DHCP client procedure, and changes to STATIC IP mode internally. STATIC IP mode removes DHCP processing time and improves connection speed.


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# Appendix H Bluetooth® LE Coexistence Feature

The Bluetooth® LE Coexistence feature is defined as follows:

- DA16200 AT command image: Bluetooth® LE Coexistence feature is disabled.
- DA16600 AT command image: Bluetooth® LE Coexistence feature is enabled (3-Pin interface).

If the Bluetooth® LE Coexistence feature or the 1-Pin interface are required, the SDK must be rebuilt. For more details see Ref. [2].


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# Appendix I Wi-Fi Passive-Scan

A client can use two scanning methods: active and passive. During an active scan, the client radio sends a probe request and receives a probe response from an AP. With a passive scan, the client radio listens for beacons periodically sent by the AP on each channel. A passive scan generally takes more time, since the client must listen and wait for a beacon, than actively probing to find an AP. Another limitation with a passive scan is that if the client does not wait long enough on a channel, then the client may miss an AP beacon.
The DA16200 (DA16600) supports active scan and passive scan. It accepts both probe responses and beacons. The DA16200 passive scan consists of frequency and time remaining on the channel, and the result is delivered to the host firmware within 10 ms.

## I.1 Passive-Scan with specified Channel and Scan-time Limit

The Wi-Fi component should be able to perform a passive Wi-Fi scan that only scans a given list of channels in given time limit. The DA16200 provides passive scan command with ATCMD.

Related command is “AT+WFPSCAN”.

## I.2 Passive-Scan Result

During a passive scan, the Wi-Fi component should be able to report each beacon signal to the host firmware within 10 ms received as a result of the passive scan. The format which is reported to host firmware is as follows:

- BSSID SSID RSSI Security Type Wi-Fi Channel

## I.3 Passive-Scan Stop

The Wi-Fi component should be able to stop an ongoing passive Wi-Fi scan and any power use associated with the scan within 100 ms of receiving a Wi-Fi beacon signal that meets the following criteria. Beacon BSSID matches given pattern AND either of the following:

- Beacon RSSI is greater than the minimum threshold, OR
- Beacon RSSI is less than the maximum threshold

When it stops scanning by condition, it prints out “+PSCAN:CONDITIONMET”.

Related commands are “AT+WFPCDTMIN”, “AT+WFCDTMAX” and “AT+WFPSTOP”.


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## I.4 Passive Scan Sequence

This section describes basic procedure for passive scan between the DA16200 (DA16600) and a PC. You can run the DA16200 AT commands on a serial terminal application on your PC. The terminal must be connected to the UART1 interface of the DA16200.

1. Enable passive scan in “config_generic_sdk.h”:

**#define __SUPPORT_PASSIVE_SCAN__**

2. Set passive scan condition using ATCMD:

**ATCMD: AT+WFPCDTMIN=**72:5d:cc:d0:82:bc,-80

3. Start passive scan using ATMD:

**ATCMD: AT+WFPSCAN=120000, 1, 3, 5**

4. Stop passive scan using ATMD:

**ATCMD: AT+WFPSTOP**

5. Check passive scan report; see Figure 14.




**Figure 14: Passive Scan Report**


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# Revision History

| Revision | Date | Description |
|---|---|---|
| 2.12 | 08-Aug-2022 | Updated Appendix for running AT command through SDIO or SPI.<br>Added 8.2 Wi-Fi Function Commands for WPA Enterprise |
| 2.11 | 14-Jun-2022 | AT+SDKVER added<br>Added 5.1 Command Format<br>Added AT+HOSTINITDONE<br>Added Appendix E<br>Added more info on AT+TZONE<br>Info updated or typo fixed: ATQ, ATB, AT+NWDHIP, AT+WFRSSI<br>Changed default status by some features are enabled<br>Added AT+CHIPNAME, AT+CALWR<br>AT+NWMQMSG : operation response added (+NWMQMSGSND)<br>Updated +WFJAP:0 and +WFDAP:0 (reason parameter is added)<br>Updated AT+NWMQBR, AT+NWMQTT<br>Updated MQTT optional configuration commands (enabled by default in SDK v3.2.3.0)<br>Added AT+NWMQCS, and CleanSession=0 Guide.<br>AT+WFJAP / AT+WFJAPA / AT+WFSAP : passphrase range added<br>AT+TRTS example updated<br>Added AT+NWMQUTS<br>Updated Note : AT+NWIP, AT+NWDHR, AT+NWDHLT, AT+NWMQMSG, <ESC>S<br>+NWMQCL updated<br>Added Appendix H<br>Added AT+NWWSC<br>Added note for <ESC>S : for ATCMD on SPI |
| 2.10 | 22-Mar-2022 | Update logo, disclaimer, copyright.<br>Added AT+TCPDATAMODE<br>Added LED/PWM/ADC/I2C related AT commands (Section 12.6, 12.7, 12.8, 12.9) |
| 2.9 | 14-Dec-2021 | AT+NWMQMSG, AT+NWMQTS : updated max length info on topic and message added<br>AT+NWDNS, HELP, ATE, ATQ : more info added |
| 2.8 | 08-Dec-2021 | Updated Section 12.2, OTP Size Reduced (8KB->2KB)<br>Updated Section 10.1.1 Data Transfer Commands<br>Updated AT+TRSSLWR in Section 10.1 Secure Socket commands<br>AT+NWMQATS, AT+NWMQDTS, AT+NWMQV311, AT+NWDHIP added<br>AT+ NWHTCSNI, AT+ NWHTCALPN, AT+NWHTCSNIDEL, AT+NWHTCALPNDEL added |
| 2.7 | 25-Nov-2021 | Title was changed. |
| 2.6 | 28-Oct-2021 | Added AT+WFAPUI: <timeout> valid value range<br>Updated Section 13.1.2<br>Updated Section 10.1.2.1<br>Updated 7 (5th parameter removed from AT+NWDHS)<br>Updated 7 (AT+NWDNS2 added)<br>AT+XTALRD, AT+FLASHDUMP : <CR><LF> is appended to data |
| 2.5 | 07-Sep-2021 | Updated table format of ATCMD (Prerequisite, example, note added) |


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| | | |
|---|---|---|
| | | Added Section 9.1 Zeroconf Commands<br>Added Section 10.2 Secure Socket Commands |
| 2.4 | 17-June-2021 | Updated MQTT Section 9.1.3, 9.1.4, and 9.1.5<br>AT+NWDHDNS deleted (not needed as WAN Port not available in Soft-AP mode)<br>Added AT+NWMQALPN, AT+NWMQSNI, AT+NWMQCSUIT, AT+SETDPMSLP1EXT, AT+DPMABNWFCCNT<br>Updated AT+WFJAP, AT+WFJAPA : Optional parameter <hidden> added<br>Updated Section 9.1.1, 9.1.2 |
| 2.3 | 01-Apr-2021 | Added OTA Update command<br>Added support for SDK V3.x.x.x |
| 2.2 | 15-Mar-2021 | Added Appendix B HTTP API Return Values<br>Added Appendix C<br>Added AT+NWMQAUTO and ATB |
| 2.1 | 13-Jan-2021 | New section added: Section 8.1 Wi-Fi Function Commands for WPA3<br>minor update (typo, or minor change done) |
| 2.0 | 08-Dec-2020 | Added additional description on the following commands<br>AT+WFSAP, AT+WFOAP, AT+WFTAP, ATF, AT+WFJAPA, AT+NWMQTT, +NWMQCL, AT+DPM<br>Added new sections:<br>7.1.4 MQTT Example: Changing Subscription Topic while running<br>7.1.5 MQTT Example: Reading Subscription Topic while running |
| 1.9 | 11-Nov-2020 | AT+NWCCRT, <ESC>C updated<br>AT+NWSNS updated<br>AT+NWHTS updated<br>AT+NWHTSS updated |
| 1.8 | 18-Aug-2020 | Added SNTP command to Section 5 Network Function Commands<br>Added Http-client command to Section 7.2 HTTP Commands<br>Added MCU FW update command using OTA to Section 7.3 OTA Commands paragraph |
| 1.7 | 30-June-2020 | Added 2.4 Configuration for MCU Wake-up<br>Correct typos and wordings |
| 1.6 | 29-Apr-2020 | Added AT+WFDIS and AT+SETDPMSLP2EXT<br>Updated MQTT commands to operate with one-port<br>Updated to process the comma in the parameters |
| 1.5 | 03-Apr-2020 | Added AT+BIDX for changing boot index in Chapter 4.<br>Added example code of MQTT command in Section 7.1.1 ~ 7.1.3<br>Updated RF Test function commands in Chapter 9.<br>Updated GPIO commands in Section 10.5 |
| 1.4 | 21-Oct-2019 | Updated 2.2 Serial Port configuration steps. Removed draft status |
| 1.3 | 15-Oct-2019 | Error correction<br>Added explanation to serial program at page 6 |
| 1.2 | 07-Oct-2019 | Editorial review and add code: UM-B-111 |
| 1.1 | 25-Jul-2019 | Added OTP Memory Address for writing MAC address in page 27 |


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| 1.0 | 03-Jul-2019 | Preliminary DRAFT Release |


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## Status Definitions

| Status | Definition |
|---|---|
| DRAFT | The content of this document is under review and subject to formal approval, which may result in modifications or additions. |
| APPROVED or unmarked | The content of this document has been approved for publication. |

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# DA16200 DA16600 AT Command

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